

LIVING DOCUMENT · UPDATED WITH EVERY CHANGE

FOX_Feature Implemented

Everything that has been built into FoxByte so far — for each feature: **what it does**, **how it works**, **how to use it**, **how to test it**, and its honest **status**.

When a feature is added or changed, this document changes in the same commit.

Product: FoxByte — serverless PostgreSQL with copy-on-write branching and a Blackbox

Last updated: 21 September 2026 · **Reflects:** `main @ 2778762` (v1.0.1, exact repository path PR #1) + the FoxByte rename and brand indirection (`chore/foxbyte`)

License: AGPL-3.0 core · Apache-2.0 client SDKs

Source of truth: `docs/FOX_Feature_Implemented.html` → `make feature-doc` → this PDF

Supersedes: `docs/foxbyte-feature-reference` (25 Aug 2026, no longer maintained)

Method: every entry was taken from a read of the source code and git history, with file references.

Status reflects the code; "Working" means the code path is complete, not that it was re-run for this edition unless the Test baseline section says so.

Change log

Newest first. Every change to FoxByte adds a row here.

Date	Feature(s)	What changed	Commit / PR
17 Sep 2026	H5, H15, H4, H2; known issues	Blackbox rules and risk flags judge the statement that runs. Both matched their patterns against <code>current_query()</code> — everything a client sent in one message — so <code>ALTER TABLE t ADD COLUMN note text DEFAULT 'drop column later'</code> was flagged and warned as dropping a column, a comment could trigger a rule, and in <code>ALTER TABLE a ADD x int; ALTER TABLE b DROP COLUMN y</code> the harmless first statement was warned or blocked for the second. Postgres gives an event trigger the command tag but not the statement's own text, so <code>ledger.sql</code> now has a tokenizer (<code>bb._sql_statements</code> : statements split at top-level semicolons, comments removed, standard, <code>E'...'</code> and dollar-quoted literals emptied, quoted identifiers kept; byte-based so multibyte text stays linear) and a per-backend position recording which statement is running, keyed by the query and <code>statement_timestamp()</code> (the same for every statement in one message, checked). The position lives in a table clients can't write and is moved by a function they can't execute — as a session setting it could have been set to line a blocked statement up with a harmless one. DDL run inside a function, a <code>D0</code> block or a trigger is recognised from <code>PG_CONTEXT</code> , which also names the statement Postgres actually ran, so it is matched by that text (closing an existing gap: a <code>D0</code> block that builds <code>'DR' 'OP COLUMN'</code> is now caught) and never moves the top-level position. Anything uncertain falls back to the whole query, so nothing that matched before stops matching. Previews (<code>fox policy check</code> , REST, MCP) match the same way. H4: flag now records the command <code>FLAGGED</code> ; <code>bb.policy</code> refuses any action but block/flag/allow (a typo used to switch a block off silently). H2: every change to <code>bb.policy</code> is recorded in the append-only <code>bb.policy_history</code> . A hang, found while testing and present on main: a blocked command after a Blackbox write in the same transaction — <code>BEGIN; CREATE TABLE a(); DROP TABLE b;</code> — hung forever, because the dblink connection recording the attempt waited for the append lock its own session held, and Postgres can't see that deadlock. The recorders now check <code>pg_locks</code> first and skip that one <code>BLOCKED</code> entry with a WARNING; the block stands and the policy evaluation is still recorded (<code>blackbox_id</code> null, as the contract already allows). docs/policy-errors.md describes what a pattern is matched against. Checked in the browser, signed in, on a throwaway branch of the owner's install carrying the new SQL (main and the running servers untouched, branch deleted afterwards): the Policies preview reports <code>drop-column</code> for a literal on main but "no rule matches" on the branch, and still reports a real drop in a multi-statement text; with the rule set to block, a statement quoting the phrase ran, a real <code>DROP COLUMN</code> and a <code>D0</code> block building it from strings were refused with BBX01, a <code>D0</code> block creating a table then dropping one returned the guardrail error at once, and the Changes page showed the literal add APPLIED, the drop FLAGGED/drop-column and the refusals BLOCKED. That session found two console bugs, fixed here: Allow & run again sent the guardrail override for a policy-rule block, so it could never work — the query API takes <code>allow_rules</code> (<code>bb.policy_allow</code> , rule ids validated) and	<code>fix/blackbox-precision</code>

the console offers **Allow rule <id> & run again**, confirmed in the browser to drop the column while a guardrail block still offers the plain override — and a command with no result rows printed its success line twice. Verified in the test VM: integration-v2 251 / 251 (24 new \$6b checks), integration 53 / 53, integration-update 51 / 51; `go test ./...` with `TestSchemaStatementMatching`, which fails if the revoke or the gate's statement matching is removed (both checked).

17 Sep 2026	M1, G1, H16; known issues	The integration suites run in a throwaway VM, and running them there found two bugs. The suites are destructive by design — <code>integration_test.sh</code> deletes main's whole Blackbox history to prove tampering is caught, restores main to an earlier point and fails HA over; all of them create accounts, keys and admin grants — and they ran in the same Lima VM as the developer's own install. That install lost its Blackbox history from before 16 Sep and was left with a dozen tables in main and 8 test accounts holding 70 working API keys. Cleaning up afterwards cannot undo a wiped ledger or a restore, and snapshotting main would leave the test run's segments in the real WAL archive, so the suites now run in their own VM. <code>scripts/test_vm.sh</code> creates <code>fox-test</code> (its own containers, storage, backups and accounts; Docker, ZFS, the test tools and Go checked against go.dev's digest), refuses the names <code>foxbyte</code> and <code>default</code> that fox would take for the real install, turns off port forwarding so the test stack never lands on the Mac's <code>localhost:8080</code>, and marks the VM; <code>scripts/lib/test_guard.sh</code>, sourced first by every suite, refuses to run without that mark and fails closed if the guard itself is missing. <code>make integration</code>, <code>integration-v2</code> and <code>integration-update</code> target it. On a fresh VM the first run gave 49 / 53, 226 / 227 and 51 / 51, and the five failures were two real bugs. G1: <code>fox start</code> launches three services that all create the auth store at once; on a new install two of them lost with <code>database is locked</code> (<code>SQLITE_BUSY</code>) and exited, so the control plane and the Agent API were down after the first start. Store creation now runs in <code>BEGIN IMMEDIATE</code> (which waits on the busy timeout) and retries a <code>BUSY</code>; <code>TestConcurrentFirstOpen</code> reproduced it (6 of 160 opens failed without the fix, none with it). H16: since agents connect through the Gateway (K5), the Gateway gave each connection a session of its own, overriding the session an agent branch is created with, so the agent's changes lost it; an agent's branch-scoped key now gets no Gateway session. Re-run on a fresh store: 53 / 53, 227 / 227, no first-start errors in the logs. With the owner's approval the leftovers were cleared from the real install: 70 test API keys revoked, a stale control plane on <code>:8090</code> stopped, two stopped test containers and the pre-cleanup snapshot branch removed.	<code>fix/test-isolation</code>
21 Sep 2026	M6, M4, M5, C8, A4, I1; known issues	A fresh install runs PostgreSQL 18, and an install made on 16 keeps running 16. The engine ran PostgreSQL 16 (Sept 2023); 18 is current, and it brings things this product's story rests on — asynchronous I/O for the hot path, page checksums on by default for a database that sells tamper-evidence, <code>uuidv7()</code>, and the <code>pg_upgrade --swap</code> machinery that makes the next major cheaper than this one. Nothing was in production, so this was the cheapest moment the move will ever have. The version was a free literal in four files that nothing kept in step; it now lives in <code>internal/branch/images.go</code> (<code>PGMajor</code>, <code>SupportedPGMajors</code>, <code>PostgresImageFor</code>, which also gives <code>brand.ImageRepo</code> its first caller — the full image name had been hardcoded, so a rebrand sweep edited it as	<code>feat/postgres-18</code>

text), and `TestPostgresMajorIsInStepEverywhere` holds the Dockerfile default, the release workflow's matrix and the Windows preload to it. **The image follows the data, not the binary:** the engine reads `PG_VERSION` from main's data directory and runs that major's image, so `fox update` onto this release leaves an install on 16 running 16 — `fox update` promises not to touch data, and moving a cluster between majors is a migration. That needs the old image to stay pullable, so the release workflow now builds one image per supported major from one parameterised Dockerfile (`ARG PG_MAJOR`), and a local build passes the major its tag names. Before every `docker run`, a branch whose `PG_VERSION` does not match the image's major is refused with both versions and the way out, where Postgres would have failed inside the container with "database files are incompatible with server", in a log nobody reads. Two smaller things fell out of reading the paths: the Windows setup probed a pre-16-Sep legacy image name to decide whether the preload was already loaded, which a newer distro never has, so every re-run loaded the 685 MB tarball again; and the published image now carries `org.opencontainers.image.source`, so its package links itself to the repository instead of starting out unlinked and private — which is what stopped the first install of `v0.1` with `unauthorized`. **Not yet proven:** the engine depends on Postgres behaviour no compiler checks (wal-g's base backups and restores, the `recovery.signal` PITR protocol, HA failback parsing `pg_controldata` by its English labels, the Blackbox's `PG_CONTEXT` string matching and event-trigger tag lists), so M6 stays Partial until `make integration`, `integration-v2` and `integration-update` have run on 18. No command yet moves an install from 16 to 18; that, and a portable `fox backup export` to take first, are the next batch. Verified here: `gofmt`, `go build`, `go vet` for macOS/Linux/Windows, `go test ./...` with six new unit tests, and the drift test checked to fail when the Dockerfile default is set back to 16.

21 Sep 2026	M3; B1, D1, H4	Two guard tests only failed on Windows, and both were about how a file is written rather than what is in it. The first green run of the merged branch failed on the <code>windows-latest</code> job alone. Paths: <code>TestDurableNamesAreNotSpeltOutTwice</code> exempts the package that defines the durable names by matching <code>internal/branch/</code> in the path, and the directory walk yields <code>..\..\internal\branch\branch.go</code> there — so the exemption matched nothing and the test reported all nine definitions in the package that owns them as duplicates. The walk now hands every rule a forward-slash path, with the separator replaced explicitly rather than by <code>filepath.ToSlash</code>, so <code>TestDurableNameExemptionsReadWindowsPaths</code> can feed it the Windows spelling and fail on every OS if it regresses — proved by reverting the helper and watching it fail on macOS. Line endings: <code>TestGeneratedFilesAreInStep</code> compares each generated file byte for byte with what <code>brand.json</code> produces, and Windows checks text files out with CRLF unless <code>.gitattributes</code> says otherwise; <code>web/src/brand.ts</code> was the one generated file with no rule, so a correct file looked like drift. The TypeScript sources are pinned to LF with the shell scripts, Go, SQL and the workflows; <code>install.ps1</code> stays CRLF and the generator keeps it that way. Reproduced locally first by converting <code>brand.ts</code> to CRLF, which failed with the same message. Neither is a defect	<code>fix/windows-ci</code>
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in the engine: both tests were right to be strict and wrong about the platform.

21 Sep 2026	E3, L1–L11 (new L11), L4, L6, L5, L9; known issues	The web console was redesigned around a FoxByte theme, and nothing mints a key behind the user's back any more. E3: the first <code>fox start</code> used to create an account called <code>local@foxbyte</code>, mint a key for it and cache it in <code>~/.fox/config</code> in plaintext — a credential nobody asked for, on an account nobody owned. Start now creates neither: it prints the connection string with <code><API_KEY></code> left to fill in and says how to make one, and the first account a person creates (web sign-up or <code>fox user create</code>) takes the admin grant on main through the new <code>auth.Store.OnFirstUser</code> hook. L11: the stylesheet was split in two — tokens (colour, a 1.2 type scale, a 4px space scale, radii, elevation, motion) and components that refer only to tokens, so the product can be reskinned from one block. The palette is the brand: an ember gradient (<code>#ff7a2f</code> → <code>#ffc24a</code>) on graphite in the dark theme, the same ember darkened to <code>#d2570c</code> for the light one; filled controls take <code>--accent-ink</code> rather than white, which was the reason the old orange buttons were hard to read. The mark is a fox head built from the same two angles as the branch graph (<code>components/brand.tsx</code>), drawn from CSS variables so it follows the theme, and shipped as <code>web/public/favicon.svg</code> for the tab. L4: the console is three cards instead of a loose stack — schema rail, a titled query card with its own toolbar, and a result with a status bar above the grid (command, rows, columns); the editor follows the theme instead of being a dark box on a white page. L5, L9: every filter moved into one bordered bar above its table, and page-level controls (branch, Refresh, Verify, Add rule) into a page header, so a page reads title-then-controls-then-content. L6: a new key is shown with the connection string already assembled around it, which is what the E3 change asks a person to do next. Tables share one style: a sticky quiet header, roomy cells, no zebra stripes, action columns that stay narrow and result grids that do not. The README opens with the logo (a light and a dark PNG, rendered from <code>docs/brand/logo.html</code>, which GitHub picks between by the reader's colour scheme), and its three console screenshots were retaken on the new theme — they still showed the purple UI of two names ago. One thing outside the UI came with it: the brand generator rewrote <code>install.ps1</code>'s generated block with LF line endings, so a file <code>.gitattributes</code> keeps as CRLF came back mixed — the block now follows whatever the file already uses. Verified: <code>gofmt</code>, <code>go build</code>, <code>go vet</code> for macOS, Linux and Windows, <code>go test ./...</code>, the TypeScript check and the production web build, plus every page driven in the browser pane in both themes and at phone width against a canned control plane.	feat/foxbyte -ui
21 Sep 2026	All; B1, D1, H4, L4; known issues	The product is FoxByte and the command is <code>fox</code> — and the names it stores no longer carry the brand, so the next rename costs one file. Renamed from OxynDB, which was renamed from VectoraDB, and each of those meant a fresh install because the name was written into the data. The names now split in two. What people read comes from <code>brand.json</code>: the product, the command, the environment variables (<code>FOX_*</code>), the state directory (<code>~/.fox</code>), the repository and module (<code>github.com/thefoxbyte/foxbyte</code>), these documents, the UI. <code>cmd/brandgen</code> turns that file into what cannot read it at runtime — Go constants, the shell library the suites source, the web UI's constants, and the generated block in each installer, since those are fetched standalone from	chore/foxbyte

GitHub — and `TestGeneratedFilesAreInStep` fails the build if any of them drifts. What is written into a database or onto disk is now brand-free and frozen: schema `bb`, roles `db_client` and `db_admin`, superuser `dbadmin`, database `appdb`, SQLSTATEs `BBX01` / `BBX02`, key prefix `key_`, cookies `dbengine_*`, pool `dbpool`, network `dbnet`, containers `pg-*`, object store `objstore` with volume `objstore-data` and bucket `wal-archive`, anchor format `ledger-anchor/1`.

`TestNoRetiredProductNames` fails on any retired name in the repository, and `TestStoredNamesAreBrandFree` fails if the product is named in the SQL the engine installs. Renaming also stopped being able to break behaviour: the database used to raise `'OxynDB guardrail: ... is blocked by policy'`, the SQL console matched that prose with a regular expression and a unit test asserted it — three places coupled by a brand string, where renaming one silently stopped the console recognising a blocked change; the message carries no product name now. The web wordmark was spelt across tags, which no text sweep catches, and is derived from the brand instead. Users keep what they have: an environment variable set under a retired prefix is still read, a state directory from a retired name is moved rather than abandoned (losing it left the object store's credentials out of step with its data), and a VM or distro from a retired name is still found. **Fixed on the way:** `setup` hung for ever when another project's container was called `minio` (the object store has a name of ours now, and the bucket step gives up with a message); containers are cleaned by label rather than by name prefix, so one left by an older name cannot keep holding ports and mounts; and renaming the object store while `AWS_ENDPOINT` still said `http://minio:9000` broke archiving, backups and restore at once — caught by the suites, now derived from one constant and pinned by a test. **New:** `fox uninstall` lists what it will remove, asks, and removes it — including binaries, VMs and state directories left by retired names. Verified: `go test ./...`, `gofmt`, `go vet` for macOS, Linux and Windows, the TypeScript check and web build, every CI script, and the three integration suites in a throwaway VM, which gained a check that a foreign `minio` container does not break `start`.

18 Sep 2026

A1, A2, A8, A9, M4, M5

Every reference names the repository exactly:

`github.com/thefoxbyte/foxbyte`. The rename wrote the organisation in lower case and the repository with a capital D, but the organisation was created as `FoxByte` and the repository as `foxbyte`. GitHub resolves any casing, so the install scripts, `fox update`'s release lookups and the links all kept working in a browser, which is exactly why it would have gone unnoticed — but the Go module path is case-sensitive, so it now matches the repository too (`module github.com/thefoxbyte/foxbyte`; the capital letter moves it ahead of other modules in import order, and `gofmt` re-sorted the two files where that mattered). Changed: 37 repository references (release downloads, the raw-install URLs, the GitHub API path and the `FOX_REPO` defaults) and 65 module-path references. Deliberately unchanged: container images stay `ghcr.io/thefoxbyte/...`, because image names must be lowercase, and `~/fox/auth.db` is the account database's file name, not the repository.

`TestNoOldProductNames` now also fails on the repository written with the wrong casing, while allowing the lowercase image names (checked by planting each form). Verified: every CI step locally — `gofmt`, `build`, `vet`, `go test ./...`, the release-checksum, release-asset, installer and checklist scripts

`fix/repo-path-foxbyte`

— and the `fox update` suite, which drives the release lookup against a fake GitHub.

18 Sep 2026	All; known issues	The product is FoxByte and its command is <code>fox</code> — renamed from VectoraDB and <code>vdb</code> before it was ever published. Nobody else ran it yet, so every identifier changed at once rather than leaving the old names behind as internals to explain later: the brand (FoxByte), the command and binaries (<code>fox</code> , <code>fox-verify</code> , release assets <code>fox-<os>-<arch></code>), environment variables (<code>FOX_*</code>), the state directory (<code>~/.fox</code>), the Go module and repository (<code>github.com/thefoxbyte/foxbyte</code> , <code>thefoxbyte/foxbyte</code>), the SQL schema, settings and roles (<code>bb.*</code> , <code>bb.actor</code> , <code>db_client</code> , <code>db_admin</code>), the policy SQLSTATEs (<code>BBX01</code> , <code>BBX02</code>), the API-key prefix (<code>key_</code>), containers (<code>pg-<branch></code>), the ZFS pool, Docker network, Postgres superuser and database, WAL bucket (<code>wal-archive</code>), the anchor format (<code>ledger-anchor/1</code>), the Lima and WSL instance names, and these documents (<code>FOX_*</code>). Feature IDs are unchanged. The rename was scripted in a fixed order, most specific name first, over tracked text files only; <code>go.sum</code> and the npm lock file were left out of the sweep because their base64 hashes can hold the old three letters by chance, and the lock file's own package name was changed by hand. No length-dependent code was affected — container names are parsed with <code>TrimPrefix</code> , not by position, though the new prefix is a character longer. <code>TestNoOldProductNames</code> now fails the build if an old name reappears anywhere in the repository, except on a line that records history by saying it was renamed from one (checked by planting each old form and watching it fail). What this breaks by design is anything outside the repository still holding old names: an existing install (its pool, containers, schema, bucket and keys) is replaced by a fresh one, MCP client configs need <code>fox mcp</code> and <code>FOX_API_KEY</code> , and releases and images start again under the new names. Verified: <code>go build</code> , <code>go test ./...</code> , <code>gofmt</code> , <code>go vet</code> for macOS, Linux and Windows, the web build, all six PDFs re-rendered, and the three integration suites in a test VM created under the new name.	chore/rename-foxbyte
18 Sep 2026	L4; known issues	The console shows a uuid as a uuid. Reported from the UI: the <code>id</code> column of every table in the user's own schema was displayed as a list of sixteen numbers. <code>pgx</code> decodes a <code>uuid</code> column into <code>[16]byte</code> , and that is what JSON makes of a byte array — <code>[195,64,152,78,...]</code> where <code>c340984e-87bf-4006-a013-72aca6fc7d77</code> belongs. Reading the whole type surface rather than that one column found three more of the same kind: an interval arrived as <code>{"Microseconds":7200000000,"Days":1,...}</code> , an <code>inet</code> carried its JSON quotes into the cell, and a <code>bytea</code> was turned into whatever its bytes looked like as text. Instead of formatting each type by hand, the console now asks <code>pgx</code>'s own type map for Postgres's text representation (<code>Map.Encode</code> with <code>TextFormatCode</code>) — what <code>psql</code> would print — for any value decoded into a byte array or a struct, which covers the geometric types, ranges, <code>macaddr</code> , time of day and hstore in the same stroke, and future ones too. Array members are rendered by their own element type, so a <code>uuid[]</code> is a list of uuids rather than a list of lists of numbers. Everything that already read well is untouched by construction: scalars, timestamps, numeric, jsonb and arrays of scalars take the paths they always did, and only a real <code>bytea</code> column is hex-encoded, since <code>pgx</code> also returns <code>[]byte</code> for a type it has no codec for (a custom enum arrives as the bytes of its text form)	fix/console-uuid-cells

and hex would make those unreadable. Two flaws in the suite came out of running it, both mine from earlier batches and both fixed rather than retried: the new bytea assertion passed its literal through as text (`decode('0102','hex')` is unambiguous), and §5b's filter counts used a fixed table name, so a second run on the same install counted the first run's entries as well — it now names its table per run, since the ledger is append-only and nothing clears it between runs. Verified: `go test ./...` with three new unit tests covering both the fixed types and the unchanged ones, `go vet` for macOS, Linux and Windows, `gofmt`, the TypeScript check, new integration-v2 §6 assertions over the REST API (267 / 267, with `make integration` 91 / 91), and in the browser against the real install — the reporter's own `app_user` table now lists `0955eb65-85ea-435d-8f48-13c51921ab4f` and its neighbours in the ID column.

18 Sep 2026

H3, K6, L5, K2; known issues

An agent can no longer be recorded as a person, and the MCP server acts as an account instead of trusting whoever starts it. H3: the Blackbox's claim is that it says who changed what, and `actor` was already non-forgable — the Gateway logs each client in as its own role, and `session_user` survives `SET ROLE`. `actor_kind` was not: it was read from `bb.actor_kind`, an ordinary session setting, so `SET bb.actor_kind = 'human'` was enough for an agent's changes to be recorded as a person's. It is derived from the login role now — an `agent-<id>` role is an agent, a per-user role is a person (one named for an email address is a person even if the address begins with "agent-"), and only the engine's own shared roles, which have no identity of their own, still take the injected value, set by engine code rather than by a client. A mechanism that looked promising was checked and dropped: `pg_settings` does not list custom placeholder GUCs at all in PostgreSQL 16, so a setting's connection-time value cannot be read back to tell the Gateway's own injection from a later `SET` — deriving from the role needs no such trick. `tool` and `session` stay client-declared by design (an `application_name` is whatever a client sends) and the Blackbox page now labels them **declared**, so the page distinguishes what the database attributes from what the client says about itself. K6: `fox mcp` took no credential — any client that could start it got all sixteen tools, and its changes were recorded against the shared role rather than an account. It now requires an API key (`FOX_API_KEY`, or `--key` for trying it by hand) and acts as that key's account, with no unauthenticated path: started without one it prints how to make a key and exits. `run_sql` records the account as the actor (`ClientQueryTextAs`), and a key scoped to one branch — the kind the Agent Branch API issues — is held to it: every call is pinned to that branch, an absent branch argument included, naming another branch is refused, and `create_branch`, `delete_branch`, `list_branches`, `blackbox_diff` and `branch_before_change` are refused outright. `docs/mcp.md`, the README, the Guide and the Docs page carry the key in their client configs. Two test flaws surfaced on the way and were fixed rather than retried: the suite's MCP helpers passed no key, and §3b read a count from main five seconds after a restart, comparing an empty string as an integer — it now waits for main over TCP (a socket probe answers from the temporary initdb server, as `waitReady` documents) and for a numeric count. Verified: `go test ./...` with three new unit tests, `go vet` for macOS, Linux and Windows, `gofmt`, the TypeScript check, the API drift check, `make integration-v2` 271 / 271 and `make`

feat/attribu
tion-mcp-
auth

integration 91 / 91 in the throwaway VM (11 new checks, including both forgery attempts) — a run in between failed 34 checks on the test VM's own Docker daemon (`setns ... No such file or directory from docker exec` against containers mid-restart, after a day of back-to-back suites); restarting the VM cleared it, which is worth knowing before reading such a run as a regression, and on the real install: on a throwaway branch upgraded to the new function, a per-user role that set `bb.actor_kind='agent'` and `bb.actor='agent-imposter'` was still recorded as `h3tester@example.com /human`, and an `agent-` role that claimed `'human'` and a chief executive's address was still recorded as `agent-h3fake /agent` — both read back from the Blackbox page, with the declared labels on tool and session. The branch, its test roles and the spare server were removed afterwards.

18 Sep 2026

D2, D3, D4, L3, K2; known issues

A restore can reach any point the archive covers, backups follow the primary, and a promoted standby archives WAL.

feat/durability-gaps

D3: `fox restore --to` always fetched the LATEST base backup, whatever point was asked for, so a target earlier than the newest backup could not be reached at all — recovery would start after it. The backup is chosen in Go now: the target time is parsed, the listing is read from a throwaway container, and the newest backup of this database (same system identifier) that had finished by then is fetched; a target no base backup precedes is refused, naming the oldest one rather than quietly restoring something later. The recovery script moved into the binary (`restore_pitr.sh` , run with `bash -c` exactly as `BranchBeforeEntry` runs its own), so installs whose image predates this get the fix from `fox update` alone, without a rebuilt image; `--to latest` is unchanged.

D2: `fox backup create` backed up `pg-main` even after a failover, when main is stopped and the promoted standby holds every write — it follows the primary pointer now and says when the primary is not `pg-main`; `fox backup list` and `fox status` no longer depend on `pg-main` running, since backups live in object storage rather than in any one container.

D4: the standby ran with no object-storage environment and no archive settings, so once promoted it archived nothing — everything written while main ran on it was outside both backups and point-in-time restore. Its container now carries the primary's wal-g environment and `archive_mode=on` , which archives only outside recovery, so nothing is double-pushed while it streams and the archive continues on the new timeline the moment it is promoted; a standby created before this change is named as unarchiving when `fox up` brings it back, because `archive_mode` can only be set at startup. **L3, K2:** time travel had no surface in the web at all, so the Dashboard gained a read-only **Base backups** panel — the earliest point reachable, each backup's time and size, and a copy-ready `fox restore --to` — over a new `GET /api/backups` ; restoring stays a CLI command, as it needs a host port and leaves a container behind. Verified: `go test ./...` with four new unit tests, `go vet` for macOS, Linux and Windows, `gofmt` , the TypeScript check, the API drift check, `make integration 91 / 91` (15 new checks, including a restore that reaches a write made after a failover) and `make integration-v2 261 / 261` in the throwaway VM, and the panel driven in the browser against the real install — where the command it offered for the oldest of 61 backups restored the database to 24 Aug 2026, choosing the right base backup and coming up without the `fox` schema, which did not exist then.

17 Sep 2026	L5, B1, G5, I8, C1, C2, C3; known issues	The Blackbox page filters and verifies, branches can be created from a branch, a continuous import can be watched and cut over from the browser, and fox vm exists. L5: the page had four of the API's eight filters and never showed the statement. It now filters by table, risk (drop, drop-column, type-change, security-definer, policy) and a since/until date range as well, fetches <code>with=session</code>, expands a row to its session, branch and full statement, and has a Verify button that reports the chain intact or names the first broken entry. Two defects came out of driving it in the browser rather than by API: typing in the table box lost the last keystroke, because the loader skipped a request while one was in flight (a request sequence number now makes a filter change always reload and drops stale answers), and the expanded statement was dark text on a dark panel, because the global <code>pre</code> rules only colour <code>pre</code> code. C1, C2: fox branch create <code><n> --from <branch></code> and <code>POST /api/branches {"name","from"}</code> (404 when the parent is unknown) copy any branch, with a parent picker on the Dashboard. The new integration checks immediately failed on a real bug: <code>zfs destroy</code> removed only the <code>main@for-<branch></code> snapshot, so a second create from the same non-main parent hit the leftover and returned 409. Delete now also destroys the clone's own origin snapshot — matched by prefix and <code>@for-<branch></code> suffix so it can never remove another branch's — and clone re-takes a stale snapshot. I8: continuous imports are readable and finishable without the CLI — <code>GET /api/replication</code>, <code>GET /api/branches/{name}/replication</code>, <code>POST .../replication/cutover</code> (409 unless the branch really is replicating), reading <code>pg_subscription</code>, <code>pg_subscription_rel</code> and <code>pg_stat_subscription</code> — behind a Continuous imports panel on the Import page with tables copied, the last message from the source and a Cut over button. Watching that panel exposed the worst bug of the batch: the import died about two minutes in, every time. The Gateway's reaper suspends a branch idle with no client connections, and a subscription's apply worker is a background worker, so a replicating branch looked idle — replication stopped silently and the branch vanished from the panel. The reaper now leaves a replicating branch running. G5: OAuth defaulted to <code>http://localhost:8080</code> and <code>http://localhost:5173</code>, neither of which a default install serves, so callbacks and the post-login redirect missed unless both URLs were set; the public URL now defaults to <code>https://localhost:8080</code> and the web origin to the public URL, and the session cookie follows the origin (Secure on https; SameSite=None only for a console on another origin, Lax otherwise). B1: fox vm status reports the engine VM's state, CPUs, memory and disk, and fox vm shell opens a shell in it — Lima on macOS, WSL on Windows, and on Linux both say there is no VM. Verified: <code>go test ./...</code> (six new unit tests), <code>go vet</code> for macOS, Linux and Windows, <code>gofmt</code>, TypeScript type-check, the API drift check (53 paths), <code>make integration 76/76</code> and <code>make integration-v2 261/261</code> in the throwaway VM, fox vm by hand on this Mac, and the whole batch driven in the browser against a spare control plane on the real install (see the test baseline).	feat/blackbox-page-gaps
17 Sep 2026	A8	A new release shows on the next fox start, not up to 6 hours later. fox update --check reported v0.8.7 while fox start said nothing. The start-time check remembered its answer for 6 hours — "up to date" included — and this	fix/test-isolation

Mac's cache held "up to date" from 15:12, before v0.8.7 was published, so every start in that window skipped GitHub. Only a found release is remembered now, so the notice still prints without a request once it has appeared; "up to date" and failed checks are not, and caches written earlier that hold "up to date" are ignored. Every other start asks for the releases list, conditional on the last ETag, so an unchanged list is a 304 that GitHub does not count against the rate limit. The unit tests' fake GitHub sent a fixed ETag and so answered "not modified" right after a publish; its ETag now follows the list, as GitHub's does. Verified: `go test ./...` with the new `TestNewReleaseShowsOnNextStart`, `go vet` for macOS, Linux and Windows, and against real GitHub using copies of this Mac's stale caches — v0.8.7 was found in 577 ms and then remembered.

17 Sep 2026

L4, H8, L9; known issues

Dropping a table from the console works, and the console lists your tables apart from FoxByte's. A user could not drop a table from the SQL console however they tried, even with `SET bb.allow_destructive=on` and a restart. Reproduced on a throwaway branch: as `db_client` the override was refused; as the same account's own role — already in `db_admin` — the guardrail let it through. The console logged every user in as the shared `db_client` role, which is never in `db_admin`, so nobody could override from it; and each run is a new connection, so a `SET` in one run was gone by the next, and the API refuses two statements in one run. Now the console logs in as the signed-in user's own role (acting as `db_client`, so data access and ownership are unchanged, as through the Gateway), and `allow_destructive` on the query request applies the override to that run. In the page an **Allow destructive changes** switch sits by Run and turns itself off after one run; a blocked change shows an **Allow & run again** button to admins and the grant command to others. Granting and revoking moved into the web: `GET|POST /api/branches/{name}/admins` and `DELETE .../admins/{email}`, with a **Who can override** panel on the Policies page — only an admin of a branch may change the list there, an unknown email is refused, and the last admin can't be removed. The schema list now hides FoxByte's `fox` tables by default, behind a **Show system tables** button (not remembered between visits, and not revealed by the filter); reload shows that it is working, refreshes the open table too, and reports errors instead of showing "No tables yet"; the list refreshes itself after DDL. Found along the way and recorded, not changed: about twelve tables in this install's `public` schema were left by test runs. Verified: `go test ./...`, `go vet` for macOS, Linux and Windows, TypeScript type-check and build, the API drift check, new integration-v2 §6 assertions (not yet run in the VM), and a live run against a spare control plane on the real stack (as above; temporary key, server and branch removed afterwards).

`fix/console-permissions`

16 Sep 2026

K5, K6; known issues

An agent can now reach the database it is given. The Agent Branch API handed back a DSN built from the container's address, which only resolves inside the VM — so on macOS and Windows an agent running on the host was given a database it could not open, while the README said "the agent connects to that dsn". Publishing a host port was the wrong fix and is deliberately refused by the design (`startContainer`: a direct connection would bypass the Gateway, its key, TLS and the Blackbox attribution), so the agent now connects the way every other client does — through the Gateway — with two

118363d (PR #39)

additions that keep it no more privileged than before. Its key is scoped to one branch (new `api_keys.scope`, with a migration, since the schema runs with `CREATE TABLE IF NOT EXISTS` and every existing install already has that table):

`Authn` refuses a scoped key outright, so it reaches neither the control plane nor the Agent API, and the Gateway opens only the branch it names. And the Gateway logs it in as the branch's own `agent-<id>` role, whose password is derived from the install secret with HMAC rather than stored or passed between processes, so `session_user` is still the agent and the recorded actor is still non-forgeable — `EnsureUserRole` is deliberately not called on that path, as it would reset the role's password and invalidate the DSN already issued. The key is revoked when the branch is deleted, but not on a reset, where the branch and the scope survive. Separately, the cap, the listing and the TTL reaper used `docker ps`, so a suspended agent branch — the usual state, since the Gateway suspends idle branches — escaped all three; they now use `docker ps -a`, and listings report `suspended` rather than claiming `ready`. Verified: `go test ./...`, `go vet` for macOS, Linux and Windows, six new unit tests (derivation is deterministic per branch and is not the install secret; the DSN parses, carries `sslmode=require` and omits the key in listings; the scope check rejects another branch, an empty target and a prefix match; a scoped key fails `userFromRequest` while an account key still passes; a store created in the pre-scope shape gains the column on open), plus new integration assertions that the DSN routes through the Gateway, that the agent's key cannot open `main` and is refused 401 by both HTTP APIs, and that it stops working once the branch is deleted. `agentDSN` and `randomPassword` had no callers left once the DSN changed, so both were removed with their tests, keeping the "never the superuser" assertion in `TestAgentGatewayDSN`. The VM suites still need a run.

16 Sep 2026

K2, L7, K6; known issues

The API description now matches the code, and the MCP server is documented. The spec is what client generators read, and it disagreed with the handlers: `/api/import` was documented as `{from, as}` while the code reads `{source, target, continuous}` and answers with an SSE stream; branch creation was documented 200 where the handler returns 201; the ledger's `since / until` filters, the multipart fields of `/api/import/file`, `/api/openapi.yaml` and all seven `/auth/*` routes were absent; and the `/blackbox` aliases existed only in prose, so a generated client got `/ledger` alone. All of that is corrected, the aliases are declared as path-item `$ref s` (one line each, so they cannot drift from the originals), and `internal/controlplane/openapi_test.go` keeps it honest: `net/http.ServeMux` exposes no way to enumerate its routes, so the test reads the `HandleFunc` literals out of `controlplane`, `auth` and `agentapi` and fails when a served route is undocumented or the spec names a path nobody serves. Two things the review list got wrong, found by reading the code rather than trusting the note: `POST /api/pipelines` genuinely returns 200, so "creates return 201" was applied only where the handler really does (branches, API keys, register); and the `FOX_WEB_ORIGIN` default on the Docs page (`http://localhost:5173`) was already correct, so it was left alone. **K6:** new `docs/mcp.md` documents all 16 tools with their arguments, the paste-ready client config, a stdio check, and the honest limits (no auth of its own, `run_sql` as the non-superuser `db_client`, `branch_before_change`

`docs/api-`
`accuracy`

taking minutes); linked from the README and the Docs page. K6 stays Partial — authentication is still open. L7: the Guide drops `make web-dev`, adds `sslmode=require` to all five connection strings and moves the agent API to `https://localhost:8088`; the Docs page grows from 7 endpoints to the full set plus an "Agents over MCP" section. Verified: `go test ./...`, `go vet` for macOS, Linux and Windows, `npm --prefix web run build`, the spec re-parsed as OpenAPI 3.1 (48 paths, every alias pointer resolving), and `scripts/test_feature_checklist.sh`.

16 Sep 2026	A4, A7, M5	Windows install verified end to end, and the summary is green again. A fresh install on a real Windows machine from v0.8.6 (after <code>wsl --unregister foxbyte</code>) exercised everything the earlier run could not: the distro was created from the prebuilt image, "Loading the preinstalled container images" ran with no apt install of Docker or btrfs tools — so the <code>ghcr.io</code> tag fix works — the engine download reported "fox-linux-amd64 from v0.8.6 verified against SHA256SUMS" (A7's first real-world proof), and setup ended with one summary carrying the working connection string and API key. Removing the installer's duplicate block had left that summary in plain text, so the engine now prints "FoxByte is running." in green itself: a Windows-only helper turns on the console's escape-sequence handling once and falls back to plain text when output is piped or the console refuses, so <code>install.log</code> never collects escape codes. A4 and M5 move to verified.	<code>fix/setup-summary-colour</code>
16 Sep 2026	A2, A4, M5	Windows install checked on a real machine; duplicate summary removed. Running the branch's installer on a Windows machine confirmed the new path: step 3 downloaded and verified the 685 MB <code>foxbyte-distro.tar.gz</code> (the old installer never fetched it), PATH was set and setup completed. It also showed the install ending with two near-identical "FoxByte is running" blocks — the engine's own summary and the installer's — so the installer now prints only its log path and leaves the summary (with the connection string and API key) to <code>fox setup</code> . What that run did not prove: it used the released v0.8.5 <code>bb.exe</code> , which predates this batch, and reused an existing distro, so the preload re-tag, the API-key summary and the btrfs storage wording (A4) were never executed. They need a run from a release containing this branch after <code>wsl --unregister foxbyte</code> , so A4 stays Partial.	<code>fix/download-integrity</code>
16 Sep 2026	A1, A2, A4, A7, M5; known issues	Downloads are verified, and Windows uses its prebuilt distro. Nothing FoxByte published was checked by its own installers, yet these files run as root inside the user's distro. Now <code>deploy/install.sh</code> , <code>deploy/install.ps1</code> and <code>fox setup</code> 's engine refresh all verify every download against the release's <code>SHA256SUMS</code> , and anything that cannot be verified stops the install (<code>FOX_NO_VERIFY=1</code> skips deliberately; a local <code>FOX_DIST</code> build has nothing to check against). The engine refresh reuses the same verified download as <code>fox update</code> through the new <code>update.Client.Release</code> , and keeps the installer-staged engine rather than installing unchecked bytes. The Windows installer now prefers <code>foxbyte-distro.tar.gz</code> (~685 MB, Docker + btrfs tools + engine + images already inside), so setup skips apt, a docker build and three registry pulls; releases without that asset (v0.7.1, v0.8.1) or an unverifiable copy fall back to the Ubuntu rootfs. The distro's Postgres image is tagged <code>ghcr.io/foxbyte/postgres-walg:16</code> — the name the engine actually runs, so the preload stops being wasted — and older distro images are re-tagged after load.	<code>fix/download-integrity</code>

Windows setup also prints the connection string with the API key (read from the guest's config), and its storage error now describes btrfs and `dbpool-storage.service` instead of the old ZFS pool. Verified: new `scripts/test_install_sh.sh` 6 / 6 (tampered, missing listing, unlisted asset, skip flag, local build) and new Pester cases, `go vet` for macOS, Linux and Windows, unit tests on every package; in the VM `make integration` 52 / 52 and `make integration-update` 51 / 51. The Windows install itself still needs a manual run.

16 Sep 2026	L8, L9, L10; A9	Batch 2 verification: the web console reviewed signed in, and fox update proven on macOS. The three pages that had only ever been type-checked were exercised in a real browser against a running stack, each answer compared with the CLI. L8 Integrity: the report matched <code>fox blackbox integrity</code> field for field; Create checkpoint anchored a new entry (and was correctly a no-op when nothing new existed); Export answered 200 <code>application/x-ndjson</code>; Branch from before a change restored a branch in 4 s that lacked exactly the table its entry created, with main untouched. L9 Policies: <code>warn</code> → <code>block</code> → <code>warn</code>, <code>disable</code> → <code>enable</code> and <code>add</code> → <code>remove</code> of a custom rule all matched <code>fox policy list</code> afterwards and reached only the selected branch; the statement check agreed with <code>fox policy check</code>; on <code>main</code>, where the account has no <code>db_admin</code>, the change was refused with a readable banner and the rule stayed <code>warn</code>. L10 sidebar: every link opened its page with the right highlight and breadcrumb (<code>/ledger</code> → <code>Changes</code>), collapse persisted across a reload and was applied before first paint, the theme toggle persisted, and no page logged a console error. Two small things recorded rather than fixed: the Integrity entry list needs a refresh to show a change recorded after the page opened, and the new-rule form pre-fills <code>Command</code> with <code>ALTER TABLE</code> as a real value that reads like a placeholder. The under-860px drawer is still unchecked — the automation could not change the page viewport. A9: updated this Mac for real twice (installed v0.8.3 with the installer → <code>fox update</code> to v0.8.4 → to v0.8.5), each time checking versions, restarted servers, containers not recreated, branch data, Blackbox, gateway, UI and the kept previous binary; Windows stays untested by choice. Test branches, the temporary key and the <code>db_admin</code> grant were all removed afterwards. Documentation only — no product change in this commit.	<code>docs/batch2-verification</code>
16 Sep 2026	A8, A9; known issues	The new-release notice now actually appears. Testing <code>fox update</code> against real releases on a Mac (v0.8.3 installed with the installer, then the published v0.8.4) showed the notice missing, while <code>fox update --check</code> reported the release correctly a moment later. (Precisely: it printed on the first start after a release was published, then stayed silent — the first fetch of a new asset is fast, repeats are throttled.) Cause: the check made two requests, and the second — downloading the release's <code>SHA256SUMS</code> — is throttled hard by GitHub for repeated downloads of the same asset. Measured from this Mac: 0.5 s the first time, then 7 s from Go and 75 s with curl and with HTTP/1.1, so the 1.5 s budget was never met (with a 30 s budget the notice did print). The start check now asks only for the releases list — newer, published, with every file this platform needs attached — which takes about 130 ms, and remembers its answer in <code>~/.fox/update-notice.json</code> for 6 hours (<code>FOX_UPDATE_CHECK_INTERVAL</code>; 0 checks every start), so repeated starts usually make no request at all. Nothing about installing changed: <code>fox update</code> still	<code>fix/update-notice-throttle</code>

downloads `SHA256SUMS` and verifies every file before anything is replaced. Verified: unit tests (new `TestAvailableMakesNoAssetRequests`, `TestNoticeIsRememberedBetweenStarts`), `go vet` for macOS, Linux and Windows; on this Mac a build stamped 0.8.3 printed the notice within the default budget (2.1 s cold, 1.2 s from the remembered answer); in the VM `make integration-update` 51 / 51 (2 new §1b checks) and `make integration` 52 / 52. A9's macOS path is now verified end to end (versions, restarted servers, containers, branch data, Blackbox, gateway, UI, kept previous binaries), and again on the released fix: v0.8.5 (which contains it) was installed with `fox update` from v0.8.4 — servers restarted on the new binary, `pg-main` not recreated, UI 200, previous binary kept — and a build of the merged code printed the notice in 2.1 s cold and 1.2 s from the remembered answer. Windows testing is deferred by the owner (16 Sep 2026).

15 Sep 2026	C3, D4, H7, I1, I5, I6, B1; known issues	Safety fixes, batch 1. C3: <code>fox branch suspend</code> refuses <code>main</code>, the HA standby, and whichever container serves <code>main</code> (the CLI allowed suspending <code>main</code>, which the gateway then couldn't wake). D4: reading the code showed the HA hazard was worse than recorded — after <code>fox ha failover</code>, <code>ha enable</code> and <code>ha disable</code> deleted the promoted standby, i.e. the live primary and every write since the failover, <code>ha failover</code> again stopped it, and <code>fox up / start</code> restarted the stale old <code>main</code> as a second writable primary. Now those three are refused after a failover, <code>fox up</code> brings the promoted standby back (recreating its container from its storage after <code>fox stop</code>) and leaves the old <code>main</code> stopped, and the new <code>fox ha failback</code> moves <code>main</code> back to its own container keeping every write: <code>main</code> is rebuilt as a replica of the standby, catches up, the standby is stopped cleanly, <code>main</code> replays to its final checkpoint and is promoted; any failure before that leaves the standby primary. H7: <code>fox blackbox revert --to</code> keeps working but its help and output now say it is a point-in-time restore of <code>main</code> (same as <code>fox restore</code>) and point to <code>branch-before</code>. I5 / I1: <code>.sql</code> and Postgres imports run the whole dump, then fail listing every statement that failed (the instance is kept) — they used to report success. I6: a JSON array or NDJSON file with an invalid document fails, naming the element or line, and loads nothing from it — an array used to stop silently at the bad element. Integration §3 fix (test-only): its reset of the <code>pit</code> table ran <code>DROP TABLE</code> without the guardrail override, so it had silently failed for a long time and passed only because the old <code>ha disable</code> threw the failover rows away; failback keeps them, which exposed it. Verified: unit tests on every package, <code>go vet</code> for macOS, Linux and Windows; in the VM <code>make integration</code> 52 / 52 (24 new checks: suspend guards, HA guards, <code>fox up</code> after a failover, failback, import errors), <code>make integration-v2</code> 214 / 214 and <code>make integration-update</code> 49 / 49 (no regressions).	<code>fix/safety-batch1</code>
15 Sep 2026	M1, M2; new document	New companion document: FOX_Checklist. <code>docs/FOX_Checklist.pdf</code> (source <code>docs/FOX_Checklist.html</code>) lists every feature in this document by the same id with its sub-features, whether each is built, partly built or not built, and what tested it — done & tested (automated), done (manual test only), partial, known issue, not started — plus a "Where we are" summary per area, suggested next steps, a "Not started" roadmap, and a note confirming FoxByte has no custom storage engine (PostgreSQL's own engine is used by design; the storage layer is the ZFS/btrfs copy-on-write drivers and wal-g durability). <code>make feature-</code>	<code>docs/feature-checklist</code>

`doc` now renders both PDFs, and first runs `scripts/test_feature_checklist.sh`, which CI also runs: it fails if a feature id is in one document and not the other, or a row's badge disagrees with its status (checked against a deliberately broken copy). Documentation and tooling only; no product change.

15 Sep 2026	new A8, A9; B1, M1, M2, M4	New-release notice on <code>fox start</code> and <code>fox update</code>. When a newer release is published, <code>fox start</code> ends with one line saying so and suggesting <code>fox update</code> — no prompt, checked in the background with a 1.5 s limit, silent when offline, off with <code>FOX_NO_UPDATE_CHECK=1</code> and on development builds. It lists releases through the GitHub API and offers only a promoted (not draft or prerelease), newer release with every file the platform needs listed in <code>SHA256SUMS</code> . <code>fox update</code> installs it in one command: downloads and checks every file first (nothing changes unless all verify), checks the new engine runs in the VM / WSL distro, stops the three servers (databases keep running), installs the engine (servers restarted on the old one if that fails), runs <code>fox up</code> , <code>fox ledger upgrade --all</code> and <code>fox start</code> on the new engine, then replaces <code>fox</code> on the computer; previous binaries are kept in <code>~/fox/updates/prev</code> . <code>--check</code> , <code>--yes</code> , <code>--version</code> . Data, containers, images, storage and settings are not touched. The only change to existing behaviour is the extra notice line after <code>fox start</code> . The release job now fails unless every file <code>fox update</code> needs is attached and listed in <code>SHA256SUMS</code> (<code>scripts/check-release-assets.sh</code> , tested in CI). The first VM run caught a real bug: <code>stdin</code> from <code>/dev/null</code> looked like a terminal, so <code>fox update </dev/null</code> silently did nothing — it now asks for <code>-yes</code> ; three test checks were also corrected (history compared by count instead of by the entries themselves, and a settings file that doesn't exist in the VM). Verified: unit tests on every package, <code>go vet</code> for macOS, Linux and Windows; in the VM <code>make integration-update 49 / 49</code> (Linux path against a local fake GitHub) and the existing suite 28 / 28 (no regressions). The macOS and Windows update paths are unit-tested but not yet run against a real release.	feat/fox-update
15 Sep 2026	new L10; L1–L9	Web console: sidebar layout for the signed-in app. The top navbar held every page in one row and was already full. Signed-in pages now use a sidebar grouped into Overview (Dashboard), Data (Console, Import, Pipelines), Blackbox (Changes, Integrity, Policies), Account (API keys) and Help (Guide, Docs), with GitHub, a light/dark switch, the signed-in account, Log out and Collapse at the bottom; it collapses to an icon rail (remembered per browser) and becomes a drawer on narrow screens. A slim top bar shows the section and page. Home and Log in keep the public navbar, which now offers "Open dashboard" to a signed-in user; Guide and Docs appear in whichever layout fits. Every address is unchanged; no API or behaviour change. TypeScript type-check and production build pass; a preview build embedded in the engine served every page (<code>/</code> , <code>/dashboard</code> , <code>/login</code> , <code>/blackbox</code>) with the new layout code. Not yet reviewed in a browser while signed in.	feat/web-sidebar-layout
15 Sep 2026	M2, M4, M5; known issues	Release checksums can no longer be dropped by the other release job. v0.8.1's <code>SHA256SUMS</code> lost the Windows distro image's entry: <code>wsl-distro.yml</code> ran first (the <code>release</code> workflow had been disabled), then a manual <code>release</code> run replaced the whole file with only the binaries' checksums — so the Windows installer would install the distro image without verifying it. Both workflows now publish through one script,	fix/release-checksums

`scripts/release-checksums.sh` , which merges its own entries into the published file by exact name, keeps the other job's, and re-downloads after uploading to confirm its entries survived (merging again if the other job uploaded at the same moment). `wsl-distro.yml` also creates the release before publishing checksums.

`scripts/test_release_checksums.sh` (run by CI) tests it against a fake `gh` : no file yet, either job finishing first (same result), a re-run replacing its own entry, exact-name matching, another job uploading at the same moment, a failed upload — 9 / 9 on macOS and on Ubuntu (gawk). v0.8.1 itself still needs `wsl-distro.yml` re-run for its tag by a repository admin to restore the distro image's checksum.

15 Sep 2026	D1, A4 (Windows distro), CI; known issues	MinIO images moved to quay.io (release fix). The <code>wsl-distro</code> release job failed: <code>minio/minio</code> and <code>minio/mc</code> no longer exist on Docker Hub, so the image preload's <code>docker pull</code> was refused — and a fresh install's <code>fox up</code> would have failed the same way (existing installs kept working from their cached images). The engine and the distro build now use <code>quay.io/minio/minio:RELEASE.2025-09-07T16-13-09Z</code> and <code>quay.io/minio/mc:RELEASE.2025-08-13T08-35-41Z</code> — the exact versions already running and tested, so nobody gets a silent MinIO upgrade. An install with the old images cached keeps using them (no pull needed); <code>FOX_MINIO_IMAGE</code> / <code>FOX_MC_IMAGE</code> override. New <code>internal/branch/images.go</code> with unit tests, including one that keeps <code>deploy/wsl-distro/build.sh</code> preloading the same names. Verified in the VM: both pinned images pull and save as the distro build does, the stack restarted on the quay.io server image against the existing data, and integration-v2 214 / 214 and the existing suite 28 / 28 pass (backups, PITR, branch-before included).	<code>fix/minio-images-quay</code>
15 Sep 2026	new H17, H18; H15, H16, B1, K1, K2, K6; known issues	Blackbox 2.0 Phase 7 — impact analysis and diff. <code>fox impact "<DDL>"</code> reports what a change would affect before it runs: the objects that depend on the table, view, index or column it changes (new read-only SQL function <code>bb.blast_radius</code> , following views built on views), the other running branches that have it (with the branch each was cloned from, read from ZFS/btrfs), the policy verdict, and a low/medium/high score with its reasons. MCP <code>execute_change</code> results now include it. <code>fox blackbox diff <a> </code> (and <code>fox branch diff</code> , which the README already advertised) finds where two branches' Blackbox histories split and lists the changes only on each side and the objects both changed — diff only, nothing is merged. New REST <code>POST /api/branches/{n}/impact</code> , <code>GET /api/ledger/diff</code> and <code>/api/blackbox/diff</code> ; MCP <code>impact</code> and <code>blackbox_diff</code> . The first VM run caught a real bug — <code>bb.blast_radius</code> pinned its search path, so unqualified table names weren't found — fixed before this commit. Verified in the VM: integration-v2 214 / 214 (31 new \$8 checks) and the existing suite 28 / 28 (no regressions).	<code>feat/blackbox-phase7-impact-diff</code>
14 Sep 2026	new H16; H9, B1, K1, K2, K5, K6	Blackbox 2.0 Phase 6 — agent provenance. Blackbox now records which agent session, task and parent session caused a change. New append-only <code>bb.agent_sessions</code> . The Agent API accepts optional <code>task_id</code> , <code>parent_session_id</code> and <code>session_id</code> when creating an agent branch — every change the agent then makes there carries them (requests without them are unchanged, response included). New MCP tool <code>execute_change</code> : previews policy rules, runs the SQL without superuser rights with the MCP session, task, parent	<code>feat/blackbox-phase6-provenance</code>

session and a sha256 call hash attached, and returns JSON (applied / blocked with the `BBX01` rule / error / preview) with the Blackbox entries it wrote. New `fox blackbox sessions` and REST `.../ledger/sessions` (also under `/blackbox`). `run_sql` is unchanged. Verified in the VM: integration-v2 183 / 183 (23 new §7 provenance checks) and the existing suite 28 / 28 (no regressions).

14 Sep 2026	M (CI); H15 tests	Windows CI fix (tests and repo config only). Two new policy unit tests assumed LF line endings, but a Windows checkout converts <code>.sql</code> and <code>.md</code> files to CRLF, so <code>TestSchemaPolicy</code> and <code>TestPolicyContractExample</code> failed on the <code>windows-ci</code> runner. The tests now normalise line endings, and <code>.gitattributes</code> pins <code>*.sql</code> to LF (like <code>*.sh</code> and <code>*.go</code>), so the SQL embedded in <code>fox</code> is the same bytes on every platform. No product change. Reproduced locally in a CRLF checkout (both tests failed as on the runner) and verified the fix there: every Go package passes with CRLF <code>.sql</code> and <code>.md</code> files, and a fresh checkout now gets LF <code>.sql</code> files.	feat/blackbox-phase5-policy
14 Sep 2026	H9 (security fix); known issues	Capture can't be switched off by a client. The Blackbox 2.0 kill switch <code>bb.v2 = 'off'</code> was a session setting any client could set, so a client could stop the capture rows (transaction id, WAL position, provenance, override use) for its own changes. It is now honoured only when set for the whole database (<code>ALTER DATABASE</code> , owner or superuser) or in a superuser's own session; a client's own <code>SET</code> — even a <code>db_admin</code> member's — is ignored. The check moved inside the capture trigger's fail-safe block. The documented kill switch and the engine's superuser use of it are unchanged. Verified in the VM: integration-v2 160 / 160 (3 new §2 checks) and the existing suite 28 / 28.	feat/blackbox-phase5-policy
14 Sep 2026	new H15, L9; H4, H9, B1, K1, K2, K6	Blackbox 2.0 Phase 5 — policy gate. Every schema change is checked against policy rules before it runs (event trigger <code>bb_policy_start</code> , after the unchanged guardrail). A warn rule lets it through with a NOTICE (SQLSTATE <code>BBX02</code>); a block rule refuses it with <code>BBX01</code> , records a <code>BLOCKED</code> Blackbox entry and an evaluation that survive the rollback, and can be overridden per rule by <code>db_admin</code> members (<code>SET bb.policy_allow</code>). <code>DETAIL</code> is the JSON defined in <code>docs/policy-errors.md</code> . Four built-in rules ship, all warn: <code>alter-column-type</code> , <code>drop-column</code> , <code>drop-index</code> , <code>grant-to-public</code> . New: <code>fox policy list check block warn enable disable add remove evaluations</code> , REST <code>/api/branches/{n}/policies...</code> (changes need <code>db_admin</code>), MCP <code>policy_check</code> , web Policies page. Verified in the VM: integration-v2 157 / 157 (50 new §6 policy checks) and the existing suite 28 / 28 (no regressions).	feat/blackbox-phase5-policy
14 Sep 2026	new H14; whole document; H2, H6, H10–H13, B1, K1, K2, K3, K4, K6, L5, L8	Schema Ledger renamed Blackbox. Everything people read now says Blackbox: this document, the README, the web console (nav "Blackbox", new address <code>/blackbox</code>), CLI help, MCP and OpenAPI descriptions, SDK descriptions and READMEs, code comments. New names were added next to the old ones, and nothing was removed: <code>fox blackbox ...</code> , <code>/api/branches/{n}/blackbox...</code> routes, MCP <code>verify_blackbox</code> / <code>blackbox_integrity</code> / <code>blackbox_entries</code> , SDK <code>blackbox()</code> / <code>verify_blackbox()</code> / <code>verifyBlackbox()</code> . Unchanged on purpose: SQL objects (<code>bb.schema_ledger</code> ...), the anchor format id, the Go package, command output lines and JSON field names. Also: the policy-gate error contract	feat/blackbox-phase5-policy

		docs/policy-errors.md (Phase 5, written and approved before any policy code: BBX01 block / BBX02 warn, JSON DETAIL, db_admin per-rule override, all default rules warn, blocked attempts recorded). Verified in the VM: integration-v2 107 / 107 (10 new §5 alias checks) and the existing suite 28 / 28 (no regressions).	
14 Sep 2026	new H13; D3, B1, K1, K2, K6, L8	Blackbox 2.0 Phase 4 — branch from just before a ledger entry. fox ledger branch-before <id> creates a new branch holding main exactly as it was the moment before that change committed: it restores the newest base backup that finished before the change and replays archived WAL up to, but not including, the change's transaction (the xid captured in Phase 2; older entries fall back to their timestamp). main is not modified; the published Postgres image is unchanged (the restore script is embedded in the engine); fox restore and fox ledger revert are untouched. New: fox ledger entries (entries with their ids — the existing ledger views are unchanged and show none), REST GET .../ledger/entries and POST .../ledger/{id}/branch, MCP ledger_entries and branch_before_change, a panel on the web Integrity page listing recent entries. Test-only fix in integration_ledger_v2.sh §2: the "base ledger still records that DDL" check counted rows by table name, so it failed when the suite ran twice without the existing suite's ledger wipe in between; it now counts only the current run's rows. Verified in the VM: integration-v2 97 / 97 (twice back-to-back, including 34 new §4 checks) and the existing suite 28 / 28 (no regressions).	feat/ledger2-phase4-branch-before
14 Sep 2026	new H10, H11, H12, L8; H2, H9, B1, K1, K2, K6, M1, M4	Blackbox 2.0 Phase 3 — tamper-evidence outside the database. Checkpoints record a Merkle root over new ledger entries in bb.ledger_checkpoints and as read-only anchor files outside the database (format published in docs/ledger-anchor-format.md), on a schedule and on demand. fox ledger integrity checks the ledger against the anchors and catches history rewritten even with a recomputed hash chain; the new open-source fox-verify does the same independently, live or from an export. New: fox ledger checkpoint integrity export, REST .../ledger/integrity checkpoint export, MCP ledger_integrity, web Integrity page, fox-verify release binaries. Verified in the VM: integration-v2 63 / 63 and the existing suite 28 / 28 (no regressions).	feat/ledger2-phase3-checkpoints
14 Sep 2026	new H9; H4, H8; test script	Blackbox 2.0 Phase 2 — capture foundation. New side table bb.ledger_ext records the transaction id, WAL position, agent task / tool-call hash and override use for every ledger row, through a fail-safe trigger with a kill switch (bb.v2 = off); installed at start and by fox ledger upgrade. The base ledger and its hash chain are untouched. Phase 1 verified in the VM (existing suite 26/28, v2 29/29) and Phase 2 verified (v2 42/42, no regressions); two false failures in integration_test.sh fixed (test-only).	56bfb81 (PR #24)
14 Sep 2026	D3, H3, H4, K5, K6, B1; new H8	Blackbox 2.0 Phase 1 — three security fixes. PITR restore uses the per-install MinIO credentials. The destructive-DDL override is honoured only for superusers and db_admin members (new fox admin grant revoke list and fox ledger upgrade; the first local user is granted automatically). Agent branches log in as their own non-superuser agent-<id> role and MCP run_sql runs as db_client (compatibility switches FOX_AGENT_SUPERUSER, FOX_MCP_SUPERUSER). New scripts/integration_ledger_v2.sh (make	201676c (PR #23)

integration-v2) pins unchanged behaviour and tests the fixes.

14 Sep 2026	Whole document	Document created. Inventory of every feature implemented as of 8b1856f : install & setup, CLI, branching & storage, durability, security, gateway, auth, Blackbox, import, pipelines, APIs/SDKs/MCP, web console, CI & releases. Known issues and the test baseline recorded.	branch docs/feature -implemented
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Test baseline

Recorded before any Blackbox 2.0 work starts, so later changes can be checked against it.

Suite	Result	Details
Go unit tests (go test ./... , host)	All pass	14 Sep 2026, go1.26.6 darwin/arm64, commit 8b1856f . Packages with tests: cmd/fox, auth, branch, controlplane, daemon, host, proxy (later also ledger, mcp, update). 15 Sep 2026 with feat/fox-update : all pass. 16 Sep 2026 with the start-notice fix: all pass. 21 Sep 2026 with the FoxByte rename: all 12 packages with tests pass, including the new internal/brand and internal/brand/gen (generated-file drift, the installers' variables, the state directory moving from a retired name, environment variables read under a retired prefix) and the repository-wide guards TestNoRetiredProductNames , TestStoredNamesAreBrandFree and TestDurableNamesAreNotSpeltOutTwice . 17 Sep 2026 with the Blackbox page and small gaps: all 10 packages pass, including the new TestVMSubcommand , TestParseReplication , TestIsOriginSnapshotFor , TestURLDefaults , TestSessionCookieSameSite and TestCanSuspend ; go vet clean for macOS, Linux and Windows. 18 Sep 2026 with the durability gaps: all pass, including TestPickBackupForTime , TestParseRestoreTime , TestRestorePITRScript and TestStandbyRunArgsArchive . Same day with attribution & MCP authentication: all pass, including TestActorKindDerivedFromRole , TestAuthenticateWithoutAKey and TestApplyScope . No test files: agentapi, ledger, mcp, secrets, tlsutil, version, web.
Integration (make integration , Lima VM)	52 / 52	14 Sep 2026. Before Phase 1 (a5a52b6): 25 passed, 3 failed — bad-key count, PITR (the minioadmin bug), append-only check. After Phase 1 (33f49ef): 26 passed, 2 failed — PITR fixed, no regressions. The remaining two were test flaws (psql prints the bad-key error twice; the append-only check discarded stderr); the protections were confirmed by hand and the checks fixed. With Phase 2 (feat/ledger2-phase2-capture): 27 passed, 1 failed — the bad-key check again, because its first fix (grep -q) broke under the script's set -o pipefail ; corrected to count matches and re-checked in the VM. With Phase 3 (feat/ledger2-phase3-checkpoints): 28 / 28 — every check passes. With Phase 4 (feat/ledger2-phase4-branch-before): 28 / 28 on the final code (also 28 / 28 on the first Phase 4 build) — no regressions. With the Blackbox rename (feat/blackbox-phase5-policy): 28 / 28. With the policy gate : 28 / 28. With the capture kill-switch fix : 28 / 28. With Phase 6 : 28 / 28. With Phase 7 : 28 / 28. With fox update (feat/fox-update): 28 / 28. With safety fixes batch 1 (fix/safety-batch1): 52 / 52 — 24 new checks (suspend and HA guards, fox up after a failover, failback,

import errors); first run 51 / 52 and second 49 / 52, both on test flaws (a container-absent check, and §3's unguarded table reset), fixed. **With the start-notice fix:** 52 / 52. **In the throwaway test VM (fix/test-isolation):** 53 / 53. **With the Blackbox page and small gaps (feat/blackbox-page-gaps):** 76 / 76 — 23 new checks (§1b hosting and login URLs, §2b branch from a branch, §5b continuous import and the reaper). The first run was 72 / 1: creating a branch from a non-main parent a second time failed on a leftover ZFS snapshot — a real bug the new checks found, fixed and re-run. **With the FoxByte rename (chore/foxbyte):** 104 / 104, including a new §0 (a foreign container named "minio" does not break the stack) and §12 (fox uninstall removes everything, keeps data with --keep-data, and finds nothing left on a second run). **With the durability gaps (feat/durability-gaps):** 91 / 91 — 15 new checks (§3b restore target selection, and six in §6 for backups and archiving after a failover). **With attribution & MCP authentication:** 91 / 91.

fox update (make
integration-update , Lima
VM)

51 / 51

15 Sep 2026, feat/fox-update . First run 41 / 45: one real bug (stdin from /dev/null treated as a terminal) and three flawed checks, all fixed; final run 49 / 49. **With safety fixes batch 1:** 49 / 49. **With the FoxByte rename:** 51 / 51. **With the start-notice fix (fix/update-notice-throttle):** 51 / 51, including two new §1b checks — the remembered answer prints with GitHub unreachable, and a forced check then prints nothing.

Blackbox 2.0 (make
integration-v2 , Lima VM)

214 / 214

14 Sep 2026, 33f49ef : §0 behaviour unchanged and §1a–§1c Phase 1 fixes all pass. With Phase 2: 42 / 42. **With Phase 3 (feat/ledger2-phase3-checkpoints):** 63 / 63, including 20 new §3 checkpoint, verifier and tamper checks and the §3b scheduler check. **With Phase 4 (feat/ledger2-phase4-branch-before):** 97 / 97, including 34 new §4 branch-before and entry-listing checks; run twice back-to-back after fixing a §2 check that wasn't repeatable (first final run 96 / 97 on that check only). **With the Blackbox rename (feat/blackbox-phase5-policy):** 107 / 107, including 10 new §5 checks that every Blackbox name matches the original. **With the policy gate:** 157 / 157, including 50 new §6 policy checks. **With the capture kill-switch fix:** 160 / 160. **With Phase 6 (feat/blackbox-phase6-provenance):** 183 / 183, including 23 new §7 provenance checks. **With Phase 7 (feat/blackbox-phase7-impact-diff):** 214 / 214, including 31 new §8 impact and diff checks. **With safety fixes batch 1:** 214 / 214. **In the throwaway test VM (fix/test-isolation):** 227 / 227. **With the console permission fixes:** 255 / 255. **With the Blackbox page and small gaps:** 261 / 261, including 6 new §5b checks for the page's filters and session column. **With the FoxByte rename:** 277 / 277. **With the durability gaps:** 261 / 261 — no regressions. **With attribution & MCP authentication (feat/attribution-mcp-auth):** 271 / 271, including 11 new §1c checks for the MCP key, its branch scope and both forgery attempts. Three earlier runs were not clean and none of it was the product: the suite's own MCP helpers passed no key, count-based §5b assertions counted the new §1c tables on main (moved to their own branch), §3b compared an empty count as an integer five seconds after a restart (it now waits for main over TCP and for a numeric answer), and one run lost 34 checks to the test VM's wedged Docker daemon.

Web console (browser, signed in)

L3–L5, L8–L10, I8, D2–D3, H3 pass

16 Sep 2026, Chrome on macOS against the running stack (v0.8.5). Every Integrity and Policies result was compared with

the CLI; the sidebar was walked link by link. **17 Sep 2026**, against a development control plane on a spare port of the real install: the Blackbox page's Verify agreed with `fox ledger verify` (270 chained entries), the table filter returned exactly the 6 rows SQL counted, risk and date filters matched their SQL counts, and an expanded row showed its session, branch and statement; the Dashboard created a branch from another branch and the child's console read the parent's row; and an import replicated a live source into a branch, showed "1 of 1 tables, streaming" in the Continuous imports panel, carried a row inserted on the source, and cut over from the panel — after which later source changes did not arrive and the copied rows stayed. Two defects were found this way and fixed (a filter request race that dropped the last keystroke, and an unreadable statement in the expanded row), as was the reaper suspending a replicating branch. Test branches, containers and the spare server were removed afterwards. **18 Sep 2026**, same arrangement: the Dashboard's new **Base backups** panel listed this install's 61 real backups with the times and sizes `wal-g` reports, flagged the newest, named 25 Aug 2026 as the earliest point reachable, and gave a copy-ready command matching each backup's own finish time; the unbounded first version filled the page, so it now shows six with "Show 55 older backups". The command it offered for the oldest backup was then run against the real install: it chose `base_000000010000000000000021` — the newest backup preceding the target, not the newest of the 61 — and the restored copy came up with no `fox` schema at all, which is correct for 24 Aug, three weeks before the Blackbox existed. The disposable container was removed afterwards. Also on 18 Sep 2026, for H3: on a throwaway branch upgraded to the new `_ctx`, a per-user role that declared itself an agent (`SET bb.actor_kind='agent'`, `bb.actor='agent-imposter'`) and an `agent-` role that declared itself a person (`'human'`, a chief executive's address) were both read back on the Blackbox page as what they really were — `h3tester@example.com /human` and `agent-h3fake /agent` — with **declared** labels on the tool column and on the session in the expanded row. The branch and its two test roles were removed afterwards. Not covered: the under-860px drawer, and there is still no automated UI test.

Windows (Pester, windows-ci.yml)	In CI	Runs on every push/PR to main.
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Feature index

Grouped by area. Badges: Working Partial Known issue

A · INSTALL & SETUP

A1	One-line installer (macOS / Linux)	Working
A2	Windows installer (install.ps1)	Partial
A3	<code>fox setup</code> on macOS (Lima)	Working
A4	<code>fox setup</code> on Windows (WSL2 + btrfs)	Partial
A5	Linux native	Working
A6	Command forwarding & file streaming	Working
A7	Always-latest engine binary	Working
A8	New-release notice on <code>fox start</code>	Working
A9	<code>fox update</code>	Partial

B · CLI & SERVICES

B1	The <code>fox</code> command set	Working
B2	Background services (daemon)	Working

C · BRANCHING & STORAGE

C1	Storage drivers (ZFS / btrfs)	Working
C2	Branch lifecycle	Working
C3	Protection of <code>main</code>	Working
C4	Private networking (no host ports)	Working
C5	Scale-to-zero & wake on connect	Working
C6	Storage quotas	Partial
C7	Capacity & status	Working
C8	Provisioning & image pull	Working

D · DURABILITY

D1	WAL archiving to object storage	Working
D2	Base backups	Working
D3	Point-in-time restore	Working
D4	HA standby & failover	Partial

E · SECURITY FOUNDATIONS

E1	Per-install generated secrets	Working
E2	TLS everywhere	Working
E3	Zero-signup local API key	Working

F · SQL GATEWAY

F1	Routing by database name	Working
F2	API-key authentication	Working
F3	Backend login (SCRAM)	Working
F4	Per-user Postgres login roles	Working
F5	Attribution session settings	Working
F6	Ctrl-C (CancelRequest) forwarding	Working
F7	Errors & insecure build tag	Working

G · ACCOUNTS & AUTH

G1	Auth store (SQLite)	Working
G2	Email & password users	Working
G3	API keys	Working
G4	Web sessions & signup control	Working
G5	OAuth (GitHub / Google)	Partial

H · BLACKBOX

H1	DDL recording	Working
H2	Hash chain, append-only, verify	Partial
H3	Attribution	Working
H4	Destructive-DDL guardrail	Partial
H5	Risk flags	Partial
H6	Ledger access (CLI/REST/MCP/UI/SDK)	Working
H7	<code>fox ledger revert --to</code>	Working
H8	Override permission & ledger upgrade	Working
H9	Blackbox 2.0 capture (xid, LSN, provenance)	Working
H10	Checkpoints & anchors	Working
H11	fox-verify (independent verifier)	Working
H12	Blackbox integrity & export	Working
H13	Branch from just before a Blackbox entry	Working
H14	Blackbox names & aliases	Working
H15	Policy gate (warn / block)	Working
H16	Agent provenance & execute_change	Working
H17	Impact analysis	Working
H18	Blackbox diff between branches	Working

I · IMPORT & MIGRATION

I1	From Postgres	Working
I2	From MySQL ≤5.7 / MariaDB	Working
I3	From MySQL 8.x	Working
I4	From MongoDB	Working
I5	From a .sql dump	Working
I6	From JSON / NDJSON	Working
I7	From CSV	Working
I8	Continuous replication & cutover	Working
I9	Guard handling during import	Working
I10	Live progress (CLI & SSE)	Working
I11	Import REST endpoints	Working
I12	Web Import page	Working
I13	Exact identifier fidelity	Working

J · ETL PIPELINES (EXPERIMENTAL)

J1	Pipeline engine	Working
J2	Data-quality tests	Working
J3	Stored pipelines & run history	Working
J4	Pipeline editor (web)	Working

K · APIS, SDKS & AGENTS

K1	Control-plane REST API	Working
K2	OpenAPI spec	Partial
K3	Python SDK	Working
K4	TypeScript SDK	Working
K5	Agent Branch API	Partial
K6	MCP server	Working

L · WEB CONSOLE

L1	Embedded single-origin UI	Working
L2	Login & signup	Working
L3	Dashboard	Working
L4	SQL console & browser	Working
L5	Blackbox page	Working
L6	API keys page	Working
L7	Landing, Guide & Docs pages	Partial
L8	Blackbox integrity page	Partial
L9	Policies page	Partial
L10	App sidebar layout	Partial
L11	Design system & FoxByte theme	Working

M · BUILD, CI & RELEASES

M1	Makefile targets	Working
M2	Linux CI	Working
M3	Windows CI	Working
M4	Releases & GHCR image	Working
M5	WSL distro / ZFS builds	Partial
M6	Postgres image	Working

Install & setup

A1 One-line installer (macOS / Linux)

Working

WHAT IT DOES	Installs the <code>fox</code> binary with one command.
HOW IT WORKS	<code>deploy/install.sh</code> detects OS (darwin/linux) and arch (arm64/amd64), downloads <code>fox-<os>-<arch></code> from the GitHub release, verifies it against the release's <code>SHA256SUMS</code> (<code>verify_file</code> , using <code>sha256sum</code> or <code>shasum</code>), installs to <code>\$PREFIX/bin</code> (sudo only if needed). On macOS it also stages <code>fox-linux-<arch></code> into <code>\$PREFIX/share/foxbyte</code> for the VM.
HOW TO USE	<code>curl -fsSL https://raw.githubusercontent.com/thefoxbyte/foxbyte/main/deploy/install.sh sh</code> · <code>env FOX_VERSION</code> , <code>FOX_REPO</code> , <code>FOX_DIST</code> (local folder), <code>FOX_PREFIX</code> (default <code>/usr/local</code>).
HOW TO TEST	Unit: <code>scripts/test_install_sh.sh</code> (run by CI) against a fake release — a verified release installs; a tampered binary, a release without <code>SHA256SUMS</code> and an asset missing from the listing each install nothing; <code>FOX_NO_VERIFY=1</code> installs anyway; a local <code>FOX_DIST</code> build still installs. Manual: <code>FOX_DIST=./dist FOX_PREFIX=\$HOME/.local sh deploy/install.sh</code> then <code>fox version</code> .
STATUS & LIMITS	Working. Every download is checked against the release's <code>SHA256SUMS</code> before it is installed, and anything that cannot be verified stops the install (<code>FOX_NO_VERIFY=1</code> skips deliberately; <code>FOX_DIST</code> is the user's own build, so there is nothing to check it against). The listing travels over the same TLS connection as the files, so this proves integrity — a complete, unaltered download — not authorship; signatures are still not built.
ADDED IN	<code>c9afda7</code> (PR #1); URLs repointed to the foxbyte org in <code>0b125f9</code> .

A2 Windows installer (install.ps1)

Partial

WHAT IT DOES	From a bare Windows machine to a running database with <code>irm ... iex</code> .
HOW IT WORKS	<code>deploy/install.ps1</code> <code>Invoke-Install</code> : (1) installs WSL (<code>wsl --install --no-distribution</code> , elevates if needed; registers a <code>RunOnce</code> resume across a reboot); (2) downloads <code>fox-windows-amd64.exe</code> , <code>fox-linux-amd64</code> , <code>foxbyte-docker-context.tar.gz</code> ; (3) downloads the Ubuntu noble WSL rootfs verified against Ubuntu's <code>SHA256SUMS</code> ; (4) adds the folder to <code>PATH</code> ; (5) runs <code>fox setup</code> . Forces TLS 1.2, BOM-free ASCII.
HOW TO USE	PowerShell: <code>irm https://raw.githubusercontent.com/thefoxbyte/foxbyte/main/deploy/install.ps1 iex</code> · <code>env FOX_VERSION</code> , <code>FOX_REPO</code> , <code>FOX_PREFIX</code> , <code>FOX_NO_SETUP</code> , <code>FOX_NO_ELEVATE</code> , <code>FOX_ROOTFS_URL</code> .
HOW TO TEST	Run on a real Windows machine on 16 Sep 2026 (from the branch, with the released v0.8.5 assets): the installer downloaded and verified the 685 MB distro image, added fox to <code>PATH</code> and completed setup. Pester: <code>deploy/install.Tests.ps1</code> (including the checksum block: listing parsing, the binary-mode <code>*name</code> form, an unlisted asset refused, a mismatch deleting the file, <code>FOX_NO_VERIFY=1</code>), <code>deploy/windows-install.Tests.ps1</code> (E2E group needs <code>FOX_TEST_E2E</code>); <code>windows-ci.yml</code> checks BOM/ASCII and PS 5.1 parsing.
STATUS & LIMITS	Works. It now downloads the prebuilt <code>foxbyte-distro.tar.gz</code> (~685 MB) so setup skips apt, the image build and the registry pulls, falling back to the Ubuntu rootfs when a release does not publish it (v0.7.1, v0.8.1) or the copy cannot be verified; every FoxByte asset is checked against <code>SHA256SUMS</code> .
ADDED IN	<code>4fc8afb</code> (PR #1); one-command flow <code>0c71da9</code> , verified downloads <code>d0c1f26</code> (PR #7); resume fix <code>288c431</code> (PR #8).

A3 fox setup on macOS (Lima)

Working

WHAT IT DOES	Creates/starts a Linux VM, installs Docker + ZFS, installs the engine and starts the stack.
HOW IT WORKS	<code>internal/host/host_darwin.go</code> <code>setupDarwin</code> : requires <code>limactl</code> ; refreshes the engine binary (A7); creates <code>template://ubuntu</code> VM if missing and runs <code>provisionGuest (zfsutils-linux docker.io)</code> ; always reinstalls the engine to <code>/usr/local/bin/fox</code> in the VM; then forwards <code>start</code> .
HOW TO USE	<code>brew install lima && fox setup</code> · env <code>FOX_LIMA_INSTANCE</code> (default <code>foxbyte</code> , falls back to default), <code>FOX_GUEST_BINARY</code> .
HOW TO TEST	No unit test for orchestration. Manual: <code>limactl delete -f foxbyte</code> ; <code>fox setup</code> ; <code>fox status</code> .
STATUS & LIMITS	Working. Provisioning only runs on VM creation (an interrupted VM isn't repaired). No image build context is staged, so if the GHCR pull fails an installed user can't fall back to a local build.
ADDED IN	<code>c9afda7</code> (PR #1); always-reinstall <code>77ef8ba</code> (PR #18).

A4 fox setup on Windows (WSL2 + btrfs)

Partial

WHAT IT DOES	Imports a dedicated <code>foxbyte</code> WSL distro, provisions it and starts FoxByte with quiet step-by-step output.
HOW IT WORKS	<code>host_windows.go</code> <code>setupWindows</code> : checks <code>wsl --status</code> ; refreshes the engine; <code>importDistro (wsl --import</code> , writes <code>/etc/wsl.conf</code> with <code>systemd</code>); <code>waitForSystemd</code> ; <code>provisionGuestWSL</code> on every run (Docker, btrfs, storage unit, preloaded images, image context, engine binary); <code>verifyBtrfs</code> ; quiet <code>start</code> with one retry; <code>finishSetup</code> . All guest output goes to <code>install.log</code> next to <code>bb.exe</code> . A <code>dbpool-storage.service</code> mounts the btrfs image before Docker at boot; <code>forwardStdin</code> waits for <code>systemd</code> on a cold distro.
HOW TO USE	<code>fox setup</code> (run by <code>install.ps1</code>) · env <code>FOX_WSL_DISTRO</code> , <code>FOX_GUEST_BINARY</code> .
HOW TO TEST	Unit: <code>host_wsl_test.go</code> (<code>TestResolveWSLDistro</code> , <code>TestWSLArgs</code> , <code>TestDecodeWSLList</code> , <code>TestWinPathToMnt</code> , <code>TestParseKernelRelease</code>). Verified on a real Windows machine (16 Sep 2026, v0.8.6, after <code>wsl --unregister foxbyte</code>): the distro was created from the prebuilt image, "Loading the preinstalled container images" ran and no apt install of Docker or btrfs tools appeared, the engine reported "fox-linux-amd64 from v0.8.6 verified against SHA256SUMS", and setup ended with a single summary carrying the working connection string (<code>postgresql://dbadmin:key_...@localhost:6432/main?sslmode=require</code>). Manual: <code>wsl --unregister foxbyte</code> ; <code>fox setup</code> ; <code>fox status</code> .
STATUS & LIMITS	Works. Fixed 16 Sep 2026: the preloaded Postgres image is now tagged with the name the engine runs (<code>ghcr.io/thefoxbyte/postgres-walg:18</code> since 21 Sep 2026, <code>:16</code> before), and a distro image built before 16 Sep is re-tagged after load, so its preload is not wasted. The final summary prints a connection string without the API key or <code>sslmode</code> . The storage-unit error text still mentions ZFS.
ADDED IN	<code>4fc8afb</code> , <code>8305e5b</code> (PR #1); quiet flow <code>0c71da9</code> + prebuilt image <code>d0c1f26</code> (PR #7); btrfs <code>fc4a1b8</code> (PR #9); initdb race fix <code>cb6655c</code> (PR #21).

A5 Linux native

Working

WHAT IT DOES	On Linux the engine runs in-process; no VM.
HOW IT WORKS	<code>host_other.go</code> : <code>fox setup</code> prints "Linux host — no VM needed. Run <code>fox start</code> ". ZFS and Docker must be installed; <code>ensureDocker</code> / <code>ensureReady</code> give instructive errors.
HOW TO USE	<code>sudo fox start</code>
HOW TO TEST	Integration suite runs on Linux (inside the Lima VM): <code>make integration</code> .
STATUS & LIMITS	Working.
ADDED IN	<code>4fc8afb</code> (PR #1).

A6 Command forwarding & host file streaming

Working

WHAT IT DOES	On macOS/Windows every engine command transparently runs inside the VM; <code>fox import --from ./file</code> works for any file on the host.
HOW IT WORKS	<code>host.Maybe (host.go)</code> forwards unless on Linux, in the guest (<code>FOX_IN_GUEST</code>) or a local command. macOS: <code>limactl shell ... env FOX_IN_GUEST=1 fox ...</code> (auto-starts the VM). Windows: <code>wsl.exe -d appdb -- env ... fox importLocalFile</code> rewrites a local file import to <code>--from - --kind <ext> -- srcname <name></code> and pipes the file over stdin.
HOW TO USE	Automatic. Env <code>FOX_GUEST_BIN</code> .
HOW TO TEST	Unit: <code>TestImportLocalFile</code> , <code>TestWSLArgs</code> . Manual: <code>limactl stop foxbyte && fox status</code> (VM auto-starts); import a CSV from the Desktop.
STATUS & LIMITS	Working. Only <code>import</code> streams files; <code>pipeline run spec.json</code> passes the path unchanged (must be readable in the VM).
ADDED IN	<code>c9afda7</code> , <code>4fc8afb</code> , <code>8305e5b</code> (PR #1); streaming <code>44cc05c</code> .

A7 Always-latest engine binary

Working

WHAT IT DOES	Re-running <code>fox setup</code> installs the newest Linux engine rather than a stale bundled copy.
HOW IT WORKS	<code>refreshEngineBinary (host.go)</code> downloads <code>fox-linux-<arch></code> from the latest (or pinned) release into <code>~/.fox/</code> with a 5-minute timeout, atomic rename and an ELF check; best-effort fallback to the installer's copy. <code>bundledLinuxBinary</code> prefers the cache.
HOW TO USE	Automatic. <code>FOX_NO_REFRESH=1</code> skips it; <code>FOX_REPO</code> / <code>FOX_VERSION</code> override the source.
HOW TO TEST	Unit: <code>host_test.go</code> — <code>TestIsELF</code> , <code>TestBundledLinuxBinaryPrefersCache</code> , <code>TestRefreshEngineBinaryDisabled</code> , <code>TestDownloadFile</code> ; <code>internal/update</code> <code>TestReleaseForSetup</code> (newest complete release, a pinned older tag, an incomplete release refused).
STATUS & LIMITS	Working, and seen on Windows on 16 Sep 2026: setup printed "Engine fox-linux-amd64 from v0.8.6 verified against SHA256SUMS" before installing it. The download is verified against the release's <code>SHA256SUMS</code> (size and SHA-256 as it streams) and an unverifiable or mismatched build is discarded in favour of the engine the installer staged — the engine runs as root inside the VM, so unchecked bytes are never installed. <code>FOX_NO_REFRESH=1</code> skips the refresh; <code>FOX_NO_VERIFY=1</code> falls back to the old unverified download.
ADDED IN	<code>77ef8ba</code> (PR #18).

A8 New-release notice on `fox start`

Working

WHAT IT DOES	When a newer release is published, <code>fox start</code> ends with one line: <i>FoxByte vX is available (you have Y). Run 'fox update' to get the new capabilities.</i> No prompt, and start is never held up.
HOW IT WORKS	New package <code>internal/update</code> . The check starts in the background when <code>start</code> begins and gets 1.5 s; the notice prints after start's own output only if the check finished in time. It lists releases through the GitHub API (not <code>/releases/latest</code> , which can point at a release still being published) and offers only one that is newer, not a draft or prerelease, and has every file this platform's update needs attached. That is a single request: it does not download <code>SHA256SUMS</code> , because GitHub throttles repeated downloads of the same release asset (measured 16 Sep 2026 from this Mac: 0.5 s the first time, then 7 s–75 s), so the first check after a new release was fast and every later one missed the budget. <code>fox update</code> still downloads <code>SHA256SUMS</code> and verifies every file before installing anything. The releases list is cached with its ETag in <code>~/.fox/update-check.json</code> , and a newer release, once found, is remembered in <code>~/.fox/update-notice.json</code> for 6 hours, so the notice keeps printing without a request. "Up to date" is never remembered, so every other start asks — a conditional request that GitHub answers with a 304 when nothing changed, which does not count against its rate limit. Runs on the computer: in <code>host.Maybe</code> around the forwarded <code>start</code> on macOS/Windows, and in <code>start</code> itself on Linux; the engine inside the VM never checks. Offline, rate limited or any error: nothing printed.
HOW TO USE	Automatic. <code>FOX_NO_UPDATE_CHECK=1</code> turns it off; <code>FOX_UPDATE_CHECK_TIMEOUT</code> (e.g. 3s); <code>FOX_UPDATE_CHECK_INTERVAL</code> sets how long a found release is remembered (default 6h; 0 checks on every start); <code>FOX_REPO</code> like the installers. Development builds (<code>0.1.0-dev</code>) never check.
HOW TO TEST	Unit: <code>internal/update/update_test.go</code> (versions, release selection, checksums, cache, a slow server doesn't delay or print, offline, turned off; <code>TestAvailableMakesNoAssetRequests</code> — the notice makes no asset request while the update path still verifies; <code>TestNoticeIsRememberedBetweenStarts</code> ; <code>TestNewReleaseShowsOnNextStart</code> — a release published after an up-to-date check shows on the next start). Integration: <code>make integration-update</code> \$1, \$1b (the remembered answer prints with GitHub unreachable; a forced check then prints nothing), \$2, \$3, \$8. Also checked for real on a Mac (installed 0.8.3, published v0.8.4): the notice prints within the default budget.
STATUS & LIMITS	Working. Releases are published as prereleases and promoted by hand (M4), so the notice appears once a release is promoted. macOS assumes the VM has the Mac's CPU for the notice (Lima's default); <code>fox update</code> checks the real one. Fixed 16 Sep 2026: against real GitHub the notice appeared only on the first start after a release was published. The check's second request (<code>SHA256SUMS</code>) is fast the first time that asset is fetched and throttled to seconds or minutes on repeats, so every later start fell past the budget. It now needs one request and is remembered for 6 hours. Fixed 17 Sep 2026: remembering "up to date" hid a new release for up to 6 hours — <code>fox update --check</code> reported v0.8.7 while <code>fox start</code> stayed silent, because its cache said "up to date" from before the release. Only a found release is remembered now, and caches holding "up to date" are ignored; checked against real GitHub with a copy of that cache (v0.8.7 found in 577 ms).
ADDED IN	<code>feat/fox-update</code> .

A9 fox update

Partial

WHAT IT DOES	Installs the newest release in one command — the new engine, restarted servers, Blackbox upgrades on running branches, and <code>fox</code> on the computer — without touching databases, branches, backups or settings. Before, updating meant re-running the installer, then <code>fox setup</code> (which left old servers running), then <code>fox blackbox upgrade --all</code> .
HOW IT WORKS	<code>internal/host/host_update.go</code> , run on the computer on every platform. (1) Downloads every file this platform needs into <code>~/.fox/updates/<tag>/</code> , each checked against <code>SHA256SUMS</code> as it streams — nothing changes unless all verify. (2) Copies the new engine into the VM / WSL distro and runs its <code>version</code> (wrong CPU or corrupt → stop). On macOS, asks for sudo now if <code>/usr/local/bin</code> isn't writable; Homebrew installs are refused. (3) <code>fox _update-guest stop-services</code> , run with the new engine, stops the three servers (they must stop: <code>start</code> leaves running ones alone); Postgres, MinIO and branch containers keep running. (4) <code>_update-guest install-engine</code> writes the engine next to the installed one and renames it over (the old one is kept in <code>~/.fox/updates/prev/fox</code> in the VM); if this fails the servers are restarted on the old engine. Windows also refreshes the Postgres image build context in the distro and next to <code>bb.exe</code> (previous kept as <code>.prev</code> ; only future image builds use it). (5) With the new engine: <code>fox up</code> (running containers left alone; main's Blackbox re-applied), <code>fox ledger upgrade --all</code> , <code>fox start</code> . (6) Replaces <code>fox</code> on the computer — macOS: write-and-rename (sudo if needed); Windows: the running <code>bb.exe</code> is renamed to <code>bb.exe.old</code> and the new one copied in — keeping the previous copy in <code>~/.fox/updates/prev/</code> , and refreshes the engine <code>fox setup</code> would install. Then checks the versions and that the control plane answers. On Linux the host binary is the engine, so steps 2–5 run on the machine itself. Never re-imports the distro, recreates storage, changes images or rewrites data files.
HOW TO USE	<code>fox update</code> (asks to confirm) · <code>--check</code> only reports · <code>--yes / -y</code> no prompt (required without a terminal) · <code>--version vX.Y.Z</code> a specific newer release. Going back: reinstall the older version with the installer (<code>FOX_VERSION=...</code>); the previous binaries are kept.
HOW TO TEST	Unit: <code>internal/host/host_update_test.go</code> (step order with a fake VM; check only, up to date, dev build, tampered download, no terminal, stdin closed, answered no and an engine that doesn't run all change nothing; a failed install restarts the old engine and leaves the host binary alone), <code>internal/update/update_test.go</code> (downloads, install with the previous copy kept, Windows-style executable swap, image context replace with path-traversal refusal), <code>cmd/fox/update_test.go</code> (flags). Integration: <code>make integration-update</code> (Lima VM, Linux path, local fake GitHub). Manual, not automated: on a Mac, <code>fox start</code> shows the notice and <code>fox update</code> replaces <code>/usr/local/bin/fox</code> and the VM engine; on Windows, the <code>bb.exe</code> swap and image context refresh.
STATUS & LIMITS	Partial: the Linux path is verified end to end in the VM, and the macOS path was run for real on 16 Sep 2026 (installed v0.8.3 with the installer, then updated to the published v0.8.4: versions, restarted servers, containers, branch data, Blackbox, gateway and UI all checked). It was run a second time for the released fix (v0.8.4 → v0.8.5) with the same checks. The Windows path is unit-tested and compiles but has not been run against a real release — deferred by the owner (16 Sep 2026). The first release that contains <code>fox update</code> must be installed the usual way. Checksums prove the files match the release, not who published it (no signatures). No automatic rollback: the previous binaries are kept for a manual one. Doesn't update the VM, WSL distro, Postgres or MinIO images. <code>fox setup</code> 's own engine download (A7) still uses <code>/releases/latest</code> without checksums.
ADDED IN	<code>feat/fox-update</code> .

CLI & services

B1 The fox command set

Working

WHAT IT DOES

One CLI for the whole product.

HOW IT WORKS

cmd/fox/main.go : host.Maybe forwarding, then a command switch with hand-rolled flag parsing (addrFlag , durFlag , restoreArg).

HOW TO USE

Command	Purpose
fox version · setup · start · stop · up · down · status · logs [svc] · psql	Lifecycle, status, logs, psql on main
fox branch create <n> [--from <branch>] list delete reset [--from p] suspend resume	Branches (C2)
fox vm [status shell]	The engine VM: state and size, or a shell inside it (macOS/Windows)
fox uninstall [--keep-data] [--yes]	Remove FoxByte from this machine: containers, storage, state, the VM and the command
fox backup create list · fox restore --to <ts latest>	Durability (D2, D3)
fox ha enable status failover disable failback	HA (D4)
fox blackbox ... (every ledger subcommand; same as fox ledger ...) · fox ledger [branch] [--limit N] · ledger verify · ledger upgrade [branch] --all · ledger checkpoint integrity export [branch] · ledger sessions [branch] · ledger diff <a> · ledger entries [branch] · ledger branch-before <id> [--as n] · ledger revert --to	Blackbox (H6–H8, H10, H12, H13)
fox import --from <src> [--as n] [--continuous] · fox import-cutover <n>	Import (I*)
fox pipeline run spec.json [--as n]	Pipelines (J1)
fox impact "<SQL>" [--branch] [--object] [--column] [--json] · fox branch diff <a> 	Impact analysis and diff (H17, H18)
fox policy [list] check "<SQL>" block warn enable disable remove <rule> add <rule> ... evaluations	Policy gate (H15)
fox user create · fox apikey create list revoke · fox admin grant revoke list [--branch]	Accounts (G2, G3, H8)
fox gateway [--addr :6432] [--idle 2m] · fox controlplane [--addr :8080] · fox serve [--addr :8088] · fox mcp	Run a service in the foreground

HOW TO TEST

Unit: cmd/fox/main_test.go — TestAddrFlag, TestDurFlag, TestRestoreArg. Behaviour via scripts/integration_test.sh .

STATUS & LIMITS

fox vm is listed as a local command (host.go) but has no implementation ("unknown command"). branch create has no --from .

ADDED IN

a7580f6 (initial); renamed foxbyte → fox in c9afda7 .

B2 Background services (daemon)

Working

WHAT IT DOES	<code>fox start</code> runs three detached TLS servers: control plane :8080 (REST + web UI), gateway :6432 (idle 2m), agent API :8088 .
HOW IT WORKS	<code>internal/daemon</code> : re-executes the binary per service, logs to <code>~/ .fox/<name>.log</code> , pidfile <code><name>.pid</code> , <code>Setsid detach</code> ; <code>Stop</code> sends SIGTERM then SIGKILL after ~3s.
HOW TO USE	<code>fox start</code> · <code>fox status</code> · <code>fox logs gateway api controlplane</code> · <code>fox stop</code>
HOW TO TEST	Unit: <code>daemon_test.go</code> TestStopTerminatesProcess. Integration §1 (stop/start).
STATUS & LIMITS	Working. No health check or restart-on-crash.
ADDED IN	<code>a7580f6</code> .

Branching & storage

C1 Storage drivers (ZFS / btrfs)

Working

WHAT IT DOES	Instant copy-on-write copies of a whole Postgres data directory. ZFS on macOS/Linux, btrfs on Windows.
HOW IT WORKS	<code>internal/branch/storage.go</code> : storage interface; <code>activeStorage()</code> picks by <code>FOX_STORAGE</code> . ZFS: pool <code>foxbyte</code> on a file (<code>FOX_ZPOOL_DEVICE</code> , size <code>FOX_ZPOOL_SIZE</code> =30G), clone = <code>zfs snapshot ... @for-<dst> + zfs clone</code> . btrfs: loop image at <code>/dbpool/branches</code> , clone = <code>btrfs subvolume snapshot</code> , new subvolumes get <code>chattr +C</code> (nodatacow).
HOW TO USE	Automatic (Windows launcher sets <code>FOX_STORAGE=btrfs</code>).
HOW TO TEST	Integration §2 (branch isolation). Manual: <code>fox branch create t1 && sudo zfs list -r foxbyte</code> .
STATUS & LIMITS	Working. Until 17 Sep 2026 ZFS <code>destroy</code> removed only the <code>main@for-<branch></code> snapshot, so creating or resetting from a non- <code>main</code> parent a second time failed on the leftover; delete now also destroys the clone's own origin snapshot (<code>isOriginSnapshotFor</code> , unit-tested), and clone re-takes a stale snapshot. btrfs snapshots the subvolume directly and was never affected.
ADDED IN	ZFS <code>a7580f6</code> ; driver split + btrfs <code>fc4a1b8</code> , nodatacow <code>81ff8f8</code> (PR #9).

C2 Branch lifecycle

Working

WHAT IT DOES	Create, list, delete, reset (start over from parent), suspend and resume isolated full copies of the database.
HOW IT WORKS	<code>branch.go</code> : Create (CHECKPOINT parent, clone, chown, start container, <code>waitReady</code> over TCP), Delete , Reset (delete + create), List , Suspend (<code>docker stop</code>), Resume , Wake → <code>EnsureRunning</code> .
HOW TO USE	<code>fox branch create qa [--from staging] · list · suspend qa · resume qa · reset qa [--from main] · delete qa</code> ; REST <code>POST /api/branches</code> with <code>{"name","from"}</code> (404 when the parent doesn't exist), <code>DELETE /api/branches/{name}</code> ; the Dashboard's create form has a parent picker.
HOW TO TEST	Unit: <code>TestContainerName</code> , <code>TestIsOriginSnapshotFor</code> . Integration §2 (isolation), §2b (<code>--from</code> copies the parent's data, leaves <code>main</code> alone, names a missing parent, and the REST 404/201), §4 (suspend/resume). Manual: the same in the browser — create a parent, add a row, create a child from it and read the row in the console.
STATUS & LIMITS	Working. A branch is created from <code>main</code> unless <code>--from / from</code> names another branch. CLI create does no name validation (REST does).
ADDED IN	<code>a7580f6</code> ; reset <code>00e7eae</code> (PR #13); TCP readiness <code>cb6655c</code> (PR #21).

C3 Protection of `main`

Working

WHAT IT DOES	Stops the primary from being deleted or reset by mistake.
HOW IT WORKS	Delete and Reset refuse <code>main</code> ; Suspend refuses <code>main</code> , the HA standby, and whichever container is serving <code>main</code> after a failover (<code>suspendRefusal</code> , so the CLI and REST both get it); <code>EnsureRunning</code> never auto-starts <code>main</code> (avoids split-brain after failover); the reaper never suspends <code>main</code> , restore or standby; ZFS reservation keeps space for <code>main</code> (C6).
HOW TO USE	Automatic.
HOW TO TEST	Unit: <code>TestSuspendRefusal</code> , <code>TestPrimaryPointerGuards</code> . Integration §4 (suspending <code>main</code> refused, <code>main</code> keeps running), §6 (suspending the promoted standby refused). Manual: <code>fox branch delete main</code> → "refusing to delete the primary branch 'main'".
STATUS & LIMITS	Working. Until 15 Sep 2026 <code>fox branch suspend main</code> was allowed from the CLI and left <code>main</code> stopped (the gateway doesn't wake it) — now refused.
ADDED IN	<code>a7580f6</code> .

C4 Private networking (no host ports)

Working

WHAT IT DOES	Branch databases are reachable only through the gateway, not directly from the host or LAN.
HOW IT WORKS	Containers <code>pg-<name></code> join docker network <code>foxbyte</code> with no published ports; the gateway dials the container IP (<code>BackendAddr</code> → <code>containerIP</code> ; main via <code>PrimaryContainer()</code>). <code>FOX_DEBUG_PORTS</code> re-publishes ports for debugging.
HOW TO USE	Automatic.
HOW TO TEST	Unit: <code>TestParsePublishedPort</code> , <code>TestDSN</code> . Manual: <code>docker ps</code> shows no <code>5432</code> mapping for <code>pg-*</code> .
STATUS & LIMITS	Working. Exceptions: <code>pg-restore</code> publishes 5433, HA standby a random port, MinIO 9000/9001.
ADDED IN	<code>797bc14</code> , <code>b664b56</code> (PR #11).

C5 Scale-to-zero & wake on connect

Working

WHAT IT DOES	Idle branches stop automatically; the next connection wakes them.
HOW IT WORKS	Gateway <code>reaper</code> (<code>proxy.go</code>) polls every idle/3 (5–30s) and suspends a branch only when <code>idle ≥ threshold</code> and <code>ActiveConnections = 0</code> . On connect, <code>EnsureRunning</code> routes, recreates or resumes the container and re-enables the guard (<code>reconcileGuard</code>).
HOW TO USE	<code>fox gateway --idle 2m</code> (default via <code>fox start</code>); <code>--idle 0</code> disables.
HOW TO TEST	Integration §7 (idle 8s: branch suspended, standby untouched). Manual: wait >2 min, then connect through the gateway.
STATUS & LIMITS	Working. Idle clocks are in memory (reset on gateway restart). Cold-start latency not measured.
ADDED IN	<code>a7580f6</code> .

C6 Storage quotas

Partial

WHAT IT DOES	Reserves space for <code>main</code> and optionally caps each branch.
HOW IT WORKS	<code>reserveMain</code> : <code>zfs set reservation</code> (default 2G) on every Init. <code>setBranchRefquota</code> : <code>zfs set refquota</code> on new branches when configured.
HOW TO USE	<code>FOX_MAIN_RESERVATION=5G FOX_BRANCH_REFQUOTA=20G fox start</code> (<code>none</code> disables).
HOW TO TEST	Manual: <code>sudo zfs get reservation,refquota dbpool/branches/main dbpool/branches/qa</code> .
STATUS & LIMITS	ZFS only; per-branch cap is opt-in (default uncapped); not implemented on btrfs.
ADDED IN	<code>21f7259</code> (PR #12).

C7 Capacity & status

Working

WHAT IT DOES	Shows services, main health, backups, branches, storage used/available.
HOW IT WORKS	<code>Status</code> , <code>Branches</code> (state, used, referenced, connections), <code>StorageInfo</code> (<code>zfs list</code> / <code>btrfs filesystem usage</code>).
HOW TO USE	<code>fox status</code> ; <code>GET /api/status</code> ; web Dashboard.
HOW TO TEST	Integration §1 (<code>mainReady</code>). Manual: <code>fox status</code> .
STATUS & LIMITS	Working. On btrfs "used" is apparent size, not the copy-on-write delta.
ADDED IN	<code>a7580f6</code> ; <code>StorageInfo</code> <code>4b0be5c</code> .

C8 Provisioning & image pull

Working

WHAT IT DOES	<code>fox up/start</code> prepares Docker, the storage pool and the Postgres image automatically.
HOW IT WORKS	<code>provision.go</code> : <code>ensureDocker</code> → <code>ensurePool</code> → <code>ensureImage</code> (inspect, else <code>docker pull</code> the image for this install's PostgreSQL major — <code>postgres-walg:18</code> for a fresh install, the major in main's <code>PG_VERSION</code> otherwise (M6) — else <code>docker build</code> from <code>FOX_IMAGE_CONTEXT</code> / <code>docker/postgres</code> with <code>--build-arg PG_MAJOR</code> set to that major, so a local build of an older install's image is still that major).
HOW TO USE	Automatic.
HOW TO TEST	Manual: <code>sudo docker rmi ghcr.io/thefoxbyte/postgres-walg:18; fox up</code> .
STATUS & LIMITS	Working; the pull needs the GHCR package to be public.
ADDED IN	<code>c9afda7</code> ; GHCR pull <code>aae9e20</code> (PR #12).

Durability

D1 WAL archiving to object storage

Working

WHAT IT DOES	Continuously ships main's write-ahead log to a local S3-compatible store, off the commit path.
HOW IT WORKS	<code>Up</code> starts MinIO (<code>quay.io/minio/minio:RELEASE.2025-09-07T16-13-09Z</code> , generated credentials) and creates bucket <code>wal-archive</code> ; main runs with <code>archive_mode=on</code> , <code>archive_command='wal-g wal-push %p'</code> , <code>archive_timeout=60</code> , <code>lz4</code> .
HOW TO USE	Automatic with <code>fox start</code> . MinIO console at <code>:9001</code> .
HOW TO TEST	Integration §3. Manual: log in to MinIO with values from <code>~/.fox/secrets.json</code> and see WAL files.
STATUS & LIMITS	Working. MinIO images unpinned; branches don't archive. Images: MinIO stopped publishing to Docker Hub, so the server and client images come from quay.io, pinned to the tested releases (<code>internal/branch/images.go</code>); an install that already has <code>minio/minio:latest</code> / <code>minio/mc:latest</code> cached keeps using them, and <code>FOX_MINIO_IMAGE</code> / <code>FOX_MC_IMAGE</code> point at a mirror.
ADDED IN	<code>a7580f6</code> ; generated credentials <code>797bc14</code> .

D2 Base backups

Working

WHAT IT DOES	Takes and lists full base backups of main.
HOW IT WORKS	<code>Backup</code> : <code>docker exec <primary> wal-g backup-push</code> , following the primary pointer (<code>PrimaryContainer</code>) so it backs up whichever container is serving main and says so when that is not <code>pg-main</code> . <code>BackupList</code> and the API's <code>Backups</code> read the listing from a throwaway container instead (<code>walgEnv</code>), so backups can be listed with main stopped or a failover in force — they live in object storage, not in any one container.
HOW TO USE	<code>fox backup create</code> · <code>fox backup list</code> · <code>GET /api/backups</code> · the Base backups panel on the Dashboard (newest 6, with the rest one click away, and a copy-ready <code>fox restore --to</code> for each).
HOW TO TEST	Integration §3, §3b (the API lists them newest-first with the newest flagged), §6 (after a failover: the backup is taken from the promoted standby and says so, one more backup is stored, and <code>backup list</code> works with main stopped).
STATUS & LIMITS	Manual only (no schedule or retention). Taking one from the primary and listing them both survive a failover. The API surface is read-only: a restore needs a host port and leaves a container behind, so it stays a CLI command.
ADDED IN	<code>a7580f6</code> ; primary-aware backups, the listing API and the Dashboard panel <code>feat/durability-gaps</code> .

D3 Point-in-time restore

Working

WHAT IT DOES	Brings up a separate copy of main at a past timestamp (or latest) on port 5433.
HOW IT WORKS	<code>Restore</code> chooses the base backup first (<code>restoreBackupName</code> : the target time is read with <code>parseRestoreTime</code> , the listing comes from a throwaway container, and <code>pickBackupForTime</code> takes the newest backup of this database — same system identifier — that had finished by then), then runs the recovery in the disposable <code>pg-restore</code> container: <code>wal-g backup-fetch <that backup></code> , <code>recovery_target_time</code> , <code>promote</code> ; <code>waitRecovered</code> polls up to 80s. The script is <code>restore_pitr.sh</code> , embedded in the binary and run with <code>bash -c as BranchBeforeEntry</code> does, so an install whose image predates the fix gets it from <code>fox update</code> alone. <code>--to latest</code> still fetches LATEST and replays everything.
HOW TO USE	<code>fox restore --to '2026-09-14 10:00:00+00'</code> or <code>--to latest</code> . The Dashboard's Base backups panel shows the earliest point reachable and a copy-ready command per backup.
HOW TO TEST	Unit: <code>TestPickBackupForTime</code> , <code>TestParseRestoreTime</code> , <code>TestRestorePITRScript</code> . Integration §3 ("PITR restores 3 rows"), §3b (a target before the newest backup reaches 3 rows and not 9, the chosen backup is named and is not the newest, a target no backup precedes is refused naming the oldest and leaves nothing behind, and <code>--to latest</code> still reaches the end of the archive), §6 (a restore reaches a write made after a failover); integration-v2 §1a (no <code>minioadmin</code> in the restore container, no hardcoded password printed).
STATUS & LIMITS	Uses the per-install MinIO credentials (fixed in Blackbox 2.0 Phase 1 — a hardcoded <code>minioadmin</code> pair used to make restore fail on fresh installs), and the printed hint no longer shows a password. A target earlier than every base backup is refused with the oldest one's time, since recovery can only start from a backup the target follows. A timestamp this binary cannot read is still passed to Postgres with LATEST as the base, as before. One restore at a time: the container is always named <code>pg-restore</code> and port 5433 is fixed.
ADDED IN	<code>a7580f6</code> ; broke with <code>797bc14</code> (PR #11); credentials fixed in <code>feat/ledger2-phase1-security</code> ; backup selection <code>feat/durability-gaps</code> .

D4 HA standby & failover

Partial

WHAT IT DOES	A single-VM demo of an asynchronous hot standby with manual failover.
HOW IT WORKS	<code>ha.go</code> : <code>HAEnable</code> (replication hba rule, <code>pg_basebackup -R</code> , start <code>pg-standby</code>); <code>HAFailover</code> (<code>pg_promote()</code>), stop old primary, write <code>standby</code> to <code>~/.fox/primary</code> so the gateway routes main to it); <code>HADisable</code> ; <code>HAStatus</code> . After a failover <code>haGuard</code> refuses <code>enable</code> , <code>disable</code> and <code>failover</code> (each would delete or stop the promoted standby, the only primary), and <code>Init</code> (<code>fox up / start</code>) brings the promoted standby up — recreating its container from its storage if <code>fox stop</code> removed it — instead of restarting the stale old main as a second primary. The standby's container carries the same object-storage environment and <code>archive_mode=on</code> as a primary (<code>standbyRunArgs</code>); <code>on</code> archives only outside recovery, so nothing is pushed while it streams — the primary is archiving those same segments — and the moment it is promoted it continues the archive on its own timeline. <code>HAFailback</code> returns to main without losing writes: main's data directory is rebuilt with <code>pg_basebackup</code> from the standby (the dataset is kept, since branches clone its snapshots), main starts as a streaming replica and catches up, the standby is stopped cleanly, main replays up to the standby's final checkpoint (read with <code>pg_controldata</code>), then main is promoted, the pointer goes back to <code>main</code> and the old standby is removed. Any failure before the promotion restarts the standby and leaves it primary.
HOW TO USE	<code>fox ha enable status failover disable failback</code> — after a failback, take <code>fox backup create</code> and <code>fox ha enable</code> for a new standby.
HOW TO TEST	Unit: <code>internal/branch/ha_test.go</code> (<code>TestHAGuard</code> , <code>TestPrimaryPointerGuards</code> , <code>TestLSNAtLeast</code> , <code>TestParseControlCheckpoint</code> , <code>TestStandbyRunArgsArchive</code>). Integration §6: streaming replica; write after failover; the promoted standby's <code>archive_mode</code> is on and its archiver is shipping segments; a backup taken from it; <code>backup list</code> with main stopped; <code>fox status</code> naming the container serving main; a point-in-time restore reaching a write made after the failover; <code>ha enable disable failover</code> and suspending the standby refused; <code>fox up</code> after the standby's container is removed recreates it and leaves the old main stopped; <code>ha failback</code> exits 0, main is primary and not in recovery, the post-failover write is on main, the standby is gone, the gateway writes to main, WAL archiving resumed. §7.
STATUS & LIMITS	Demo: no automatic detection, fencing or multi-host. Failback briefly stops writes (between stopping the standby and promoting main). Fixed 15 Sep 2026: <code>ha enable</code> or <code>ha disable</code> after a failover deleted the live primary and the writes made since, and <code>fox up</code> restarted the old main as a second primary. Fixed 18 Sep 2026: a promoted standby archived no WAL, so nothing written while main ran on it could be backed up or restored — it archives now, and a standby created before this change is named as unarchiving when <code>fox up</code> brings it back (<code>archive_mode</code> is a postmaster setting, so only a rebuild adds it).
ADDED IN	<code>a7580f6</code> ; btrfs support <code>fc4a1b8</code> .

Security foundations

E1 Per-install generated secrets

Working

WHAT IT DOES	Every install gets its own Postgres and MinIO passwords — nothing hardcoded.
HOW IT WORKS	<code>internal/secrets : Load()</code> reads or creates <code>~/.fox/secrets.json</code> (0600, atomic write) with <code>crypto/rand</code> values (<code>pg_password</code> , <code>minio_user</code> , <code>minio_password</code>); env overrides are applied but never written. Used by the engine, gateway backend login and role password sync.
HOW TO USE	Automatic. Env <code>FOX_PG_PASSWORD</code> , <code>FOX_MINIO_USER</code> , <code>FOX_MINIO_PASSWORD</code> .
HOW TO TEST	Indirect (TestDSN isolates HOME). Manual: <code>cat ~/.fox/secrets.json</code> in the VM.
STATUS & LIMITS	Working, except PITR ignores it (D3). Process-local caching only.
ADDED IN	797bc14 (PR #11).

E2 TLS everywhere

Working

WHAT IT DOES	Gateway, control plane and agent API all serve TLS with zero setup; clients can use <code>sslmode=require</code> .
HOW IT WORKS	<code>internal/tlsutil : EnsureCert</code> uses <code>FOX_TLS_CERT / _KEY</code> or generates a self-signed ECDSA P-256 cert (10 years, SAN localhost/127.0.0.1/::1) under <code>~/.fox/tls</code> with a cross-process file lock; min TLS 1.2. Gateway answers <code>SSLRequest</code> with <code>'S'</code> and upgrades; GSS gets <code>'N'</code> .
HOW TO USE	<code>psql "postgresql://dbadmin:<key>@localhost:6432/main?sslmode=require"; curl -k https://localhost:8080/api/status</code> .
HOW TO TEST	Unit: <code>TestReadStartupSkipsSSL</code> , <code>TestReadStartupUpgradesTLS</code> . Manual: <code>openssl s_client -connect localhost:8080</code> .
STATUS & LIMITS	Working. TLS is optional at the gateway (a <code>sslmode=disable</code> client still connects, sending its key in cleartext). Servers fall back to plain HTTP if the cert can't load.
ADDED IN	0bd52f3 ; file lock b664b56 (PR #11).

E3 Keys are made by a person, not by start

Working

WHAT IT DOES	<code>fox start</code> creates no account and no API key. It points at the web UI, and the person who will use the key signs in and makes it.
HOW IT WORKS	Nothing is minted at start. The first account created on an install — through the web sign-up or <code>fox user create</code> — gets the admin grant on main, through <code>auth.Store.OnFirstUser</code> (the auth package cannot reach the engine, so the engine sets the hook) and <code>branch.AdminForFirstAccount</code> . Keys are made by <code>fox apikey create</code> or the API keys page, and shown once.
HOW TO USE	<code>fox start</code> , then create your account at <code>https://localhost:8080</code> and make a key on the API keys page.
HOW TO TEST	Integration §1 (a fresh install has no account and no key until one is created; the first account can override the guardrail). Manual: fresh HOME, <code>fox start</code> , and no key anywhere.
STATUS & LIMITS	Working. Until 21 Sep 2026 the first start created an account of its own (<code>local@foxbyte</code>) with a key cached in <code>~/.fox/config</code> in plaintext, and printed it: convenient, but a credential nobody asked for, belonging to an account nobody owned. Both are gone, and so is the cached key.
ADDED IN	dcc32ab (PR #11); replaced by sign-in on <code>feat/foxbyte-ui</code> .

SQL gateway

F1 Routing by database name

Working

WHAT IT DOES	One Postgres endpoint (:6432); the database name you connect to is the branch.
HOW IT WORKS	<code>proxy.Serve</code> → <code>handle</code> : reads the <code>StartupMessage</code> (protocol 3.0, ≤1 MiB), <code>EnsureRunning(target)</code> , rewrites <code>database</code> → <code>foxybyte</code> and <code>user</code> → login role, then relays bytes both ways. No <code>dbname</code> → <code>main</code> .
HOW TO USE	<code>psql "postgresql://dbadmin:<key>@localhost:6432/qa?sslmode=require"</code>
HOW TO TEST	Unit: <code>TestBuildParseStartupRoundtrip</code> . Integration §1, §6.
STATUS & LIMITS	Working. No pooling or rate limiting.
ADDED IN	<code>a7580f6</code> .

F2 API-key authentication

Working

WHAT IT DOES	Every client must use a <code>key_</code> API key as its Postgres password.
HOW IT WORKS	<code>requestClientPassword</code> asks for a cleartext password (inside TLS), <code>VerifyKey</code> checks it; bad key → <code>28P01 invalid API key</code> . <code>AuthenticationOk</code> is sent only after the backend login succeeds.
HOW TO USE	<code>PGPASSWORD=key_...</code> <code>psql "postgresql://dbadmin@localhost:6432/main?sslmode=require"</code>
HOW TO TEST	Integration §1 (bad key rejected, key accepted).
STATUS & LIMITS	Working.
ADDED IN	<code>e42e792</code> ; auth store <code>235ff41</code> .

F3 Backend login (SCRAM)

Working

WHAT IT DOES	The gateway logs in to the branch itself, so clients never see the database password.
HOW IT WORKS	<code>backendAuth</code> (<code>auth.go</code>) supports cleartext, md5 and SCRAM-SHA-256 using the per-install password; failures → <code>08006</code> .
HOW TO USE	Automatic.
HOW TO TEST	Exercised by integration §1, §6, §8 (no unit test for SCRAM).
STATUS & LIMITS	Working.
ADDED IN	<code>e42e792</code> .

F4 Per-user Postgres login roles

Working

WHAT IT DOES	Each user logs in as a role named after their email, so the ledger's actor (<code>session_user</code>) can't be forged; permissions stay those of non-superuser <code>db_client</code> (RLS and GRANTS apply).
HOW IT WORKS	<code>loginUser</code> → <code>branch.EnsureUserRole</code> : <code>CREATE ROLE <email> LOGIN NOSUPERUSER ... IN ROLE db_client</code> , <code>ALTER ROLE ... SET role = db_client</code> ; cached per process; falls back to <code>db_client</code> on error.
HOW TO USE	Automatic. Check with <code>select session_user, current_user;</code> → <code><email></code> , <code>db_client</code> .
HOW TO TEST	Manual query above (no automated test).
STATUS & LIMITS	Working. Roles share the per-install password (the gateway authenticates the user, not Postgres).
ADDED IN	<code>96c8c31</code> (role split <code>aad84e4</code>).

F5 Attribution session settings

Working

WHAT IT DOES	Tags every session with who, what kind, which branch and a session id for the Blackbox.
HOW IT WORKS	<code>ledgerOptions</code> appends <code>-c bb.actor=<email> bb.actor_kind=human agent bb.branch=bb.session=<random> ; agent</code> when the branch name starts with <code>agent-</code> .
HOW TO USE	Automatic. <code>SHOW bb.actor;</code>
HOW TO TEST	Integration §8 (DDL attributed to the human key).
STATUS & LIMITS	Working; <code>actor_kind</code> , <code>branch</code> , <code>session</code> are ordinary settings a client could change (the actor itself comes from F4).
ADDED IN	<code>e42e792</code> .

F6 Ctrl-C (CancelRequest) forwarding

Working

WHAT IT DOES	Pressing Ctrl-C in psql cancels a running query through the gateway.
HOW IT WORKS	<code>relayStartupCaptureKey</code> records <code>BackendKeyData</code> (<code>pid:secret</code> → backend address); <code>forwardCancel</code> replays a cancel packet to the right backend.
HOW TO USE	<code>select pg_sleep(60);</code> then Ctrl-C → "canceling statement due to user request".
HOW TO TEST	Unit: <code>TestReadStartupHandlesCancel</code> . Manual as above.
STATUS & LIMITS	Working; the key map lives in one gateway process.
ADDED IN	<code>4758737</code> / <code>4cccbd6</code> (PR #17).

F7 Errors & insecure build tag

Working

WHAT IT DOES	Failures reach clients as real Postgres errors; the no-auth bypass can't exist in release builds.
HOW IT WORKS	<code>sendError</code> writes FATAL <code>ErrorResponses</code> (<code>28P01</code> , <code>3D000</code> branch missing, <code>08006</code>). <code>FOX_GATEWAY_NOAUTH</code> only works in binaries built with <code>-tags insecure</code> .
HOW TO USE	Connect to a missing branch → <code>FATAL: branch "x" does not exist</code> .
HOW TO TEST	Unit: <code>TestSendError</code> .
STATUS & LIMITS	Working.
ADDED IN	<code>a7580f6</code> ; insecure tag <code>1884445</code> (PR #12).

Accounts & auth

G1 Auth store (SQLite)

Working

WHAT IT DOES	Stores users, API keys, sessions, OAuth identities, pipelines and runs for a single-tenant install.
HOW IT WORKS	<code>internal/auth : OpenFromEnv (FOX_DB , default ~/.fox/auth.db in the VM)</code> , pure-Go SQLite with WAL + <code>busy_timeout</code> so all three services share it. The schema is created and migrated inside <code>BEGIN IMMEDIATE</code> , retried on <code>SQLITE_BUSY</code> , because <code>fox start</code> launches all three services together and on a new install they create the file at the same moment.
HOW TO USE	Automatic.
HOW TO TEST	Unit: <code>TestVerifyKeyConcurrent</code> , <code>TestConcurrentFirstOpen</code> (four stores opening a new file at once, 40 rounds; without the fix 6 of 160 opens failed).
STATUS & LIMITS	Working. Every signed-in user can see every branch (no RBAC). Fixed 17 Sep 2026: on a new install the first <code>fox start</code> could leave the control plane and the Agent API down — they lost the race to create the store and exited with <code>database is locked (SQLITE_BUSY)</code> .
ADDED IN	235ff41 (PR #1); WAL c2dd024 .

G2 Email & password users

Working

WHAT IT DOES	Create an account and log in with email + password.
HOW IT WORKS	<code>CreateUser</code> (lower-cased email, bcrypt), <code>Login</code> (generic "invalid credentials"); 8-char minimum enforced in the CLI and <code>/auth/register</code> .
HOW TO USE	<code>fox user create you@x.com ; POST /auth/register , POST /auth/login .</code>
HOW TO TEST	Unit: <code>TestPasswordLogin</code> . Integration §1.
STATUS & LIMITS	Working. No password change/reset, user list or delete.
ADDED IN	235ff41 .

G3 API keys

Working

WHAT IT DOES	Non-interactive credentials for REST (Bearer / <code>X-API-Key</code>) and the gateway.
HOW IT WORKS	Secret <code>key_</code> + 24 random bytes; only the sha256 is stored; shown once; per-user list/revoke; <code>last_used</code> updated at most once a minute.
HOW TO USE	<code>fox apikey create <email> [name] · list · revoke ; GET/POST /api/keys , DELETE /api/keys/{id} ; web Keys page.</code>
HOW TO TEST	Unit: <code>TestAPIKeys</code> . Integration §1.
STATUS & LIMITS	Working. Revoking an unknown id still returns success.
ADDED IN	235ff41 .

G4 Web sessions & signup control

Working

WHAT IT DOES	Browser login sessions; self-signup can be closed.
HOW IT WORKS	30-day session token in HttpOnly cookie <code>dbengine_session</code> ; <code>Authn</code> accepts cookie, Bearer or <code>X-API-Key</code> . <code>FOX_SIGNUP=closed</code> makes register return 403; <code>GET /auth/providers</code> tells the UI what's available.
HOW TO USE	<code>/auth/login , /auth/logout , /auth/me ; FOX_SIGNUP=closed fox start .</code>
HOW TO TEST	Unit: <code>TestSessions</code> . Manual: <code>curl -k https://localhost:8080/auth/providers .</code>
STATUS & LIMITS	Working. Signup is open by default: anyone who can reach <code>:8080</code> can register.
ADDED IN	235ff41 .

G5 OAuth (GitHub / Google)

Partial

WHAT IT DOES	"Continue with GitHub / Google" sign-in.
HOW IT WORKS	<code>oauth.go</code> : state cookie, code exchange, identity fetch, <code>upsertOAuth</code> links by email; redirects to <code>FOX_WEB_ORIGIN/dashboard</code> .
HOW TO USE	Set <code>FOX_GITHUB_CLIENT_ID/_SECRET</code> or <code>FOX_GOOGLE_CLIENT_ID/_SECRET</code> . Nothing else is needed on a default install; set <code>FOX_PUBLIC_URL</code> when the engine is reached at another address, and <code>FOX_WEB_ORIGIN</code> only when the console is hosted separately.
HOW TO TEST	Integration §1 (the GitHub authorize URL carries <code>redirect_uri=https%3A%2Flocalhost%3A8080%2Fauth%2Foauth%2Fgithub%2Fcallback</code> with nothing set, and a password login gets a <code>Secure</code> , <code>SameSite=Lax</code> cookie). Unit: <code>TestURLDefaults</code> , <code>TestSessionCookieSameSite</code> . Manual, and still untested: a real OAuth app.
STATUS & LIMITS	The defaults now match what the engine serves (<code>urlDefaults</code> : public URL <code>https://localhost:8080</code> , web origin = public URL), so only the client ID and secret are required. Untested against a real GitHub or Google app.
ADDED IN	<code>235ff41</code> .

Blackbox (the record of schema changes, formerly the Schema Ledger)

The ledger lives inside each branch database (`bb.schema_ledger`), installed into `main` by `InstallLedger` during `Init` and inherited by every branch through the copy-on-write clone. Source: `internal/ledger/ledger.sql`.

H1 DDL recording

Working

WHAT IT DOES	Records every schema change — who, with which tool, on which branch, the statement, its status and risk.
HOW IT WORKS	Event triggers: <code>ddl_command_end</code> → <code>log_ddl_end</code> (one row per object, incl. GRANT/REVOKE and function DDL); <code>sql_drop</code> → <code>log_ddl_drop</code> . Skips objects in schema <code>fox</code> , COMMENT, CREATE EXTENSION, event triggers, CREATE SCHEMA, sequences and <code>_pkey</code> indexes. Columns: id, at, actor, actor_kind, tool, session, branch, command_tag, object_type, object_identity, statement, status, risk, prev_hash, row_hash.
HOW TO USE	Automatic for all clients (gateway, console, psql, ORMs, imports).
HOW TO TEST	Integration §8 ("ledger captures DDL, attributed to the human key").
STATUS & LIMITS	Working. Not captured: DML, role/database/tablespace changes (Postgres doesn't fire event triggers for those); some dropped object types (triggers, policies, extensions).
ADDED IN	<code>e42e792</code> ; GRANT/REVOKE/function DDL <code>db9dc0c</code> (PR #13).

H2 Hash chain, append-only & verify

Partial

WHAT IT DOES	Makes the ledger tamper-evident: rows can't be edited/deleted normally, and any break in the chain is detected.
HOW IT WORKS	<code>chain_row</code> BEFORE INSERT sets <code>row_hash = sha256(prev_hash id at(UTC) ... risk)</code> under <code>pg_advisory_xact_lock</code> . <code>bb_append_only</code> / <code>bb_no_truncate</code> triggers + REVOKE block UPDATE/DELETE/TRUNCATE. <code>LedgerVerify</code> recomputes every hash and link.
HOW TO USE	<code>fox ledger verify [branch]</code> ; GET <code>/api/branches/{name}/ledger/verify</code> ; MCP <code>verify_ledger</code> ; SDK <code>verify_ledger()</code> . Also <code>fox blackbox verify</code> , <code>.../blackbox/verify</code> , MCP <code>verify_blackbox</code> , SDK <code>verify_blackbox</code> / <code>verifyBlackbox</code> (H14).
HOW TO TEST	Integration §8 (plain DELETE blocked; 0 broken rows). Manual: <code>fox ledger verify</code> → "ledger intact — N rows...".
STATUS & LIMITS	A superuser can bypass the triggers (<code>session_replication_role=replica</code>) and rewrite the whole chain so this check still passes. Since Phase 3, <code>fox ledger integrity</code> and <code>fox-verify</code> (H10–H12) catch that by checking against anchors kept outside the database. Changes to <code>bb.policy</code> are now recorded, with who made them, in the append-only <code>bb.policy_history</code> (17 Sep 2026); that history is not hash-chained or anchored, so a superuser can still rewrite it unseen.
ADDED IN	<code>db9dc0c</code> , <code>d14d9a1</code> (PR #13).

H3 Attribution

Working

WHAT IT DOES	Records who made each change and whether it was a human or an agent.
HOW IT WORKS	<code>bb._ctx</code> derives both attribution fields from the login role, which a client cannot change (<code>SET ROLE</code> leaves <code>session_user</code> alone): <code>actor = session_user</code> for a per-user role (F4) or an agent role, falling back to the injected <code>bb.actor</code> only for the engine's own shared roles (<code>foxbyte</code> , <code>db_client</code>); <code>kind = agent</code> for an <code>agent-<id></code> role, <code>human</code> for a per-user role — one named for an email address is a person even if the address starts with "agent-" — and for the shared roles the injected <code>bb.actor_kind</code> . <code>tool = application_name</code> and <code>session = bb.session</code> (or the backend pid) stay as the client declares them, and are labelled declared on the Blackbox page. Console sets <code>actor</code> to the signed-in email; agent branches get <code>bb.actor='agent-<id>'</code> database defaults; MCP passes its key's account.
HOW TO USE	Automatic.
HOW TO TEST	Unit: <code>TestActorKindDerivedFromRole</code> (the kind is derived, and the session setting is only the shared-role fallback). Integration §8 (<code>actor_kind = human</code>); integration-v2 §1c: a human session that sets <code>bb.actor_kind='agent'</code> is still recorded as human, and an agent session that sets <code>'human'</code> is still recorded as an agent.
STATUS & LIMITS	Non-forgable where an identity exists: gateway users log in as their own role, each agent branch as its own <code>agent-<id></code> role, and the kind now follows the role rather than a session setting — until 18 Sep 2026 a client could <code>SET bb.actor_kind</code> and be recorded as the other kind, which is exactly what this record exists to prevent. The engine's own shared roles (console, MCP, imports) have no identity of their own, so their sessions still carry an injected actor and kind — set by engine code, not by the client. Tool and session are declared by the client by design (an <code>application_name</code> is whatever a client sends) and are shown as declared.
ADDED IN	<code>e42e792</code> ; non-forgable <code>96c8c31</code> (PR #16); kind derived from the role <code>feat/attribution-mcp-auth</code> .

H4 Destructive-DDL guardrail

Working

WHAT IT DOES	Refuses <code>DROP TABLE</code> and <code>DROP SCHEMA</code> before they run, and still records the attempt.
HOW IT WORKS	<code>ddl_command_start</code> → <code>guard_ddl_start</code> looks up <code>bb.policy</code> (seeded <code>DROP TABLE/SCHEMA</code> → block), honours <code>bb.allow_destructive</code> only when <code>bb._may_override()</code> is true (H8), and raises <code>insufficient_privilege</code> with a HINT; a <code>dblink</code> insert records status <code>BLOCKED</code> in its own transaction so it survives the rollback. Imports/pipelines disable the trigger temporarily (I9).
HOW TO USE	Automatic: <code>DROP TABLE t;</code> → "FoxByte guardrail: DROP TABLE is blocked by policy".
HOW TO TEST	Integration §8 (DROP blocked; BLOCKED row recorded). integration-v2 §6b: <code>flag</code> runs the command and records it <code>FLAGGED</code> ; an unknown action is refused; a blocked DROP TABLE after a write in the same transaction returns the unchanged error at once instead of hanging.
STATUS & LIMITS	Since Phase 1 the override is honoured only for superusers and members of <code>db_admin</code> (H8); anyone else stays blocked. Each use of the override is recorded in <code>bb.ledger_ext.override_used</code> (H9). <code>block</code> , <code>flag</code> (runs, recorded <code>FLAGGED</code>) and <code>allow</code> (runs, as if unlisted) work, and any other action is refused (<code>policy_action_valid</code> , <code>NOT VALID</code> so older rows are left alone); every change is recorded in <code>bb.policy_history</code> . No CLI/API/UI to manage <code>bb.policy</code> yet. Fixed 17 Sep 2026: <code>flag</code> did nothing, a typo such as <code>blok</code> silently let the command through, and a blocked command after a Blackbox write in the same transaction (<code>BEGIN; CREATE TABLE ...; DROP TABLE ...; </code>) hung the client forever — the <code>dblink</code> connection recording the attempt waited for the append lock its own session held. That attempt now isn't recorded (a WARNING says so) and the block stands. Broader, rule-based checks with <code>warn/block</code> and a machine-readable error are the policy gate (H15), which runs after this guardrail and leaves it unchanged.
ADDED IN	<code>e42e792</code> .

H5 Risk flags

Working

WHAT IT DOES	Marks risky changes so they stand out.
HOW IT WORKS	Regexes on the statement that ran set status <code>FLAGGED</code> with risk <code>type-change</code> , <code>drop-column</code> or <code>security-definer</code> ; drops get risk <code>drop</code> , blocked attempts and commands the guardrail marks <code>flag</code> get <code>policy</code> . The statement comes from <code>bb._statement_texts</code> (<code>ledger.sql</code>): the query as sent is split into statements with comments removed and string literals emptied, and a position per backend — in a table clients cannot write, moved by a function they cannot call — says which one is running. DDL run from inside a function or <code>D0</code> block is judged by the statement Postgres reports running (<code>PG_CONTEXT</code>). When the statement can't be established, the whole query is used, as before.
HOW TO USE	Automatic; filter with <code>?risk=</code> / <code>?status=FLAGGED</code> .
HOW TO TEST	integration-v2 §6b (17 Sep 2026, 24 checks): the tokenizer on comments, nested comments, standard, <code>E'...'</code> and dollar-quoted literals, quoted identifiers and multibyte text; a literal mentioning drop column is not flagged; in add-then-drop only the drop is. Unit: <code>internal/ledger/statement_test.go</code> pins the safeguards (checked by deliberately removing two of them).
STATUS & LIMITS	Working. Still text matching on the statement, not dependency analysis (impact analysis, H17, does that). Query strings over 256 KB, and runs whose statements can't be lined up with the commands that fire, fall back to the whole query.
ADDED IN	<code>e42e792</code> ; security-definer <code>db9dc0c</code> .

H6 Blackbox access (CLI / REST / MCP / UI / SDK)

Working

WHAT IT DOES	Read the ledger from anywhere.
HOW IT WORKS	CLI <code>Ledger</code> (psql table, default 50 rows). REST <code>GET /api/branches/{name}/ledger</code> with filters actor, table, risk, status, kind, since, until, limit (≤ 2000), offset. MCP <code>changes</code> . Web Ledger page (L5). SDK <code>ledger()</code> .
HOW TO USE	<code>fox ledger qa --limit 20</code> ; <code>curl -k -H "Authorization: Bearer \$KEY"</code> <code>"https://localhost:8080/api/branches/main/ledger?status=BLOCKED"</code> Blackbox names: <code>fox</code> <code>blackbox [branch]</code> , <code>GET /api/branches/{name}/blackbox</code> , <code>web /blackbox</code> , SDK <code>blackbox()</code> (H14).
HOW TO TEST	No automated test of the endpoints; manual curl above.
STATUS & LIMITS	Working. Any signed-in user can read any branch's ledger. No export format.
ADDED IN	<code>e42e792</code> ; MCP <code>d62c676</code> (PR #14).

H7 `fox ledger revert --to`

Working

WHAT IT DOES	A point-in-time restore of <code>main</code> into a disposable container on port 5433 — the same as <code>fox restore</code> . Nothing is reverted: <code>main</code> and every branch stay as they are. The help and the command's output say so, and point to <code>fox blackbox branch-before <id></code> (H13) for a branch holding <code>main</code> as it was just before a change.
HOW IT WORKS	Calls <code>branch.Restore(ts)</code> (D3). No branch argument.
HOW TO USE	<code>fox blackbox revert --to '<timestamp>' latest</code> (also <code>fox ledger revert</code>)
HOW TO TEST	Unit: <code>TestUsageDescribesRevertAndFailback</code> (the help describes a restore of <code>main</code> , not a revert of a branch). The restore itself: integration §3 (D3).
STATUS & LIMITS	Working. The name is kept so scripts don't break; its description no longer claims it reverts a branch (it did until 15 Sep 2026). Inherits D3's limit: always restores from the latest base backup.
ADDED IN	<code>e42e792</code> .

H8 Override permission (db_admin) & ledger upgrade

Working

WHAT IT DOES	Decides who may override the destructive-DDL guardrail, and applies the current ledger definition to branches that already exist.
HOW IT WORKS	<code>ledger.sql</code> creates role <code>db_admin</code> and <code>bb._may_override()</code> — true for superusers or members of <code>db_admin</code> , checked on <code>session_user</code> (the login identity a client can't change). <code>fox admin grant</code> ensures the user's per-user role and runs <code>GRANT db_admin (internal/branch/security.go , cmd/fox/admin.go)</code> . Roles live in each branch's Postgres, so without <code>--branch</code> a change applies to main and every running branch; new branches inherit main's. The first local user created by <code>fox start</code> is granted automatically. <code>fox ledger upgrade</code> re-runs the idempotent <code>ledger.sql</code> ; main is also upgraded on every <code>fox start</code> .
HOW TO USE	<code>fox admin grant you@x.com · fox admin list · fox admin revoke you@x.com --branch qa · fox ledger upgrade --all</code> or the Who can override panel on the Policies page (<code>GET POST /api/branches/{name}/admins</code> , <code>DELETE .../admins/{email}</code>).
HOW TO TEST	integration-v2 §1b: non-admin override refused with a HINT, admin override works, revoke restores the block, superuser override (imports/pipelines) unaffected. §6: the REST list says whether the caller is an admin, a non-admin can't grant (403), unknown accounts and non-admins are 404. Live on 17 Sep 2026: an unknown email was 404 and removing the only admin was refused (409).
STATUS & LIMITS	Verified in the VM (integration-v2 §1b, 14 Sep 2026). Existing installs: the owner runs <code>fox admin grant <email></code> once (only brand-new installs grant the first user automatically), and existing branches need <code>fox ledger upgrade --all</code> . Suspended branches are skipped until running. From the web, only an admin of a branch may grant or revoke there (otherwise anyone could grant themselves), and the last admin can't be removed from the web.
ADDED IN	Blackbox 2.0 Phase 1 (<code>feat/ledger2-phase1-security</code>).

H9 Blackbox 2.0 capture (transaction, WAL position, provenance)

Working

WHAT IT DOES	Records extra context for every ledger row: the transaction id and WAL position of the change, the agent task and tool-call hash when the session sets them, and whether the destructive-DDL override was in effect. It is the foundation for branching from just before a change (Phase 4) and agent provenance (Phase 6).
HOW IT WORKS	New file <code>internal/ledger/ledger_v2.sql</code> , which never alters an object <code>ledger.sql</code> owns. Side table <code>bb.ledger_ext</code> , keyed by ledger row id, is filled by an <code>AFTER INSERT</code> trigger <code>bb_ext_capture</code> that reads <code>txid_current()</code> , <code>pg_current_wal_insert_lsn()</code> and the settings <code>bb.task</code> , <code>bb.parent_session</code> , <code>bb.call_hash</code> and <code>bb.allow_destructive</code> . Each row carries <code>ext_hash</code> (<code>bb._ext_hash</code>). Append-only like the ledger; clients may read, not write. Fail-safe: any capture error becomes a WARNING and the user's change still goes through. Kill switch: <code>ALTER DATABASE foxbyte SET bb.v2 = 'off'</code> (needs the database owner or a superuser), or <code>SET bb.v2 = 'off'</code> in a superuser's own session. A client — including a <code>db_admin</code> member — setting it in its own session is ignored (<code>bb._capture_disabled</code>), so nobody can hide the capture details of their own changes. Installed on main at every <code>fox start</code> (best effort — never stops start-up) and by <code>fox ledger upgrade</code> .
HOW TO USE	Automatic. Tag a session: <code>PGOPTIONS='-c bb.task=task-42' psql ...</code> . Inspect: <code>SELECT * FROM bb.ledger_ext ORDER BY ledger_id DESC LIMIT 5</code> ; Installed version: <code>SELECT bb.ledger_v2_version()</code> ; The Agent API and MCP <code>execute_change</code> fill the task, parent session and call hash automatically (H16).
HOW TO TEST	integration-v2 §2: one capture row per ledger row; xid and LSN present; task and call hash captured; hash verifies; override use recorded; a failing capture doesn't block DDL; kill switch (database-wide, and a superuser session); a client's own <code>SET bb.v2 = 'off'</code> — also as <code>db_admin</code> — does not skip capture; append-only; clients can't write; base chain still verifies.
STATUS & LIMITS	Verified in the VM (integration-v2 §2, 14 Sep 2026). Ledger rows written before 2.0 was installed have no capture row. Blocked attempts are inserted through a separate dblink session, so their capture carries that session's transaction and no task. Existing branches need <code>fox ledger upgrade --all</code> . Capture hashes are covered by checkpoint anchors (H10).
ADDED IN	Blackbox 2.0 Phase 2 (<code>feat/ledger2-phase2-capture</code>).

H10 Checkpoints & anchors

Working

WHAT IT DOES	Takes tamper-evidence outside the database. At intervals it records a Merkle root over the newest ledger entries — in the database and as a read-only anchor file somewhere else — so rewritten, deleted or wiped history can be proven later.
HOW IT WORKS	<code>bb.ledger_checkpoints (ledger_v2.sql)</code> is append-only, and a trigger refuses any checkpoint that doesn't continue the last one (<code>from_id = last to_id + 1, prev_root = last root</code>). The engine (<code>internal/branch/checkpoint.go</code>) reads the new entries, refuses if their hash chain is already broken, and builds the anchor with <code>internal/ledger/integrity.go</code> : each leaf is <code>sha256(0x00 recomputed row hash "I" capture hash)</code> , each node <code>sha256(0x01 left right)</code> . It records the checkpoint, then writes <code>~/.fox/anchors/<branch>/<to_id>.json</code> with mode 0444, never overwriting. A scheduler in the control plane checkpoints running branches every <code>FOX_CHECKPOINT_INTERVAL</code> (10m) or once <code>FOX_CHECKPOINT_ENTRIES</code> (500) are waiting; <code>FOX_CHECKPOINTS=off</code> disables it. <code>FOX_ANCHOR_DIR</code> moves anchors (e.g. to a write-once mount); <code>FOX_ANCHOR_IMMUTABLE=1</code> adds <code>chattr +i</code> . The format is published in <code>docs/ledger-anchor-format.md</code> .
HOW TO USE	<code>fox ledger checkpoint [branch]</code> · <code>POST /api/branches/{name}/ledger/checkpoint</code> · web Integrity page → Create checkpoint
HOW TO TEST	Unit: <code>internal/ledger/integrity_test.go</code> . integration-v2 §3: anchor written read-only in the published format; no-op when nothing is new; the database refuses an out-of-sequence checkpoint; checkpoints are append-only. §3b: the scheduler anchors new entries on its own.
STATUS & LIMITS	Verified in the VM (integration-v2 §3 and §3b, 14 Sep 2026). Anchors are only as trustworthy as where they're stored — the default directory is on the same host, so point <code>FOX_ANCHOR_DIR</code> at storage the database's operators can't rewrite. Entries written since the last checkpoint are protected by the hash chain only. Cloud write-once anchor writers (S3 Object Lock etc.) are not built (commercial scope). A cloned branch inherits its parent's checkpoint rows but not its anchor files.
ADDED IN	Blackbox 2.0 Phase 3 (<code>feat/ledger2-phase3-checkpoints</code>).

H11 fox-verify — independent verifier

Working

WHAT IT DOES	A separate open-source program anyone can run to check a ledger without trusting FoxByte, its engine or the database.
HOW IT WORKS	<code>cmd/fox-verify</code> reads every ledger entry over a plain Postgres connection (read access through the gateway with an API key is enough) or from a <code>fox ledger export</code> file, recomputes every row hash and Merkle root with the standard-library code in <code>internal/ledger/integrity.go</code> , and compares them with the anchor files. Exit 0 = intact, 1 = tampered, 2 = usage or connection error; <code>--json</code> prints the full report. Built and released as <code>fox-verify-<os>-<arch></code> next to <code>fox</code> .
HOW TO USE	<code>fox-verify --dsn 'postgres://dbadmin:<api-key>@localhost:6432/main?sslmode=require' --anchors ~/.fox/anchors/main</code> · offline: <code>fox ledger export main > main.jsonl</code> then <code>fox-verify --export main.jsonl --anchors ./anchors/main</code>
HOW TO TEST	Unit tests (shared verification code). integration-v2 §3: agrees with integrity on a live branch and on an export; exits 1 on a forged edit.
STATUS & LIMITS	Verified in the VM (integration-v2 §3 and §3b, 14 Sep 2026). Needs a copy of the branch's anchor files. Reads the whole ledger into memory.
ADDED IN	Blackbox 2.0 Phase 3 (<code>feat/ledger2-phase3-checkpoints</code>).

H12 Blackbox integrity & export

Working

WHAT IT DOES	Answers "has this branch's schema history been tampered with?" and exports the full ledger for audits.
HOW IT WORKS	<code>Integrity</code> (<code>internal/branch/checkpoint.go</code>) loads every entry and the branch's anchor files and runs the same checks as <code>fox-verify</code> : the hash chain, the anchor sequence, and each anchor's entry count, Merkle root and last entry. It notes when the database's own checkpoint list disagrees with the anchors (the anchors win). Unlike <code>fox ledger verify</code> , which only checks the chain inside the database, it catches a rewritten chain. Export writes every entry with its capture columns as JSON lines.
HOW TO USE	<code>fox ledger integrity [branch]</code> (exit 1 when tampered) · <code>fox ledger export [branch] > ledger.jsonl</code> · REST <code>GET .../ledger/integrity</code> , <code>GET .../ledger/export</code> · MCP <code>ledger_integrity</code> · web Integrity page
HOW TO TEST	integration-v2 §3: an untampered branch is INTACT via CLI, REST, MCP and <code>fox-verify</code> ; a forged edit with a rewritten chain fools the in-database check but is TAMPERED here; so are a deleted newest anchored entry and a wiped ledger; deleting the database's checkpoint rows leaves it INTACT because the anchors are the source of truth.
STATUS & LIMITS	Verified in the VM (integration-v2 §3 and §3b, 14 Sep 2026). Reads the whole ledger on each check (fine at typical sizes; very large ledgers would need streaming).
ADDED IN	Blackbox 2.0 Phase 3 (<code>feat/ledger2-phase3-checkpoints</code>).

H13 Branch from just before a Blackbox entry

Working

WHAT IT DOES	Creates a new branch holding main exactly as it was the moment before a chosen schema change — to inspect what things looked like, or to recover from a bad change — without touching main.
HOW IT WORKS	<code>BranchBeforeEntry</code> (<code>internal/branch/branch_before.go</code>) reads the entry and its captured transaction id and WAL position (H9) from the current primary, and refuses BLOCKED entries (the change never happened), sources other than main (only the primary archives WAL) and names already in use. It switches main to a new WAL segment and waits for the archiver, so the change's commit is in object storage. It lists base backups (<code>wal-g backup-list --json --detail</code> in a throwaway container) and picks the newest backup <i>of the same database</i> (system identifier) that finished before the change — by WAL position, or by time for entries without capture. It creates empty storage for the new branch and runs the stock image with an embedded script (<code>internal/branch/restore_before.sh</code> , passed with <code>bash -c</code> , so the published image needs no new file): <code>backup-fetch</code> of that backup, <code>recovery_target_xid</code> (or <code>recovery_target_time</code>) with <code>recovery_target_inclusive = off</code> , then promote. Once promoted it checks the entry is absent, clears the recovery settings (<code>ALTER SYSTEM RESET</code>), restarts the data directory as an ordinary branch container and re-enables the guardrail. Any failure removes the partial branch. One runs at a time.
HOW TO USE	Find the entry id with <code>fox ledger entries [--limit N]</code> (REST <code>GET /api/branches/main/ledger/entries</code> , MCP <code>ledger_entries</code> ; the existing ledger views don't show ids and are unchanged), then <code>fox ledger branch-before 612 [--as fix-orders]</code> (default name <code>main-before-612</code>) · REST <code>POST /api/branches/main/ledger/612/branch</code> with optional <code>{"name": "..."} (201; 400 invalid or BLOCKED, 404 no such entry, 409 name taken, 422 no usable base backup)</code> · MCP <code>branch_before_change</code> · web Integrity page → "Branch from before a change". Connect through the gateway like any branch; remove it with <code>fox branch delete</code> .
HOW TO TEST	Unit: <code>internal/branch/branch_before_test.go</code> and <code>ledger_entries_test.go</code> (entry listing, entry parsing, <code>xid/time</code> targets, <code>backup-list</code> parsing, base-backup choice incl. another database's backups, WAL-archive check, names, the embedded script). integration-v2 §4: the entry id is listed by CLI, REST and MCP; by transaction id via the CLI — rows written before the change (even after the backup) are present, the change and later changes are not, the ledger stops before the entry, no recovery settings left, reachable through the gateway with the guardrail active, main untouched and its integrity result unchanged; by timestamp via MCP for an entry without capture; via REST; refusals for an existing name, a BLOCKED entry, an unknown entry, a non-main source and a bad id, with nothing left behind.
STATUS & LIMITS	Verified in the VM (integration-v2 §4, 14 Sep 2026: by transaction id, by timestamp, via CLI, REST and MCP). Takes minutes, not seconds (base backup fetch plus WAL replay), and the REST call waits for it. Needs a base backup that finished before the change (<code>fox backup create</code> ; schedule backups to make older changes reachable). It excludes the change's whole transaction, so other statements in the same transaction are excluded too. Entries recorded before Blackbox 2.0 capture use their timestamp, which also excludes anything that committed after the change started. Only main can be the source; after an HA failover the promoted standby doesn't archive WAL, so later changes can't be branched before. The new branch keeps main's checkpoint rows up to that point but not its anchor files, so <code>fox ledger integrity</code> on it notes the difference.
ADDED IN	Blackbox 2.0 Phase 4 (<code>feat/ledger2-phase4-branch-before</code>).

H14 Blackbox names & aliases

Working

WHAT IT DOES	Blackbox is the product name for the record of schema changes that was called the Schema Ledger. Every surface gains a Blackbox name; every original name keeps working, so no script, integration, stored record or anchor breaks.
HOW IT WORKS	CLI: <code>fox blackbox ...</code> runs the same handler as <code>fox ledger ...</code> (<code>cmd/fox/main.go</code>). REST: <code>blackboxAlias</code> (<code>internal/controlplane/blackbox.go</code>) sits inside the auth gate and serves <code>/api/branches/{name}/blackbox[/...]</code> with the <code>.../ledger[/...]</code> handler. MCP: <code>verify_blackbox</code> , <code>blackbox_integrity</code> and <code>blackbox_entries</code> are listed next to and resolve to <code>verify_ledger</code> , <code>ledger_integrity</code> , <code>ledger_entries</code> (<code>resolveTool</code> in <code>internal/mcp/mcp.go</code>). SDKs: Python <code>blackbox()</code> / <code>verify_blackbox()</code> , TypeScript <code>blackbox()</code> / <code>verifyBlackbox()</code> call the original routes, so they also work against older servers. Web: <code>nav "Blackbox"</code> at <code>/blackbox</code> ; <code>/ledger</code> still opens the same page. Kept as they were: SQL objects (<code>bb.schema_ledger</code> , <code>bb.ledger_ext</code> , <code>bb.ledger_checkpoints</code> , <code>bb._ledger_hash</code>), the Go package <code>internal/ledger</code> , the anchor format <code>ledger-anchor/1</code> and <code>docs/ledger-anchor-format.md</code> file name, output lines such as <code>ledger integrity: INTACT</code> , JSON fields such as <code>last_ledger_id</code> , the export file name.
HOW TO USE	<code>fox blackbox · fox blackbox integrity main · curl -k -H "Authorization: Bearer \$KEY" https://localhost:8080/api/branches/main/blackbox/verify · MCP blackbox_integrity · db.verify_blackbox("main")</code>
HOW TO TEST	Unit: <code>internal/controlplane/blackbox_test.go</code> (<code>TestBlackboxToLedgerPath</code> , <code>TestBlackboxAlias</code>), <code>internal/mcp/mcp_test.go</code> (<code>TestBlackboxToolAliases</code>). integration-v2 §5: CLI <code>verify/entries/integrity</code> , REST <code>verify/list/entries</code> and the auth requirement, and MCP listing, integrity and verify all match the original names.
STATUS & LIMITS	Verified in the VM (integration-v2 §5, 14 Sep 2026; existing suite 28 / 28). The OpenAPI spec documents the alias in the ledger tag and info description rather than repeating every path, so generated clients only get the <code>/ledger</code> operations. The screenshot file is still <code>docs/screenshots/ledger.png</code> and shows the old page title until recaptured.
ADDED IN	Blackbox rename (<code>feat/blackbox-phase5-policy</code>).

H15 Policy gate (warn / block)

Working

WHAT IT DOES	Checks every schema change against policy rules before it runs. A warn rule lets the change through and tells the client why it is risky; a block rule refuses it. Tools and agents get a machine-readable answer they can act on, defined in the published contract <code>docs/policy-errors.md</code> .
HOW IT WORKS	<code>internal/ledger/policy.sql</code> , installed after the 2.0 objects by <code>EnsureLedgerV2</code> (every <code>fox start</code> , <code>fox blackbox upgrade</code>). Rules live in <code>bb.policy_rules</code> (rule id, command tag, optional case-insensitive regex over the statement, <code>warn block</code> , reason, hint, enabled); changes go to the append-only <code>bb.policy_rule_history</code> with who made them. The event trigger <code>bb_policy_start</code> fires on <code>ddl_command_start</code> after the guardrail <code>bb_guard_start</code> (name order). Warn : an evaluation row and a NOTICE with SQLSTATE <code>BBX02</code> . Block : a <code>BLOCKED</code> Blackbox entry (risk <code>policy</code>) and an evaluation written through dblink so they survive the rollback, then an ERROR with <code>BBX01</code> ; blocks win over warnings. DETAIL is one-line JSON (<code>v</code> , <code>rule_id</code> , <code>action</code> , <code>command</code> , <code>matched</code> , <code>reason</code> , <code>hint</code> , <code>override</code> , <code>evaluation_id</code> , <code>blackbox_id</code> , <code>impact</code>). Override : superusers and <code>db_admin</code> members may <code>SET bb.policy_allow = '<rule ids>'</code> ; it is recorded as an <code>allowed</code> evaluation and in the capture row's <code>override_used</code> ; <code>bb.allow_destructive</code> does not override policy. Fail-safe : any error while matching or recording becomes a WARNING and never stops a statement a warn rule matched; a block still refuses (with <code>evaluation_id null</code>). The gate ignores the client-settable <code>bb.v2</code> ; a superuser turns it off with <code>ALTER EVENT TRIGGER bb_policy_start DISABLE</code> . Rule changes through REST need <code>db_admin</code> on that branch (<code>IsAdmin</code>); clients can read rules and evaluations but not write them. The existing guardrail (<code>bb.policy</code> , <code>DROP TABLE/SCHEMA</code> , SQLSTATE <code>42501</code>) is unchanged.
HOW TO USE	<code>fox policy · fox policy check "ALTER TABLE orders DROP COLUMN note"</code> (exit 1 if a rule would block) · <code>fox policy block drop-column · fox policy add no-orders-drop --command "ALTER TABLE" --pattern 'orders' --block --reason "orders is shared"</code> · <code>fox policy evaluations</code> · REST <code>GET POST /api/branches/{n}/policies</code> , <code>PUT DELETE .../policies/{rule}</code> , <code>POST .../policies/check</code> , <code>GET .../policies/evaluations</code> · MCP <code>policy_check</code> · web Policies page. In an application: catch SQLSTATE <code>BBX01</code> and parse the error's DETAIL as JSON.
HOW TO TEST	integration-v2 §6b (statement matching, previews, blocks in multi-statement runs, <code>D0</code> blocks, clients can't move the position). Unit: <code>internal/ledger/statement_test.go</code> , <code>internal/ledger/policy_test.go</code> (TestSchemaPolicy — both SQLSTATEs, fires after the guardrail, built-ins ship as warn, ignores <code>bb.v2</code> , touches no existing object; TestParsePolicyDetail; TestPolicyContractExample — the contract's own example parses; TestCommandTag; TestValidRuleID) and <code>internal/branch/policy_test.go</code> (validation, rule SQL quoting, error messages, formatting). integration-v2 §6: warn (runs, <code>BBX02</code> , contract JSON, evaluation) for all four built-ins; block (<code>BBX01</code> , statement didn't run, JSON with ids, <code>BLOCKED</code> entry and evaluation survive the rollback, <code>HINT</code> names the override); non-admin override, <code>bb.v2=off</code> and <code>bb.allow_destructive</code> don't bypass; admin override runs and is recorded; REST list / 200 / 403 / 404 / 409 / check / evaluations; CLI check and exit code; MCP <code>policy_check</code> ; clients can't write rules; fail-safe; kill switch; custom and disabled rules; rule history; invalid pattern; built-in can't be removed; append-only evaluations; guardrail error unchanged; reinstall keeps rule changes; hash chain still verifies.
STATUS & LIMITS	Verified in the VM (integration-v2 §6, 14 Sep 2026; existing suite 28 / 28). Patterns match the statement being run, with comments removed and string literals emptied (see H5 and <code>docs/policy-errors.md</code>); where that statement can't be established — DDL inside a function is matched by the statement Postgres reports, a query over 256 KB or a run that falls out of step — the whole query is used, as before. A <code>D0</code> block that builds DDL from strings is now caught. Rules are a guard against mistakes, not a security boundary against a client determined to evade a text pattern. <code>fox policy check</code> guesses the command tag from the statement's first words; the gate itself uses the real tag. <code>TRUNCATE</code> is not covered (it doesn't fire DDL event triggers). Imports and pipelines run through the gate too (with the built-in warn rules that only adds notices). Rules are per branch; new branches copy main's. The <code>impact</code> field is always null until impact analysis (Phase 7).
ADDED IN	Blackbox 2.0 Phase 5 (<code>feat/blackbox-phase5-policy</code>).

H16 Agent provenance & execute_change

Working

WHAT IT DOES	Answers "which agent run, for which task, spawned by what, made this change?" for every change an agent makes — and gives agents one tool that runs a change with that provenance attached, checks policy first, and says exactly what happened.
HOW IT WORKS	<code>internal/ledger/provenance.sql</code> (installed by <code>EnsureLedgerV2</code>) adds <code>bb.agent_sessions</code> : session id, agent, parent session, task, tool, start time; append-only, written only by the engine, readable by clients. A session's changes carry its id in the Blackbox entry's <code>session</code> and the task, parent session and call hash in the capture row (H9), which checkpoints anchor (H10). Agent API: <code>POST /agents/{id}/branch</code> with <code>{"task_id", "parent_session_id", "session_id"}</code> creates the branch, records the session (tool <code>agent-api</code>) and sets <code>bb.session</code> , <code>bb.task</code> and <code>bb.parent_session</code> as the branch database's defaults, so the agent's own connections are attributed without it doing anything; if recording fails the branch is removed. Invalid ids (not 1–200 printable characters) get 400 before any branch is made. Without those fields the original code path and response are used. MCP <code>execute_change</code> (<code>branch.ExecuteChange</code>): one session per MCP server process (agent = the client's name from <code>initialize</code>); previews the policy rules (H15); unless <code>dry_run</code> , records the session, connects as the non-superuser <code>db_client</code> , sets the actor, <code>actor_kind = agent</code> , tool <code>mcp</code> , session, task, parent session and <code>bb.call_hash</code> (sha256 over the tool, branch, sql, task and parent session), runs the SQL, and returns JSON: <code>status</code> (<code>applied</code> <code>blocked</code> <code>error</code> <code>preview</code>), <code>command</code> , <code>session_id</code> , <code>call_hash</code> , <code>policy_preview</code> , <code>notices</code> (with the parsed BBX02 detail), <code>policy</code> (the BBX01 rule), <code>error</code> and <code>blackbox_entries</code> . Blocked and error results are also flagged <code>isError</code> . <code>run_sql</code> is unchanged. Its result also carries <code>impact</code> , the impact report for the statement when its target is recognised (H17).
HOW TO USE	<code>curl -k -H "Authorization: Bearer \$KEY" -X POST -d '{"task_id":"JIRA-42","parent_session_id":"planner-1"}' https://localhost:8088/agents/alice/branch · MCP <code>execute_change {"sql": "ALTER TABLE orders ADD COLUMN note text", "task_id": "JIRA-42"} (add "dry_run": true to preview) · fox blackbox sessions [branch] · REST GET /api/branches/{n}/blackbox/sessions · SQL: SELECT s.id, s.actor, a.task_id, a.parent_session_id FROM bb.schema_ledger s JOIN bb.agent_sessions a ON a.session_id = s.session</code></code>
HOW TO TEST	Unit: <code>internal/ledger/provenance_test.go</code> (<code>TestSchemaProvenance</code>) and <code>internal/branch/provenance_test.go</code> (<code>TestValidProvenanceID</code> , <code>TestNewSessionID</code> , <code>TestCallHash</code> , <code>TestProvenanceSQL</code> , <code>TestClassifyExecError</code> , <code>TestFormatAgentSessions</code>). integration-v2 §7: sessions table installed, clients can't write it; Agent API without provenance keeps its response keys and sets no session; with provenance returns and records session / task / parent, and the agent's own DDL over its DSN carries them; an invalid task id is a 400 with no branch made; <code>execute_change</code> applies and returns the entry, which carries agent, tool, session, task, parent and call hash; the MCP session is recorded; runs without superuser rights; a warn rule gives BBX02 in <code>notices</code> ; a block gives status <code>blocked</code> with the rule and its BLOCKED entry, and the change didn't run; <code>dry_run</code> runs nothing; the call hash is stable for the same arguments and differs otherwise; a database error is reported; an invalid task id is refused; CLI and REST sessions; the hash chain verifies.
STATUS & LIMITS	Verified in the VM (integration-v2 §7, 14 Sep 2026; existing suite 28 / 28). Provenance is supplied by the caller and not verified — it records what the agent said its task and parent were; the agent's identity itself stays non-forgeable (login role, H3). Agent API provenance is set per branch, so all connections to that agent branch share one session; start a new branch for a new session. An MCP session lasts as long as the MCP server process. A session recorded on an agent branch is removed together with that branch. <code>bb.agent_sessions</code> itself isn't anchored by checkpoints (the ids it explains are). No web view yet. Fixed 17 Sep 2026: since agents connect through the Gateway (K5) the Gateway gave every connection a session of its own, which overrode the branch's — an agent's changes carried a random session instead of the one it was created with. An agent's branch-scoped key now gets no Gateway session, so the branch's applies (integration-v2 §7 caught it).
ADDED IN	Blackbox 2.0 Phase 6 (<code>feat/blackbox-phase6-provenance</code>).

H17 Impact analysis

Working

WHAT IT DOES	Before a schema change runs, shows what it would affect: what depends on the object, which other branches have it, what the policy gate would say, and how risky it is overall — for people (<code>fox impact</code>) and for agents (MCP <code>impact</code> , and inside every <code>execute_change</code> result).
HOW IT WORKS	Target: <code>ledger.ParseTarget</code> reads the statement to find the table, view, materialized view, index or sequence it changes and, where relevant, the column and kind of change (<code>drop</code> , <code>drop-column</code> , <code>alter-column-type</code> , <code>rename</code> , <code>rename-column</code> , <code>truncate</code> are destructive; <code>alter</code> , <code>alter-column</code> , <code>create-index</code> , <code>grant</code> , <code>revoke</code> are not). Quoted and schema-qualified names work; <code>--object</code> / <code>--column</code> name it directly. Dependents: the read-only SQL function <code>bb.blast_radius(target, column, depth)</code> (<code>internal/ledger/impact.sql</code> , installed by <code>EnsureLedgerV2</code> , runs with the caller's rights and search path, so an unqualified name resolves as in the caller's own statement) walks <code>pg_depend</code> for normal and automatic dependencies and reports each dependent's type, identity, owning relation, depth and the view it was reached through: views and materialized views (and views on those, up to depth 5, capped at 10), constraints including foreign keys from other tables, indexes, triggers, row-level security policies, defaults, owned sequences and SQL-standard functions. For a column, only what uses that column. Every reported name is schema-qualified, and an index is reported against the table it indexes. Branches: every other running branch is checked for the object (and column) without waking suspended ones, with the branch it was cloned from (ZFS <code>origin</code> , btrfs parent UUID). Policy: the rules the statement would trigger (H15). Score (<code>scoreImpact</code>): destructive 5, otherwise 1; each dependent view, function or foreign key from another table +3, trigger or policy +2, index or other +1; deeper than checked +3; $\geq 10k$ rows +1, $\geq 1M$ +3; each other running branch with it +1 (max 5); a blocking rule +5, a warning +2. 15+ high, 5–14 medium, below low; every point is listed as a reason.
HOW TO USE	<code>fox impact "ALTER TABLE orders DROP COLUMN note"</code> · <code>fox impact --object orders_summary --json</code> · <code>fox impact "DROP TABLE orders" --branch qa</code> · REST POST <code>/api/branches/{n}/impact</code> with <code>{"sql": ...}</code> or <code>{"object": ..., "column": ...}</code> · MCP <code>impact</code> · SQL: <code>SELECT bb.blast_radius('public.orders', 'note')</code>
HOW TO TEST	Unit: <code>internal/ledger/impact_test.go</code> (TestSchemaImpact — read-only, capped depth, caller's rights; TestParseTarget — 27 statements; TestTargetDestructive) and <code>internal/branch/impact_test.go</code> (TestScoreImpact, TestBranchParentParsing — ZFS and btrfs output, TestFormatImpact). integration-v2 §8: a table's views (depth 1 and 2, via), the foreign key from another table and its index; a column's dependents only; quoted names; non-destructive changes; a missing object; a statement without a target asks for <code>--object</code> ; a view's dependents; the depth limit; the object on a branch cloned after it existed (with its parent) and not on one without it; the policy verdict in the score; REST and MCP agree with the CLI; <code>execute_change</code> includes the impact.
STATUS & LIMITS	Verified in the VM (integration-v2 §8, 15 Sep 2026; existing suite 28 / 28). The target is read from the statement text: one object per statement (the first of a multi-object DROP), and statements on functions, types, schemas or whole-schema grants aren't recognised — name the object with <code>--object</code> . Dependencies PostgreSQL doesn't record aren't seen: application queries, functions with plain-string bodies (only SQL-standard <code>BEGIN ATOMIC</code> bodies are tracked), dynamic SQL, other databases. Row counts are the planner's estimate. The score is a guide, not a guarantee. The policy error DETAIL's <code>impact</code> field stays null (the gate doesn't run this inside the database); use <code>fox impact</code> or <code>execute_change</code> . No web view yet.
ADDED IN	Blackbox 2.0 Phase 7 (<code>feat/blackbox-phase7-impact-diff</code>).

H18 Blackbox diff between branches

Working

WHAT IT DOES	Shows which schema changes distinguish two branches: where they split, what was changed only on each since, and which objects were changed on both (likely conflicts). Diff only — nothing is merged or changed.
HOW IT WORKS	<code>branch.DiffLedgers</code> (<code>internal/branch/diff.go</code>) reads both branches' Blackbox histories in id order (starting a suspended branch first, as other Blackbox commands do). A branch is a copy-on-write clone, so it begins with its parent's history entry for entry; the fork point is the longest run of entries with the same id and the same recomputed hash. Everything after it is listed for each side (id, time, command, object, status, actor). Objects changed on both sides are listed with their entry ids; BLOCKED entries don't count, since those changes never happened. Each branch's parent is read from the storage layer.
HOW TO USE	<code>fox blackbox diff main qa</code> · <code>fox branch diff main qa --json</code> · REST GET <code>/api/blackbox/diff?a=main&b=qa</code> (also <code>/api/ledger/diff</code>) · MCP <code>blackbox_diff</code>
HOW TO TEST	Unit: <code>internal/branch/impact_test.go</code> <code>TestBuildDiff</code> (fork point, same id with different content, both-changed objects, BLOCKED ignored, self diff, no shared history, text output). <code>integration-v2 §8</code> : a branch cloned from main, changes on each side — the parent, shared history, changes only on main, only on the branch, the object both changed; <code>fox branch diff</code> matches; text output; a branch against itself; REST on both paths; unknown branch 404; MCP.
STATUS & LIMITS	Verified in the VM (<code>integration-v2 §8</code> , 15 Sep 2026; existing suite 28 / 28). It compares recorded schema changes, not the schemas themselves: changes made while Blackbox wasn't recording (before it was installed, or with the capture turned off by a superuser) don't appear, and data changes are out of scope. Branches that weren't cloned from each other (e.g. two imports) have no shared history, so everything shows on both sides. Rewritten history on one branch moves the fork point earlier — check <code>fox blackbox integrity</code> . Reads both histories in full.
ADDED IN	Blackbox 2.0 Phase 7 (<code>feat/blackbox-phase7-impact-diff</code>).

Import & migration

Every import lands in a **new branch** (a copy-on-write clone of `main` whose `public` schema is wiped first), so a failed import never touches `main`. An import with 0 tables fails loudly. Code: `internal/branch/import.go`. Only the MongoDB path has end-to-end automated coverage (integration §9–10).

I1 From Postgres

Working

WHAT IT DOES	Copies a Postgres (or compatible) database, schema and data, into a new branch.
HOW IT WORKS	<code>loadPostgres : pg_dump <dsn> --no-owner --no-acl --no-comments psql</code> inside the target container. <code>psql</code> 's errors are counted: if any statement failed, the import fails listing them (see I5).
HOW TO USE	<code>fox import --from postgres://u:pw@host:5432/db [--as name] ; REST POST /api/import ; web Import.</code>
HOW TO TEST	Manual: start <code>postgres:18</code> (or any older major) on network <code>dbnet</code> with a 2-row table, import, <code>SELECT count(*) = 2</code> .
STATUS & LIMITS	Working; no automated test of the Postgres source itself (the error handling is shared with I5 and tested there); depends on <code>pg_dump</code> version compatibility — a statement the target can't run now fails the import instead of being skipped silently. <code>pg_dump</code> runs in the engine image, and an older <code>pg_dump</code> cannot dump a newer server, so the newest source this accepts is the engine's own major: moving a fresh install to PostgreSQL 18 (21 Sep 2026) raised that from 16 to 18. (The test note named the network <code>foxbyte</code> ; it has been <code>dbnet</code> since the names were frozen.)
ADDED IN	<code>44cc05c</code> .

I2 From MySQL ≤5.7 / MariaDB

Working

WHAT IT DOES	Migrates MySQL/MariaDB with type mapping.
HOW IT WORKS	<code>loadMySQL</code> builds a local <code>pgloader</code> image on first use and runs <code>pgloader <dsn> postgresql://...@pg-<target></code> ; success judged by table count.
HOW TO USE	<code>fox import --from mysql://u:pw@host:3306/db --as legacy (or mariadb://)</code>
HOW TO TEST	Manual: <code>mariadb:11</code> container with a table, import, query rows.
STATUS & LIMITS	Working; no automated test.
ADDED IN	<code>b842a28</code> ; hardening <code>c929eb5</code> .

I3 From MySQL 8.x

Working

WHAT IT DOES	Imports MySQL 8+ (whose auth <code>pgloader</code> can't handle).
HOW IT WORKS	<code>mysqlNeedsNative</code> detects ≥8; <code>loadMySQLNative</code> generates <code>CREATE TABLE</code> from <code>information_schema</code> (type map: <code>json</code> → <code>jsonb</code> , decimal kept, else text) and streams <code>mysqldump --no-create-info</code> INSERTs; per-table progress.
HOW TO USE	<code>fox import --from mysql://root:pw@host:3306/db --as m8</code>
HOW TO TEST	Manual: <code>mysql:8</code> with <code>json</code> + decimal columns; check <code>information_schema.columns</code> .
STATUS & LIMITS	Working; no automated test. Only columns, NOT NULL and PK are carried over (no other indexes, FKs or defaults).
ADDED IN	<code>c929eb5</code> .

I4 From MongoDB

Working

WHAT IT DOES	Turns each collection into a typed Postgres table with the exact collection name.
HOW IT WORKS	<code>loadMongo</code> : list collections with <code>mongosh</code> , export each as NDJSON (ObjectId → hex, Date → ISO), then load via the JSON path (I6).
HOW TO USE	<code>fox import --from mongodb://host/db --as events</code>
HOW TO TEST	Integration §9, §10. Manual: <code>mongo:7</code> with a nested document; <code>\d</code> shows jsonb for objects, numeric for numbers.
STATUS & LIMITS	Working (the only import path with end-to-end tests).
ADDED IN	<code>b842a28</code> ; relational tables <code>f2c964e</code> .

I5 From a .sql dump

Working

WHAT IT DOES	Runs a SQL dump into a new branch.
HOW IT WORKS	<code>loadSQL</code> pipes the file into <code>psql</code> in the container; branch named <code>import-<file></code> . The whole dump runs; <code>psql</code> 's <code>ERROR</code> / <code>FATAL</code> lines are collected (<code>psqlErrorCollector</code>) and, if there were any, the import fails with the count and the first 20 — the new instance is kept with everything that did load.
HOW TO USE	<code>fox import --from ./dump.sql ; POST /api/import/file ; web Import.</code>
HOW TO TEST	Unit: <code>TestPsqlErrorCollector</code> . Integration §11: a dump with one failing statement exits 1, lists it, and the rest loaded; a clean dump exits 0.
STATUS & LIMITS	Working. Any failed statement fails the import, including harmless ones (e.g. a setting only a newer Postgres knows) — they are listed so you can decide. Until 15 Sep 2026 failing statements were ignored and the import reported success.
ADDED IN	<code>44cc05c</code> .

I6 From JSON / NDJSON

Working

WHAT IT DOES	Loads a JSON array or NDJSON into a table with columns inferred from top-level keys.
HOW IT WORKS	<code>loadJSON</code> stages documents as jsonb, then <code>relationalizeJSON</code> picks a type per key (<code>_id</code> text; string→text; number→numeric; boolean; nested/mixed→jsonb), column order <code>_id</code> then alphabetical. <code>streamJSONDocs</code> reads the array or lines; an invalid document stops the load, rolls back everything from that file, and fails the import naming the array element or line number.
HOW TO USE	<code>fox import --from ./people.ndjson</code>
HOW TO TEST	Unit: <code>TestJSONColumn</code> , <code>TestSQLIdent</code> , <code>TestStreamJSONDocs</code> . Integration §10; §11 (an array with an invalid 2nd element exits 1 naming it; a valid array imports).
STATUS & LIMITS	Working. Until 15 Sep 2026 a JSON array import stopped silently at the first element it couldn't decode and reported success.
ADDED IN	<code>44cc05c</code> ; relationalization <code>f2c964e</code> .

I7 From CSV

Working

WHAT IT DOES	Loads a CSV into a table (one text column per header).
HOW IT WORKS	<code>loadCSV</code> : header → <code>CREATE TABLE</code> , rows → <code>\copy ... FORMAT csv</code> .
HOW TO USE	<code>fox import --from ./people.csv</code>
HOW TO TEST	Manual: <code>printf 'id,Name\n1,a\n' > p.csv</code> , import, <code>SELECT "Name" FROM p</code> .
STATUS & LIMITS	All columns are text; a header containing a newline won't parse.
ADDED IN	<code>44cc05c</code> .

I8 Continuous replication & cutover

Working

WHAT IT DOES	Copies a Postgres source's schema and keeps streaming changes for a zero-downtime switch; cutover stops replication.
HOW IT WORKS	<code>ImportContinuousTo</code> : schema-only dump, <code>CREATE PUBLICATION bb_pub</code> on source (best effort), <code>CREATE SUBSCRIPTION bb_sub</code> on target. <code>ImportCutover / ImportCutoverTo</code> drop the subscription. <code>replication.go</code> reads the progress back — <code>pg_subscription</code> , <code>pg_subscription_rel</code> (<code>srsbstate='r'</code> = initial copy done) and <code>pg_stat_subscription</code> — as <code>ReplicationStatus</code> for one branch and <code>Replicating</code> for all running ones; <code>CutoverReplication</code> refuses a branch with no subscription (<code>ErrNotReplicating</code>) so an API cutover can't silently succeed against the wrong branch. The Gateway's reaper leaves a replicating branch running (<code>canSuspend</code>).
HOW TO USE	CLI: <code>fox import --from postgres://... --continuous --as live</code> then <code>fox import-cutover live</code> . Web: tick Continuous replication on the Import page, then watch the Continuous imports panel (tables copied, last message from the source) and press Cut over . REST: <code>GET /api/replication</code> , <code>GET /api/branches/{name}/replication</code> , <code>POST .../replication/cutover</code> .
HOW TO TEST	Integration §5b: a real source with <code>wal_level=logical</code> , initial copy 1 of 1 tables, a source insert streaming across, the branch surviving a short-idle reaper while a plain branch is suspended, cutover returning <code>standalone</code> , later source changes not arriving, and a second cutover (and a cutover of a branch that never replicated) refused 409. Unit: <code>TestParseReplication</code> , <code>TestCanSuspend</code> .
STATUS & LIMITS	Working, and verified in the browser end to end (import, panel, streamed row, cutover). Publication and subscription names are fixed, so one source per branch. Progress is what Postgres reports: table counts and the last message time, not a byte-level lag.
ADDED IN	<code>b842a28</code> ; status, cutover API and panel <code>feat/blackbox-page-gaps</code> .

I9 Guard handling during import

Working

WHAT IT DOES	Lets bulk loads run DDL the guardrail would block, while the ledger keeps recording — and can't leave a branch unguarded after a crash.
HOW IT WORKS	<code>setGuard</code> disables/enables <code>bb_guard_start</code> around the load (deferred); loaders set <code>bb.allow_destructive=on</code> ; <code>reconcileGuard</code> re-enables the guard whenever <code>EnsureRunning</code> restarts a branch.
HOW TO USE	Automatic.
HOW TO TEST	Manual: stop the engine mid-import, <code>fox branch resume <target></code> , <code>SELECT evtenabled FROM pg_event_trigger WHERE evtname='bb_guard_start' → 0</code> .
STATUS & LIMITS	Working.
ADDED IN	<code>b842a28</code> ; crash-safe <code>1884445</code> (PR #12).

I10 Live progress (CLI & SSE)

Working

WHAT IT DOES	Shows a live migration log and, where the source supports it, a progress bar.
HOW IT WORKS	<code>Progress{Log, Step}</code> ; the control plane streams Server-Sent Events <code>log</code> , <code>progress</code> , <code>done</code> , <code>error</code> . Step events from MySQL 8 (per table), MongoDB (per collection), pipeline models.
HOW TO USE	<code>curl -N -k -H "Authorization: Bearer \$KEY" -H 'Content-Type: application/json' -d '{"source":"mongodb://mongo-src/shop"}' https://localhost:8080/api/import</code>
HOW TO TEST	Manual curl above; watch <code>event: log / event: progress</code> .
STATUS & LIMITS	psql subprocess output goes to engine logs, not SSE.
ADDED IN	<code>f2c964e</code> .

I11 Import REST endpoints

Working

WHAT IT DOES	API entry points for imports (auth required).
HOW IT WORKS	<code>POST /api/import</code> body {source, target, continuous} — connection strings only (postgres, mysql, mariadb, mongodb). <code>POST /api/import/file</code> multipart <code>file</code> + <code>target</code> , kind from extension. Both stream SSE.
HOW TO USE	<code>curl -N -k -H "Authorization: Bearer \$KEY" -F file=@people.csv -F target=ppl https://localhost:8080/api/import/file</code>
HOW TO TEST	Manual curl; without a key → 401.
STATUS & LIMITS	Working. Target name not validated against the branch-name rule; OpenAPI documents the wrong field names (K2).
ADDED IN	44cc05c , b842a28 .

I12 Web Import page

Working

WHAT IT DOES	"Migrate a database" form with a live migration console.
HOW IT WORKS	<code>web/src/pages/Import.tsx</code> : connection-string panel (continuous checkbox for postgres), file upload (<code>.sql/.csv/.json/.ndjson/.jsonl</code>), SSE console with progress, result with links to Console and Ledger.
HOW TO USE	<code>https://localhost:8080/import</code>
HOW TO TEST	Manual: upload a CSV and check the result panel.
STATUS & LIMITS	Footer mentions SQLite imports via the CLI, but no SQLite loader exists.
ADDED IN	44cc05c ; console f2c964e (PR #3).

I13 Exact identifier fidelity

Working

WHAT IT DOES	Source names like <code>leaseAiChats</code> or <code>userId</code> keep their exact case.
HOW IT WORKS	<code>sqlIdent</code> quotes and truncates to 63 bytes (tables, columns); <code>sanitizeIdent</code> only for branch names.
HOW TO USE	Automatic; quote camelCase in SQL: <code>SELECT "userId" ...</code>
HOW TO TEST	Unit: <code>TestSQLIdent</code> . Integration §10.
STATUS & LIMITS	Working.
ADDED IN	f2c964e .

ETL pipelines (experimental)

J1 Pipeline engine

Working

WHAT IT DOES	dbt-style extract → land raw → transform with SQL models → quality tests, into a fresh branch.
HOW IT WORKS	<code>RunPipeline (pipeline.go)</code> : delete + re-create the target branch, import the source, rename <code>public</code> → <code>raw</code> , run each model in order as <code>CREATE TABLE VIEW public."name" AS ... with {{ ref('m') }} / {{ source('t') }}</code> templating, then tests.
HOW TO USE	<code>fox pipeline run spec.json [--as name]</code> (default <code>pl-<file></code> ; exit 1 on test failure).
HOW TO TEST	Unit: <code>TestRenderSQL</code> , <code>TestTestQuery</code> . Integration §9.
STATUS & LIMITS	No dependency graph (listed order), no incremental models, no scheduling. A run destroys any existing branch with the target name.
ADDED IN	<code>f2c964e</code> .

J2 Data-quality tests

Working

WHAT IT DOES	Assertions after transform: <code>not_null</code> , <code>unique</code> , <code>accepted_values</code> , <code>row_count_min</code> , <code>custom</code> .
HOW IT WORKS	<code>testQuery</code> builds a "count violating rows" query; a test passes when the count is 0. Failures mark the run failed but keep the data.
HOW TO USE	<code>"tests": [{"type":"unique","model":"stg","column":"city"}]</code>
HOW TO TEST	Unit: <code>TestTestQuery</code> . Integration §9 (failing <code>row_count_min</code> → exit 1, data kept).
STATUS & LIMITS	Working.
ADDED IN	<code>f2c964e</code> .

J3 Stored pipelines & run history

Working

WHAT IT DOES	Save pipelines per user, run them over the API, keep run history.
HOW IT WORKS	SQLite tables <code>pipelines</code> , <code>pipeline_runs</code> ; routes <code>GET/POST /api/pipelines</code> , <code>GET/PUT/DELETE /api/pipelines/{id}</code> , <code>GET .../runs</code> (last 50), <code>POST .../run</code> (SSE).
HOW TO USE	<code>curl -k -H "Authorization: Bearer \$KEY" -d '{"name":"shop etl","spec":{"...}}' https://localhost:8080/api/pipelines</code>
HOW TO TEST	Manual: create, run, <code>GET .../runs</code> shows <code>success</code> .
STATUS & LIMITS	No endpoint to read a run's log; a run whose process dies stays <code>running</code> .
ADDED IN	<code>f2c964e</code> .

J4 Pipeline editor (web)

Working

WHAT IT DOES	Create, edit, run and inspect pipelines in the browser.
HOW IT WORKS	<code>Pipelines.tsx</code> (list) and <code>PipelineEditor.tsx</code> (source, model cards, test cards, Load example, Save, Run with streaming console, run history).
HOW TO USE	<code>https://localhost:8080/pipelines</code> → New pipeline → Load example → Save → Run.
HOW TO TEST	Manual with the integration §9 Mongo data.
STATUS & LIMITS	Delete on the list has no confirmation.
ADDED IN	<code>f2c964e</code> .

APIs, SDKs & agents

K1 Control-plane REST API

Working

WHAT IT DOES

HOW IT WORKS

internal/controlplane/server.go ; store.Authn on /api/* ; CORS for configured/localhost origins.

Routes	Purpose
POST /auth/register login logout , GET /auth/me providers , GET /auth/oauth/{p}/{callback}	Accounts (public)
GET /api/openapi.yaml	Spec (public)
GET /api/status	Health, branches, agents, HA, storage
GET POST /api/branches , DELETE /api/branches/{n} , POST .../suspend resume	Branches
POST /api/branches/{n}/query	SQL as db_client , 10s, ≤1000 rows, attributed to the user
GET /api/branches/{n}/ledger , .../ledger/verify	Blackbox
GET .../ledger/integrity , POST .../ledger/checkpoint , GET .../ledger/export	Ledger integrity (H10, H12)
GET /api/branches/{n}/ledger/entries , POST .../ledger/{id}/branch	Entries with ids; branch from before a ledger entry (H13)
GET /api/branches/{n}/ledger/sessions	Agent sessions (H16)
POST /api/branches/{n}/impact , GET /api/ledger/diff , GET /api/blackbox/diff	Impact analysis and diff (H17, H18)
GET POST /api/branches/{n}/policies , PUT DELETE .../policies/{rule} , POST .../policies/check , GET .../policies/evaluations	Policy gate; changes need db_admin (H15)
/api/branches/{n}/blackbox[/...]	Every .../ledger[/...] route under its Blackbox name (H14)
POST /api/import , /api/import/file	Import (SSE)
/api/pipelines...	Pipelines
GET POST /api/keys , DELETE /api/keys/{id}	API keys

HOW TO USE

HOW TO TEST

STATUS & LIMITS

ADDED IN

curl -k -H "Authorization: Bearer \$KEY" https://localhost:8080/api/status

Unit: TestNameValidation. Integration §1; integration-v2 §0 (ledger/verify columns).

Working. Not versioned (/api/ , not /api/v1/).

a7580f6 ; TLS 0bd52f3 .

K2 OpenAPI spec

Working

WHAT IT DOES	Machine-readable API description for client generators.
HOW IT WORKS	<code>openapi.yaml</code> (OpenAPI 3.1) embedded and served at <code>/api/openapi.yaml</code> ; also documents agent routes.
HOW TO USE	<code>curl -k https://localhost:8080/api/openapi.yaml</code>
HOW TO TEST	<code>go test ./internal/controlplane:TestOpenAPIMatchesRoutes</code> compares the spec's paths with the routes the code registers (it reads the <code>HandleFunc</code> literals, because <code>net/http.ServeMux</code> cannot list what was registered), and <code>TestOpenAPIAliasRefsResolve</code> checks every alias pointer. Manual: <code>curl -k https://localhost:8080/api/openapi.yaml</code> .
STATUS & LIMITS	Matches the code: <code>/api/import</code> takes <code>{source, target, continuous}</code> and answers with an SSE stream, <code>/api/import/file</code> documents its multipart fields, branch and API-key creation answer 201, the ledger's <code>since / until</code> filters are described, and <code>/api/openapi.yaml</code> plus the seven <code>/auth/*</code> routes are documented. The <code>/blackbox</code> aliases are declared as path-item <code>\$ref s</code> , so a generated client gets both spellings with no second copy to keep in step. Adding a route without documenting it now fails the build. Not checked automatically: whether a described body or response still matches its handler.
ADDED IN	52b1fc5 / cc5cbd7 (PR #15); corrected on <code>docs/api-accuracy</code> .

K3 Python SDK `foxbyte`

Working

WHAT IT DOES	Dependency-free client for the control plane (Apache-2.0).
HOW IT WORKS	<code>clients/python/foxbyte: FoxByte(api_key, base_url, verify_tls); status, branches, create_branch, delete_branch, suspend, resume, query, ledger, verify_ledger</code> . Blackbox names <code>blackbox()</code> and <code>verify_blackbox()</code> call the same routes as <code>ledger()</code> and <code>verify_ledger()</code> (H14).
HOW TO USE	<code>pip install ./clients/python; db = FoxByte(api_key="key_...", verify_tls=False); db.query("qa","select 1")</code>
HOW TO TEST	No tests; manual snippet.
STATUS & LIMITS	Not on PyPI; no import/pipeline/key/agent methods.
ADDED IN	52b1fc5.

K4 TypeScript SDK `@foxbyte/client`

Working

WHAT IT DOES	The same client for Node and browsers (Apache-2.0).
HOW IT WORKS	<code>clients/typescript/src/index.ts: new FoxByte(apiKey, baseUrl); same methods (createBranch, verifyLedger,...) using fetch</code> . Blackbox names <code>blackbox()</code> and <code>verifyBlackbox()</code> call the same routes as <code>ledger()</code> and <code>verifyLedger()</code> (H14).
HOW TO USE	<code>cd clients/typescript && npm install && npm run build; local self-signed cert needs NODE_TLS_REJECT_UNAUTHORIZED=0</code> .
HOW TO TEST	No tests; manual Node script.
STATUS & LIMITS	Not published to npm.
ADDED IN	52b1fc5.

K5 Agent Branch API

Working

WHAT IT DOES	A disposable database per AI agent over HTTPS on <code>:8088</code> .
HOW IT WORKS	<code>internal/agentapi</code> : GET <code>/healthz</code> (public), GET <code>/agents</code> , POST <code>/agents/{id}/branch</code> (creates agent- <code><id></code> , returns DSN), DELETE <code>/agents/{id}/branch</code> . Cap <code>FOX_AGENT_MAX</code> (50); TTL reaping with <code>FOX_AGENT_TTL</code> . Optional body <code>{"task_id", "parent_session_id", "session_id"}</code> records Blackbox provenance for the agent's changes and adds <code>session_id</code> to the response; without it nothing changes (H16).
HOW TO USE	<code>curl -k -H "Authorization: Bearer \$KEY" -X POST https://localhost:8088/agents/alice/branch</code>
HOW TO TEST	Integration §5 (usable via its DSN; the key dies with the branch); integration-v2 §1c (own non-superuser role, can't disable triggers or override, gateway-routed DSN, key refused for another branch and by both HTTP APIs). Unit: <code>TestAgentRolePassword</code> , <code>TestAgentGatewayDSN</code> , <code>TestGatewayHostPort</code> , <code>TestScopeAllows</code> , <code>TestScopedKeys</code> , <code>TestScopeColumnAddedToExistingStore</code> .
STATUS & LIMITS	The DSN is routed through the Gateway, so an agent running on the host can open it — the branch's Postgres still publishes no host port, because a direct connection would bypass the Gateway, its key, TLS and the attribution. Its password is an API key scoped to that one branch (<code>api_keys.scope</code>): the control plane and the Agent API refuse it, and the Gateway opens only the named branch with it, so a leaked agent credential reaches neither another agent's data nor the account's branches. The Gateway logs the agent in as the branch's own agent- <code><id></code> role, whose password is derived from the install secret rather than stored, so <code>session_user</code> stays the agent and attribution remains non-forgeable. The key is shown once and revoked when the branch is deleted (a reset keeps it). The cap, the listing and the TTL reaper now count suspended branches too. <code>FOX_AGENT_SUPERUSER=1</code> still restores the old container-IP superuser DSN. Agent branches created before this change have no agent role (recreate them), and <code>FOX_GATEWAY_HOSTPORT</code> is needed if the Gateway is not on <code>localhost:6432</code> .
ADDED IN	<code>a7580f6</code> ; cap/TTL <code>e720c4e</code> (PR #12); reachability on <code>fix/agent-reachability</code> .

K6 MCP server

Working

WHAT IT DOES	Lets agent frameworks manage agent branches, run SQL and read/verify the ledger via the Model Context Protocol.
HOW IT WORKS	<code>internal/mcp</code> : JSON-RPC 2.0 over stdio (<code>initialize</code> , <code>tools/list</code> , <code>tools/call</code>). Tools: <code>create_branch</code> , <code>list_branches</code> , <code>delete_branch</code> , <code>run_sql</code> , <code>changes</code> , <code>verify_ledger</code> , <code>ledger_integrity</code> , <code>ledger_entries</code> , <code>branch_before_change</code> (H13). Blackbox names <code>verify_blackbox</code> , <code>blackbox_integrity</code> and <code>blackbox_entries</code> are listed too and run the same tools (H14). <code>policy_check</code> previews policy rules for a statement (H15). <code>execute_change</code> runs SQL with agent provenance and a policy preview and returns structured JSON (H16). <code>impact</code> and <code>blackbox_diff</code> expose impact analysis and branch diff (H17, H18).
HOW TO USE	Configure an MCP client to launch command <code>fox</code> with args <code>["mcp"]</code> and an <code>env</code> block carrying <code>FOX_API_KEY</code> (<code>--key <key_></code> also works, but a key on the command line is visible in the process list). Make one with <code>fox apikey create you@example.com mcp</code> .
HOW TO TEST	Unit: <code>TestAuthenticateWithoutAKey</code> , <code>TestApplyScope</code> . integration-v2 §1c (<code>run_sql</code> role, <code>list_branches</code> without passwords, compat switch; no key exits 1 saying how to make one; an invalid key is refused; <code>run_sql</code> records the key's account as the actor; a scoped key's call lands in its own branch and not main, another branch is refused, and the tools that reach further are refused), §3 (<code>ledger_integrity</code>), §4 (<code>ledger_entries</code> lists the entry id; <code>branch_before_change</code> listed, refuses a BLOCKED entry, restores by timestamp). Manual: pipe <code>initialize</code> , <code>tools/list</code> , <code>tools/call</code> <code>run_sql</code> JSON lines into <code>fox mcp</code> .
STATUS & LIMITS	The server requires an API key and acts as that account: <code>run_sql</code> runs as the non-superuser <code>db_client</code> role but is recorded with the key's account as the actor (<code>ClientQueryTextAs</code>), and a key scoped to one branch pins every call to it, refuses another branch by name, and refuses <code>create_branch</code> , <code>delete_branch</code> , <code>list_branches</code> , <code>blackbox_diff</code> and <code>branch_before_change</code> outright. There is no opt-out: without a valid key the server prints how to make one and exits. <code>FOX_MCP_SUPERUSER=1</code> still restores the superuser role for compatibility. Beyond the key there is no per-call authorization or rate limiting, and anyone who can read the client config can read the key — it is still meant for a local agent runtime. <code>list_branches</code> no longer shows passwords. Client setup and the full 16-tool reference are in <code>docs/mcp.md</code> , linked from the README and the Docs page. <code>create_branch</code> returns the same Gateway-routed, branch-scoped DSN as the Agent API (K5), so an agent driven over MCP can reach its database from the host.
ADDED IN	<code>d62c676</code> / <code>ed5a379</code> (PR #14); authentication <code>feat/attribution-mcp-auth</code> .

Web console

L1 Embedded single-origin UI

Working

WHAT IT DOES	The engine serves the React app itself at <code>https://localhost:8080</code> — no separate web server.
HOW IT WORKS	<code>web/embed.go</code> (<code>//go:build embedui</code>) embeds <code>web/dist</code> ; <code>serveUI</code> falls back to <code>index.html</code> . <code>make web-build</code> builds with same-origin API URLs; Linux release binaries use <code>-tags embedui</code> . <code>make web-dev</code> is deprecated (hot-reload only).
HOW TO USE	<code>fox start</code> → open <code>https://localhost:8080</code> (accept the self-signed cert).
HOW TO TEST	Manual: open <code>/dashboard</code> directly and refresh.
STATUS & LIMITS	Working. <code>make web-dev</code> defaults to <code>http://localhost:8080</code> and needs <code>VITE_API_URL=https://localhost:8080</code> .
ADDED IN	4927095 ; React app 84741df .

L2 Login & signup

Working

WHAT IT DOES	Log in / sign up tabs, OAuth buttons when configured, redirect to Dashboard.
HOW IT WORKS	<code>/login</code> ; <code>RequireAuth</code> guards private routes; API client redirects to <code>/login</code> on 401.
HOW TO USE	<code>https://localhost:8080/login</code>
HOW TO TEST	Manual sign-up and logout.
STATUS & LIMITS	Working (OAuth caveat G5).
ADDED IN	235ff41 .

L3 Dashboard

Working

WHAT IT DOES	Live tiles (primary, branches, agent DBs, gateway, replica, storage), a branch table with create (from any parent), suspend, resume, delete and a copyable connection string, and a Base backups panel: how far back this install can go, with a copy-ready restore command per backup.
HOW IT WORKS	Polls <code>/api/status</code> and <code>/api/branches</code> every 3s; offline banner when the API is unreachable. The backups panel reads <code>/api/backups</code> once when the page opens (backups change when someone takes one, not every three seconds) and shows the newest 6, with the rest behind "Show N older backups".
HOW TO USE	<code>https://localhost:8080/dashboard</code>
HOW TO TEST	Manual: create a branch and see it appear; read the backups panel against <code>fox backup list</code> . Checked in the browser against a real install's 61 backups.
STATUS & LIMITS	Working. The panel is read-only: taking a backup and restoring stay CLI commands (a restore needs a host port and leaves a container to query).
ADDED IN	84741df ; backups panel feat/durability-gaps .

L4 SQL console & browser

Working

WHAT IT DOES	Schema sidebar, row browser (100/page), structure and index tabs, SQL editor (Cmd/Ctrl+Enter), row-as-JSON. FoxByte's own tables (the <code>bb</code> schema: Blackbox, policies, agent sessions) are hidden whenever the console opens; a Show system tables button at the end of the list shows or hides them, and the filter never reveals them on its own. An Allow destructive changes switch lets an admin run one blocked change.
HOW IT WORKS	<code>Console.tsx</code> → <code>POST /api/branches/{name}/query</code> , which logs in as the signed-in user's own role (a member of <code>db_client</code> acting as it, as the Gateway does), so the guardrail sees who is asking. <code>allow_destructive</code> applies <code>bb.allow_destructive=on</code> to that run's session only, and the switch turns itself off afterwards. <code>GET /api/branches/{name}/admins</code> tells the page whether the user may override. Reload refreshes the list and the open table and shows errors instead of an empty schema; the list also refreshes after DDL. Each value is rendered by <code>cell</code> : scalars, timestamps, jsonb and arrays of scalars as they were, and for the types <code>pgx</code> decodes into a byte array or a struct — <code>uuid</code> , <code>bytea</code> , <code>interval</code> , <code>time of day</code> , <code>inet</code> , <code>macaddr</code> , the geometric types, <code>ranges</code> , <code>hstore</code> — Postgres's own text form, asked of <code>pgx</code> 's type map (<code>Map.Encode</code> with <code>TextFormatCode</code>), which is what <code>psql</code> would print. Array members are rendered by their own element type, so a <code>uuid[]</code> is a list of uuids.
LAYOUT	Redesigned 21 Sep 2026 (L11): a page header with the branch picker, then three cards — the schema rail, the query card (title bar, editor, toolbar with the destructive switch and Run query), and the result (a status bar with the command tag, row and column counts above the grid). The editor follows the theme rather than being dark in both. A browsed table keeps its own header, tabs and a pager aligned to the right of the row count.
HOW TO USE	<code>https://localhost:8080/console</code>
HOW TO TEST	Unit: <code>TestCellRendersPostgresTypes</code> (<code>uuid</code> , <code>uuid[]</code> , <code>bytea</code> , <code>interval</code> , <code>time of day</code> , <code>inet</code> , <code>macaddr</code> , and the types that must not change), <code>TestCellTimestamp</code> , <code>TestCellUnknownType</code> . integration-v2 §6: the value rendering above over the REST API; the console runs as the signed-in user; a non-admin's override is refused; an admin is blocked without the switch and allowed with it. Run live on 17 Sep 2026 against a spare control plane on a throwaway branch: <code>session_user</code> was the account, <code>DROP TABLE</code> was blocked without the switch and went through with it. Pages type-checked and built; not reviewed in a browser after this change.
STATUS & LIMITS	Working. 10s timeout, 1000-row cap. Each run is its own connection, so a <code>SET</code> typed in one run does not reach the next — the switch is the way to override. Only the <code>bb</code> schema is treated as system: tables in <code>public</code> left by test runs still show as ordinary tables. Fixed 17 Sep 2026 (found testing Blackbox precision in the browser): for a Blackbox policy block (BBX01) the Allow & run again button sent <code>allow_destructive</code> , which only overrides the guardrail, so an admin got the same error back; the panel now names the rule and its button sends <code>allow_rules</code> (<code>bb.policy_allow</code> for that rule), and commands with no result rows no longer print their success line twice. Fixed 18 Sep 2026 (reported from the UI): a <code>uuid</code> column — every <code>id</code> in the user's own tables — was displayed as a list of 16 numbers, because <code>pgx</code> decodes <code>uuid</code> to <code>[16]byte</code> and that is what JSON makes of it; the same pass fixed intervals showing as <code>{"Microseconds":...}</code> , <code>inets</code> carrying their quotes and <code>bytea</code> arriving as mangled text. Only a real <code>bytea</code> column is hex-encoded: <code>pgx</code> also hands back <code>[]byte</code> for a type it has no codec for, such as a custom enum, and those stay readable.
ADDED IN	<code>84741df</code> ; override, system group and reload on <code>fix/console-permissions</code> ; value rendering on <code>fix/console-uuid-cells</code> .

L5 Blackbox page

Working

WHAT IT DOES	Browse, filter and verify the Blackbox per branch, and read the statement behind any entry.
HOW IT WORKS	<code>Ledger.tsx</code> : branch, status (APPLIED/FLAGGED/BLOCKED), kind (human/agent), actor, table, risk (drop, drop-column, type-change, security-definer, policy) and since/until date filters — every filter the API has — with "Clear filters"; infinite scroll 25 rows at a time; columns Time, Actor, Tool, Change, Status. A row expands to its session, branch and full statement (the list is fetched <code>with=session</code>). Verify calls <code>.../ledger/verify</code> and reports the chain intact or the first broken entry, linking to Integrity. A request sequence number (<code>seqRef</code>) means a filter change always reloads and a stale response is dropped. Nav label and page title "Blackbox"; reachable at <code>/blackbox</code> and, as before, <code>/ledger</code> (H14).
HOW TO USE	<code>https://localhost:8080/blackbox</code>
HOW TO TEST	Integration-v2 §5b (the <code>with=session</code> column set, default columns unchanged, table and table+risk filters, and a date range). Manual: in the browser, filter by table and risk, set a To date, expand a row, and press Verify — checked against the same counts from SQL.
STATUS & LIMITS	Working. Dates are whole days in the browser's locale, sent as <code>since / until</code> (the To date covers the day to 23:59:59). Filtering is server-side; the visible rows are whatever the filters return.
ADDED IN	<code>e42e792</code> ; filters, verify and row detail <code>feat/blackbox-page-gaps</code> .

L6 API keys page

Working

WHAT IT DOES	Create (shown once, copy button), list and revoke API keys. Since E3 this is where an install's first key comes from, so a new key is shown twice: on its own, and inside the connection string it belongs in.
HOW IT WORKS	<code>/keys</code> → <code>/api/keys</code> . The connection string is assembled in the page from the host the console was opened on and the gateway's port, so it is correct for a remote install as well as localhost.
HOW TO USE	<code>https://localhost:8080/keys</code>
HOW TO TEST	Manual create + revoke.
STATUS & LIMITS	Working.
ADDED IN	<code>235ff41</code> .

L7 Landing, Guide & Docs pages

Working

WHAT IT DOES	Public landing page, a developer guide (install, connect, CRUD, branches, agents, PDF download) and a docs page (CLI, REST, env tables with live status buttons).
HOW IT WORKS	<code>Landing.tsx</code> , <code>Guide.tsx</code> , <code>Docs.tsx</code> .
HOW TO USE	<code>https://localhost:8080/</code> , <code>/guide</code> , <code>/docs</code>
HOW TO TEST	Type-check and build (<code>npm --prefix web run build</code>); read line by line against the code, 16 Sep 2026.
STATUS & LIMITS	Refreshed: the Guide drops <code>make web-dev</code> for <code>https://localhost:8080</code> , every connection string carries <code>sslmode=require</code> , and the agent-API examples use <code>https://localhost:8088</code> with a note on <code>-k</code> . The Docs page lists the whole REST surface (Blackbox and its aliases, policies, impact, diff, import, pipelines, keys, auth, agents), points at <code>GET /api/openapi.yaml</code> , and gains an "Agents over MCP" section linking <code>docs/mcp.md</code> . The <code>FOX_WEB_ORIGIN</code> default it shows (<code>http://localhost:5173</code>) was checked against <code>internal/auth/auth.go</code> and is correct. The landing page is unchanged.
ADDED IN	<code>84741df</code> , <code>e9210fb</code> ; refreshed on <code>docs/api-accuracy</code> .

L8 Blackbox integrity page

Working

WHAT IT DOES	Verify a branch against its anchors, create a checkpoint, download the ledger export, and branch main from just before a ledger entry — from the browser.
HOW IT WORKS	<code>web/src/pages/Integrity.tsx</code> (route <code>/integrity</code> , nav "Integrity") calls the integrity, checkpoint and export endpoints and shows INTACT / TAMPERED, entry and checkpoint counts, problems and notes. The "Branch from before a change" panel lists main's 20 newest entries (<code>GET .../ledger/entries</code>) with a Select button (disabled for BLOCKED), takes an entry id and optional name and calls <code>POST .../ledger/{id}/branch</code> (H13), showing progress while the restore runs.
HOW TO USE	<code>https://localhost:8080/integrity</code>
HOW TO TEST	TypeScript type-check and production build pass. Reviewed signed in on 16 Sep 2026 (Chrome, macOS): on a test branch the report matched <code>fox blackbox integrity</code> field for field (INTACT, 61 entries, 61 hash-chained, 2 checkpoints, 2 anchored, 59 unanchored, same notes); Create checkpoint anchored a new entry (checkpoints 1 → 2, confirmed on disk) and was correctly a no-op when nothing new existed; Export (JSONL) answered 200 <code>application/x-ndjson</code> with one JSON line per entry; Branch from before a change restored <code>main-before-3649</code> in 4 s from the right base backup — the branch lacked the table that entry created, its Blackbox stopped at the previous entry, and main was untouched. No automated UI test.
STATUS & LIMITS	Working — exercised in a browser against a running stack (16 Sep 2026): verify, checkpoint, export and branch-before all matched the CLI. Note: the entry list is loaded once, so a change recorded after the page opened appears only after a refresh.
ADDED IN	Blackbox 2.0 Phase 3 (<code>feat/ledger2-phase3-checkpoints</code>); branch-before panel in Phase 4 (<code>feat/ledger2-phase4-branch-before</code>).

L9 Policies page

Working

WHAT IT DOES	Shows a branch's policy rules, lets a <code>db_admin</code> switch each between warn and block, enable or disable it, and add or remove custom rules; previews which rules a statement would trigger; lists recent warnings, blocks and overrides. Who can override lists a branch's <code>db_admin</code> members and lets an admin grant or revoke by email.
HOW IT WORKS	<code>web/src/pages/Policies.tsx</code> (route <code>/policies</code> , nav "Policies") over the <code>/api/branches/{n}/policies...</code> endpoints (H15). A non-admin sees the server's 403 message on any change.
HOW TO USE	<code>https://localhost:8080/policies</code>
HOW TO TEST	TypeScript type-check and production build pass; the endpoints it calls are covered by integration-v2 §6. Reviewed signed in on 16 Sep 2026 on a test branch: the four built-in rules matched <code>fox policy list</code> ; <code>warn</code> → <code>block</code> → <code>warn</code> , <code>disable</code> → <code>enable</code> , and <code>add</code> → <code>remove</code> of a custom rule each matched the CLI afterwards and never touched <code>main</code> ; Check a statement returned the same verdict as <code>fox policy check</code> (<code>ALTER TABLE</code> , <code>WARN alter-column-type</code> , same reason); on <code>main</code> , where the account holds no <code>db_admin</code> , a change was refused with a readable banner naming the grant command, and the rule stayed <code>warn</code> . No automated UI test.
STATUS & LIMITS	Working — exercised in a browser against a running stack (16 Sep 2026): rule changes, the statement check and the non-admin refusal all matched the CLI, and writes reached only the branch selected. Minor: the new-rule form pre-fills Command with <code>ALTER TABLE</code> as a real value, which reads like a placeholder and is easy to type after by mistake.
ADDED IN	Blackbox 2.0 Phase 5 (<code>feat/blackbox-phase5-policy</code>).

L10 App sidebar layout

Partial

WHAT IT DOES	Gives the signed-in app a left sidebar with its pages grouped into sections, so adding pages doesn't overflow a single navbar; keeps the public navbar for the marketing pages.
HOW IT WORKS	<code>web/src/components/AppLayout.tsx</code> : <code>APP_NAV</code> lists the sections and pages (a new page is one entry), each with an inline SVG icon from <code>components/icons.tsx</code> (no icon library). The active page is highlighted (including the old <code>/ledger</code> address and nested pages such as <code>/pipelines/{id}</code>), and a top bar shows "section / page". The footer has GitHub, the light/dark switch, the account, Log out and Collapse; the collapsed state is stored in <code>localStorage</code> (<code>bb.sidebar.collapsed</code> , ignored if storage is unavailable). Under 860px the sidebar is an off-canvas drawer, opened from the top bar and closed by choosing a page, tapping outside or Escape. <code>web/src/main.tsx</code> routes Home and Log in through <code>PublicLayout</code> , the app pages through <code>SignedInLayout</code> (redirects to Log in when signed out), and Guide/Docs through <code>HelpLayout</code> (sidebar when signed in, public navbar otherwise). The logo and theme toggle moved to <code>components/brand.tsx</code> (still exported from <code>main.tsx</code>). Styles use the existing light/dark design tokens.
HOW TO USE	Sign in at <code>https://localhost:8080</code> — the app opens with the sidebar. Collapse with the button at the bottom; on a phone use the menu button top left.
HOW TO TEST	TypeScript type-check and production build pass. A preview control plane built with <code>-tags embedui</code> served <code>/</code> , <code>/dashboard</code> , <code>/login</code> and <code>/blackbox</code> (200) with the sidebar code in the bundle. Reviewed signed in on 16 Sep 2026 (Chrome, macOS): Dashboard, Console, Import, Pipelines, Changes, Integrity, Policies, API keys, Guide and Docs each opened with the right section highlighted and breadcrumb, <code>/ledger</code> highlighted Changes, collapse survived a reload (<code>app-shell collapsed</code> , <code>bb.sidebar.collapsed=1</code> , applied before first paint), the theme toggle set and kept <code>data-theme</code> , and no console errors appeared on any page. Still manual: narrow the window below 860px and open/close the drawer (menu button, tapping outside, Escape, choosing a page). No automated UI test.
STATUS & LIMITS	Working — reviewed in a browser while signed in (16 Sep 2026): every sidebar link opened its page with the right section highlighted and breadcrumb (including <code>/ledger</code> → Changes), collapse persisted across a reload with no flicker on first paint, and the theme toggle applied and persisted. Not checked: the under-860px drawer — the automation could not change the page viewport, so it needs a manual look. Visual only: routes, addresses, APIs and page contents are unchanged. The breadcrumb shows the section and page name, not deeper context such as the pipeline being edited.
ADDED IN	<code>feat/web-sidebar-layout</code> .

L11 Design system & FoxByte theme

Working

WHAT IT DOES	Gives the whole web console one set of colours, type sizes, spacing steps, radii and motion, and one mark — so a page is laid out from the same vocabulary as every other page, and the product can be reskinned from a single block.
HOW IT WORKS	<code>web/src/styles.css</code> is two layers. The top is tokens: a 1.2 type scale (<code>--t-xs ... --t-3xl</code>), a 4px space scale (<code>--s-1 ... --s-8</code>), <code>--radius-sm/--radius/--radius-lg/--radius-pill</code> , elevation, <code>--speed</code> , a focus <code>--ring</code> , and the palette per theme. Everything below refers to tokens — no raw hexes, no picked pixel values. The palette is the brand: ember <code>#ff7a2f</code> → fox amber <code>#ffc24a</code> on graphite for dark, the same ember darkened to <code>#d2570c</code> on paper for light; <code>--accent-ink</code> is the text colour on an accent fill (dark on orange in the dark theme, white in the light one), which is why the filled buttons are legible in both. <code>--editor-bg</code> lets the SQL editor follow the theme while a terminal transcript stays dark. The mark is <code>Mark()</code> in <code>web/src/components/brand.tsx</code> : a fox head drawn from the two angles the branch graph uses, filled from a gradient whose stops are <code>var(--grad-a) / var(--grad-b)</code> and whose id is per-instance, so two marks on a page do not share one gradient. <code>Wordmark()</code> picks out the last capitalised part of the product name from <code>brand.json</code> rather than from markup. <code>web/public/favicon.svg</code> is the same head on a graphite tile with the two colours written out, because a favicon cannot read CSS variables. <code>docs/brand/</code> holds the README's logo: one page, <code>logo.html</code> , rendered to a light and a dark PNG (<code>docs/brand/README.md</code> has the two commands), which GitHub picks between by the reader's colour scheme.
HOW TO USE	Style a new page with the tokens: <code>var(--s-*)</code> for spacing, <code>var(--t-*)</code> for type, <code>.page-head</code> for the title and its controls, <code>.filterbar</code> for filters, <code>.table-wrap</code> for a table, <code>.empty</code> for nothing-to-show. Reskinning is the token block; renaming is <code>brand.json</code> (see <code>docs/branding.md</code>).
HOW TO TEST	TypeScript check and production build. Driven in a browser against a canned control plane on 21 Sep 2026: Dashboard, Console (query, result, browsed table), Blackbox, Policies, Integrity, API keys (including a freshly minted key), Import, Pipelines, Login and the landing page, each in both themes, and the console at 375px wide.
STATUS & LIMITS	Working. Visual only — no route, API or behaviour change. Not yet reviewed against a running engine with real data (the canned control plane stands in for one), and the emoji icons on the landing page's feature cards are still emoji rather than drawn marks.
ADDED IN	<code>feat/foxybyte-ui</code> ; the palette and mark replace the purple theme the product carried through both earlier names.

Build, CI & releases

M1 Makefile targets

Working

WHAT IT DOES	One place to build, test and release.
HOW IT WORKS	<code>build</code> , <code>vet</code> , <code>fmt</code> , <code>test</code> , <code>vm-build</code> (Lima, embedui), <code>integration</code> , <code>integration-v2</code> and <code>integration-update</code> (in the throwaway <code>fox-test</code> VM; <code>test-vm</code> , <code>test-vm-stop</code> , <code>test-vm-delete</code> manage it), <code>web-build</code> , <code>web-dev</code> (deprecated), <code>release</code> (5 <code>fox</code> + 5 <code>fox-verify</code> binaries + docker context), <code>release-linux</code> , <code>wsl-distro</code> , <code>wsl-zfs</code> , <code>feature-doc</code> (renders this PDF and <code>FOX_Checklist.pdf</code> , after checking both list the same features).
HOW TO USE	<code>make test</code> · <code>make integration</code> · <code>make release VERSION=x.y.z</code> · <code>make feature-doc</code>
HOW TO TEST	CI runs <code>build/vet/test/gofmt</code> . On 17 Sep 2026 all three integration suites ran in a new <code>fox-test</code> VM: <code>integration</code> 53 / 53, <code>integration-v2</code> 227 / 227, <code>integration-update</code> 51 / 51, after two bugs they found were fixed (G1, H16). The guard was checked on the real VM without running a suite: each stops with exit 2 before its first action, and a missing guard stops it too.
STATUS & LIMITS	Working. <code>release</code> prints an outdated ZFS-bundle note. The integration suites are destructive by design (they wipe Blackbox history, restore main to an earlier point, fail HA over, create accounts and keys), so they run only in a VM that <code>scripts/test_vm.sh</code> has marked (<code>/etc/fox-test-instance</code>): <code>scripts/lib/test_guard.sh</code> refuses everywhere else. The test VM never uses the names <code>foxbyte</code> or <code>default</code> (fox would take it for the real install) and forwards no ports to the Mac. It needs about 4 GiB of memory while running; <code>make test-vm-stop</code> frees it.
ADDED IN	<code>a7580f6</code> ; <code>feature-doc</code> added with this document.

M2 Linux CI

Working

WHAT IT DOES	Every push to main and every PR is built and tested on Linux.
HOW IT WORKS	<code>.github/workflows/ci.yml</code> : <code>go build</code> , <code>go vet</code> , <code>go test</code> , <code>gofmt check</code> . Also runs <code>scripts/test_release_checksums.sh</code> , which checks the release checksum merge never drops another job's entries, and <code>scripts/test_check_release_assets.sh</code> , which checks the release job's test that every file <code>fox update</code> needs was published, and <code>scripts/test_feature_checklist.sh</code> , which checks <code>FOX_Checklist.html</code> lists exactly the features in this document.
HOW TO USE	Automatic on PRs.
HOW TO TEST	Open a PR and check the Actions tab.
STATUS & LIMITS	The integration suite does not run in CI (needs a VM).
ADDED IN	<code>797bc14</code> (PR #11).

M3 Windows CI

Working

WHAT IT DOES	Tests the Windows build and installer on every push/PR.
HOW IT WORKS	<code>windows-ci.yml</code> : <code>build/vet/test</code> , cross-compile, Pester (both files), <code>install.ps1</code> encoding + PS 5.1 parse; Linux job checks <code>make release assets</code> .
HOW TO USE	Automatic.
HOW TO TEST	Actions tab.
STATUS & LIMITS	No end-to-end WSL install test (needs nested virtualisation). Fixed 21 Sep 2026: this job is the only one that runs the repository's guard tests on backslash paths and a CRLF checkout, and both failed there while every other job was green — a path exemption that matched nothing, and a generated file that looked like drift because Git had rewritten its line endings. Worth reading a Windows-only failure as a question about the platform before the code.
ADDED IN	<code>4fc8afb</code> (PR #1); path and line-ending fixes on <code>fix/windows-ci</code> .

M4 Releases & GHCR image

Working

WHAT IT DOES	A <code>v*</code> tag publishes binaries, checksums and a multi-arch Postgres image.
HOW IT WORKS	<code>release.yml</code> : <code>make release</code> , verify assets (including <code>fox-verify</code>), create (pre)release, <code>SHA256SUMS</code> , upload; image job builds <code>docker/postgres</code> for amd64+arm64 once per supported PostgreSQL major (a matrix over <code>16</code> and <code>18</code> , passed as <code>PG_MAJOR</code>) and pushes <code>ghcr.io/theofabyte/postgres-walg:<major></code> and <code>:<major>-<tag></code> , labelled with <code>org.opencontainers.image.source</code> so the package links itself to the repository. Checksums go through <code>scripts/release-checksums.sh</code> , which merges into the release's existing <code>SHA256SUMS</code> instead of replacing it, so the distro image's entry from <code>wsl-distro.yml</code> is kept whichever job finishes first. A last step, <code>scripts/check-release-assets.sh</code> , fails the job unless every file <code>fox update</code> needs (A9) is attached and listed in <code>SHA256SUMS</code> ; a Go test keeps its list in step with the updater's.
HOW TO USE	Push a tag <code>vX.Y.Z</code> (or run manually with a tag).
HOW TO TEST	Check the release page assets, and <code>docker pull ghcr.io/theofabyte/postgres-walg:18</code> and <code>:16</code> anonymously — the package must be public, or every install fails with <code>unauthorized</code> (it did on 21 Sep 2026, on <code>v0.1</code>).
STATUS & LIMITS	Working. GHCR package must be made public manually the first time.
ADDED IN	<code>8cafdb6</code> (PR #2); image job <code>aae9e20</code> (PR #12).

M5 WSL distro & ZFS builds

Partial

WHAT IT DOES	Builds a prebuilt WSL distro image; legacy ZFS module bundle for WSL kernels.
HOW IT WORKS	<code>wsl-distro.yml</code> / <code>make wsl-distro</code> (Ubuntu + Docker + btrfs + engine + saved images); <code>wsl-zfs.yml</code> / <code>make wsl-zfs</code> (OpenZFS for the WSL kernel). It creates the release if needed, then publishes its checksum through the same <code>scripts/release-checksums.sh</code> as <code>release.yml</code> .
HOW TO USE	Tag push (distro) or manual dispatch.
HOW TO TEST	Actions artifacts; a fresh Windows install from v0.8.6 imported the image and loaded its preinstalled containers.
STATUS & LIMITS	Confirmed end to end on Windows (16 Sep 2026): the installer downloaded and verified the distro image, and a fresh install imported it — setup skipped the apt work entirely and loaded the preinstalled images. It now does that on every install, and the build tags its Postgres image with the name a fresh install runs (<code>ghcr.io/theofabyte/postgres-walg:18</code> since 21 Sep 2026; a test keeps it in step with <code>PGMajor</code>) so the preload is used. The setup-side "already loaded" probe checked the pre-16-Sep legacy name, which a newer distro never carries, so every re-run of setup loaded the whole tarball again; it now probes the image the preload actually holds. Still open: the image is 685 MB (a fallback to the Ubuntu rootfs covers slow or failed downloads), and the ZFS bundle is legacy (Windows uses btrfs).
ADDED IN	<code>d0c1f26</code> (PR #7); <code>8305e5b</code> (PR #1).

M6 Postgres image

Partial

WHAT IT DOES	Stock PostgreSQL plus wal-g for archiving, backups and PITR. A fresh install runs PostgreSQL 18 ; an install created on 16 keeps running 16.
HOW IT WORKS	<code>docker/postgres/Dockerfile</code> : <code>ARG PG_MAJOR=18</code> , <code>FROM postgres:\${PG_MAJOR}-bookworm</code> , <code>wal-g v3.0.9</code> , <code>restore-entrypoint.sh</code> . One Dockerfile serves every supported major; bookworm is kept for all of them because the wal-g release binary is built against an older glibc. The majors live in one place, <code>internal/branch/images.go</code> (<code>PGMajor</code> for a fresh install, <code>SupportedPGMajors</code> for what fox can still run), and <code>TestPostgresMajorIsInStepEverywhere</code> fails if the Dockerfile default, the release workflow's matrix or the Windows preload disagrees with them. The image follows the data, not the binary (<code>pgImage</code> , <code>internal/branch/pgversion.go</code>): the engine reads <code>PG_VERSION</code> from main's data directory and runs the image for that major, so <code>fox update</code> onto a binary that ships 18 leaves a 16 install on 16 — moving a cluster between majors is a migration, not an update. <code>FOX_PG_IMAGE</code> overrides it. Before every <code>docker run</code> , <code>checkDataMajor</code> compares the branch's <code>PG_VERSION</code> with the image's major and refuses a mismatch with both versions named and the way out, instead of leaving Postgres to fail inside the container with "database files are incompatible with server".
HOW TO USE	Pulled automatically (C8). <code>FOX_PG_IMAGE=<ref></code> for a registry mirror.
HOW TO TEST	Unit: <code>TestResolvePGImageFollowsTheData</code> , <code>TestCheckDataMajor</code> , <code>TestImageMajor</code> , <code>TestParsePGVersion</code> , <code>TestBuildImageArgsPassTheMajor</code> , <code>TestPostgresMajorIsInStepEverywhere</code> (checked to fail when the Dockerfile default is set back to 16). <code>docker run --rm ghcr.io/thefoxbyte/postgres-wal-g:18 wal-g --version</code> . The three suites in the throwaway VM on 18.
STATUS & LIMITS	Partial until the suites have run on 18. Moved to 18 on 21 Sep 2026: the code, the version guard and the drift test are done and unit-tested, but the engine leans on Postgres behaviour no compiler checks — wal-g's <code>backup-push</code> / <code>backup-fetch</code> , the <code>recovery.signal</code> PITR protocol, <code>pg_controldata</code> output parsed by its English label in HA failback, the Blackbox's <code>PG_CONTEXT</code> string matching and event-trigger tag lists — and that is only proven by <code>make integration</code> , <code>integration-v2</code> and <code>integration-update</code> on an 18 cluster. wal-g's own suite covers PostgreSQL 14–18 since Aug 2026, but v3.0.9 predates that job. No command yet moves an existing install from 16 to 18: until one exists the way is <code>fox uninstall</code> and a fresh install. No extensions added (pgvector is not included).
ADDED IN	<code>a7580f6</code> ; GHCR name <code>aae9e20</code> ; PostgreSQL 18 and the per-major images on <code>feat/postgres-18</code> .

Test matrix

Every automated test and the feature it covers. Run: `make test` (unit, host) · `make integration` (Lima VM) · Pester via `windows-ci.yml`.

Test file	Tests	Covers
<code>cmd/fox/main_test.go</code>	<code>TestAddrFlag</code> , <code>TestDurFlag</code> , <code>TestRestoreArg</code> , <code>TestUsageDescribesRevertAndFallback</code>	B1, H7
<code>internal/branch/ha_test.go</code>	<code>TestSuspendRefusal</code> , <code>TestHAGuard</code> , <code>TestPrimaryPointerGuards</code> , <code>TestLSNAtLeast</code> , <code>TestParseControlCheckpoint</code>	C3, D4
<code>internal/branch/import_errors_test.go</code>	<code>TestPsqlErrorCollector</code> , <code>TestStreamJSONDocs</code>	I1, I5, I6
<code>internal/auth/auth_test.go</code>	<code>TestPasswordLogin</code> , <code>TestAPIKeys</code> , <code>TestVerifyKeyConcurrent</code> , <code>TestSessions</code> , <code>TestScopedKeys</code> , <code>TestScopeColumnAddedToExistingStore</code> , <code>TestConcurrentFirstOpen</code>	G1–G4, K5
<code>internal/branch/agent_access_test.go</code> , <code>internal/proxy/scope_test.go</code>	<code>TestAgentRolePassword</code> , <code>TestAgentGatewayDSN</code> , <code>TestGatewayHostPort</code> , <code>TestScopeAllows</code> , <code>TestLedgerOptionsLeavesAgentSessionToTheBranch</code>	K5, K6, H16
<code>internal/branch/branch_test.go</code>	<code>TestParsePublishedPort</code> , <code>TestDSN</code> , <code>TestAgentBranchName</code> , <code>TestContainerName</code>	C2, C4, K5
<code>internal/branch/branch_before_test.go</code>	<code>TestParseEntryJSON</code> , <code>TestRecoveryTarget</code> , <code>TestParseBackups</code> , <code>TestPickBaseBackup</code> , <code>TestWalArchived</code> , <code>TestValidateBeforeName</code> , <code>TestEntryQuery</code> , <code>TestRestoreBeforeScript</code>	H13
<code>internal/branch/ledger_entries_test.go</code>	<code>TestEntriesQuery</code> , <code>TestFormatLedgerEntries</code>	H13
<code>internal/ledger/impact_test.go</code> , <code>internal/branch/impact_test.go</code>	<code>TestSchemaImpact</code> , <code>TestParseTarget</code> , <code>TestTargetDestructive</code> , <code>TestScoreImpact</code> , <code>TestBranchParentParsing</code> , <code>TestFormatImpact</code> , <code>TestBuildDiff</code>	H17, H18
<code>internal/ledger/provenance_test.go</code> , <code>internal/branch/provenance_test.go</code>	<code>TestSchemaProvenance</code> , <code>TestValidProvenanceID</code> , <code>TestNewSessionID</code> , <code>TestCallHash</code> , <code>TestProvenanceSQL</code> , <code>TestClassifyExecError</code> , <code>TestFormatAgentSessions</code>	H16
<code>internal/ledger/statement_test.go</code>	<code>TestSchemaStatementMatching</code>	H5, H15, H4, H2
<code>internal/ledger/policy_test.go</code> , <code>internal/branch/policy_test.go</code>	<code>TestSchemaPolicy</code> , <code>TestParsePolicyDetail</code> , <code>TestPolicyContractExample</code> , <code>TestCommandTag</code> , <code>TestValidRuleID</code> , <code>TestValidatePolicyRule</code> , <code>TestPolicyRuleSQL</code> , <code>TestFriendlyPolicyErr</code> , <code>TestFormatPolicy</code>	H15
<code>internal/controlplane/blackbox_test.go</code> , <code>internal/mcp/mcp_test.go</code>	<code>TestBlackboxToLedgerPath</code> , <code>TestBlackboxAlias</code> , <code>TestBlackboxToolAliases</code>	H14
<code>internal/controlplane/openapi_test.go</code>	<code>TestOpenAPIMatchesRoutes</code> , <code>TestOpenAPIAliasRefsResolve</code>	K2
<code>internal/branch/security_test.go</code>	<code>TestTruthyEnv</code> , <code>TestSuperuserSwitchesDefaultOff</code>	K5, K6

<code>internal/ledger/integrity_test.go</code> , <code>internal/ledger/ledger_v2_test.go</code>	<code>TestRowHashMatchesSQLFormula</code> , <code>TestFormatAt</code> , <code>TestMerkleRoot</code> , <code>TestVerifyIntactAndUnanchoredTail</code> , <code>TestVerifyDetectsTampering</code> (6 cases), <code>TestVerifyDetectsAnchorGap</code> , <code>TestCheckChain</code> , <code>TestBuildAnchorRejectsBadRanges</code> , <code>TestAnchorFiles</code> , <code>TestDecodeRowJSON</code> , <code>TestQueries</code> , <code>TestSchemaV2IsAdditive</code> , <code>TestSchemaV2Safety</code>	H9–H12
<code>internal/branch/pipeline_test.go</code>	<code>TestSQLIdent</code> , <code>TestRenderSQL</code> , <code>TestTestQuery</code> , <code>TestJSONColumn</code>	I6, I13, J1, J2
<code>internal/controlplane/server_test.go</code>	<code>TestNameValidation</code>	K1
<code>internal/daemon/daemon_test.go</code>	<code>TestStopTerminatesProcess</code>	B2
<code>internal/host/host_test.go</code>	<code>TestIsELF</code> , <code>TestBundledLinuxBinaryPrefersCache</code> , <code>TestRefreshEngineBinaryDisabled</code> , <code>TestDownloadFile</code>	A7
<code>internal/update/update_test.go</code>	<code>TestParseVersion</code> , <code>TestCompareVersions</code> , <code>TestRequiredAssets</code> , <code>TestCandidates</code> , <code>TestChecksums</code> , <code>TestResolveAndDownload</code> , <code>TestReleaseListCache</code> , <code>TestBackgroundCheck</code> , <code>TestShouldCheck</code> , <code>TestInstallBinary</code> , <code>TestSwapExecutable</code> , <code>TestReplaceDirFromTarGz</code> , <code>TestReleaseWorkflowPublishesUpdateAssets</code>	A8, A9, M4
<code>internal/host/host_update_test.go</code> , <code>cmd/fox/update_test.go</code>	<code>TestUpdateFlow</code> , <code>TestUpdateChangesNothingUnlessReady</code> (8 cases), <code>TestUpdateInstallFailureRestartsOldEngine</code> , <code>TestParseUpdateArgs</code>	A9
<code>scripts/test_feature_checklist.sh</code>	same feature ids in both documents; no duplicates; valid statuses; badge matches status	M1, M2
<code>scripts/test_check_release_assets.sh</code>	complete release; missing attachment; unlisted checksum; binary-mode and CRLF lines; no SHA256SUMS	M4
<code>internal/host/host_wsl_test.go</code>	<code>TestResolveWSLDistro</code> , <code>TestWSLArgs</code> , <code>TestDecodeWSLList</code> , <code>TestWinPathToMnt</code> , <code>TestParseKernelRelease</code> , <code>TestImportLocalFile</code>	A4, A6
<code>internal/proxy/proxy_test.go</code>	<code>TestBuildParseStartupRoundtrip</code> , <code>TestReadStartupSkipsSSL</code> , <code>TestReadStartupUpgradesTLS</code> , <code>TestReadStartupHandlesCancel</code> , <code>TestSendError</code>	F1, E2, F6, F7
<code>deploy/install.Tests.ps1</code> , <code>deploy/windows-install.Tests.ps1</code>	Arch/asset URLs, PATH, encoding, WSL independence, elevation/reboot probes, checksum verification, E2E (opt-in)	A2

Integration section (<code>scripts/integration_test.sh</code>)	Assertions	Covers
§1 Fresh start + auth	unauthenticated API → 401; main ready; gateway rejects bad key; accepts API key	B2, G2, G3, F2, K1
§2 Branch isolation	branch sees its row; main doesn't have the table	C1, C2
§1b Hosting & login URLs	<code>fox vm</code> on Linux says there is no VM, its shell fails with a reason, an unknown subcommand is rejected; a password login sets a <code>Secure</code> , <code>SameSite=Lax</code> cookie; with no URLs set the GitHub authorize link calls back to <code>https://localhost:8080</code>	B1, G3, G5
§2b Branch from a branch	CLI <code>--from</code> copies the parent's data and leaves main alone; a missing parent is named; REST 404 for a missing parent and 201 naming the parent, with the copied row readable	C1, C2, K1
§5b Continuous import	replication starts into its branch; initial copy 1 of 1 tables; the rows are there; a source insert streams across; the branch survives a short-idle reaper while a plain branch is suspended and is still streaming; the list includes it; cutover returns <code>standalone</code> and it stops replicating; later	I8, C3, K1

source changes don't arrive and the data stays; a second cutover and a cutover of a branch that never replicated are refused 409

§3 Time-travel / PITR	backup + restore --to latest → 3 rows	D1–D3
§3b PITR target selection	the API lists backups newest-first with the newest flagged; a restore says which backup it starts from and it is not the newest; a point before the newest backup reaches 3 rows and not 9; a point before every backup is refused, names the oldest and leaves no container; --to latest still reaches the end of the archive	D2, D3, K1
§4 Suspend / resume	exited after suspend; running after resume; suspending main refused, main keeps running	C2, C3
§5 Agent branch API	POST returns DSN; table created and read	K5
§6 HA	1 streaming replica; write via gateway after failover; the promoted standby archives WAL and its archiver is shipping; a backup taken from it (named as such) adds one to the stored backups; backup list works with main stopped; fox status names the container serving main; a restore reaches a write made after the failover; ha enable disable failover and suspending the standby refused after it; fox up recreates the promoted standby and leaves the old main stopped; ha fallback : exit 0, main primary and not in recovery, pointer back on main, post-failover write kept, standby removed, gateway writes to main, WAL archiving resumed	C3, D4, F1
§7 Reaper regression	standby still running; idle branch exited	C3, C5
§8 Blackbox	DDL attributed human; DROP blocked; BLOCKED recorded; DELETE blocked; chain intact	H1–H4
§9 ETL pipeline	exit 0; raw tables; staged values; failing test → exit 1 with data kept	I4, J1, J2
§11 Import errors	a .sql dump with a failing statement exits 1, lists it, and the rest loaded; a clean dump exits 0; a JSON array with an invalid 2nd element exits 1 naming it; a valid array imports	I5, I6
§10 Schema fidelity	exact userId column; nested → jsonb; model query returns 1 row	I4, I6, I13
Integration section (scripts/integration_ledger_v2.sh)		
	Assertions	Covers
§0 Behaviour unchanged	guardrail error text; REST ledger and verify columns; row_hash formula (SQL vs Python); CLI verify intact	H2, H4, H6, K1
§1a Restore credentials	restore returns 3 rows; no hardcoded password printed; no minioadmin in the restore container	D3
§1b Override permission	ledger upgrade; non-admin refused with HINT; admin allowed; revoke restores the block; superuser override still works	H4, H8, F4
§1c Least privilege, attribution & MCP auth	agent DSN uses its own non-superuser role; DB usable; attributed to the agent; can't disable triggers or override; MCP list hides passwords; MCP run_sql as db_client ; compat switch; fox mcp without a key exits 1 saying how to make one and naming the env var; an invalid key is refused; run_sql records the key's account; a scoped key's call lands in its own branch and not main, another branch is refused, and the wider tools are refused; a human claiming to be an agent is recorded as human and an agent claiming to be human as an agent	K5, K6, H3
§2 Blackbox 2.0 capture	installed at start; upgrade idempotent; capture row per ledger row with xid/LSN; task and call hash; hash verifies; override recorded; failing capture doesn't block DDL; kill switch (database-wide, and a superuser session); a client's own SET bb.v2 = 'off' — also as	H9, H2

`db_admin` — does not skip capture; append-only; clients can't write; base chain intact

§3 Checkpoints & verifier	anchor written read-only in the published format; no-op when nothing new; out-of-sequence checkpoint refused; append-only; untampered branch INTACT via CLI, REST, MCP, fox-verify (live and export); forged edit with rewritten chain, deleted newest anchored entry and wiped ledger all TAMPERED; anchors outrank the database's checkpoint rows	H10–H12
§3b Scheduled checkpoints	with a 15s interval the control plane anchors new entries on its own	H10
§8 Impact & diff	impact on a table (views at depth 1 and 2 via the first, the foreign key from another table, its index, score), a column (only what uses it), quoted names, non-destructive, missing object, no target → <code>--object</code> , a view, depth limit, branch presence with parent, policy in the score, REST and MCP match the CLI, <code>execute_change</code> includes it; diff of a clone: parent, shared history, only-on-each-side, object both changed, <code>fox branch diff</code> matches, text output, self diff, REST on both paths, 404, MCP	H17, H18
§7 Agent provenance	sessions table installed, not client-writable; Agent API without provenance unchanged; with provenance: session / task / parent returned, recorded, and carried by the agent's own DDL; invalid id → 400, no branch; <code>execute_change</code> : applied with entry carrying agent / tool / session / task / parent / call hash, session recorded, no superuser rights, BBX02 warning, BBX01 block with its BLOCKED entry, dry run, stable call hash, database error, invalid id; CLI and REST sessions; chain verifies	H16, K5, K6
§6 Policy gate & console values	a uuid, a uuid[], an interval, an inet and a bytea read as Postgres writes them, while numeric, jsonb, arrays and text are unchanged; built-ins installed as warn; fires after the guardrail; warn runs with BBX02 and contract JSON for all four built-ins; block refuses with BBX01, JSON ids, BLOCKED entry and evaluation surviving the rollback, HINT; non-admin <code>bb.policy_allow</code> , session <code>bb.v2=off</code> and <code>bb.allow_destructive</code> don't bypass; admin override runs and is recorded (evaluation + <code>override_used</code>); REST list / 200 / 403 / 404 / 409 / check / evaluations; CLI check + exit code; MCP <code>policy_check</code> ; clients can't write rules; fail-safe; kill switch; custom / disabled rules; history; invalid pattern; built-in not removable; append-only; guardrail error unchanged; reinstall keeps changes; chain verifies	H15, H4, H9
§5b Blackbox page	<code>with=session</code> adds the session column and the default columns are unchanged; a table filter returns only that table's entries; table + risk narrows further; a date range covering today returns them and one ending in 2000 returns none	L5, K1
§5 Blackbox names	<code>fox blackbox verify entries integrity</code> , REST <code>.../blackbox/verify</code> , <code>.../blackbox</code> (same columns), <code>.../blackbox/entries</code> , 401 without a key, MCP listing and <code>blackbox_integrity</code> / <code>verify_blackbox</code> all match the original names	H14
§4 Branch before an entry	entry has a captured xid; its id is listed by <code>fox ledger entries</code> , REST and MCP; CLI branch-before by xid: rows before the change present, the change and later changes absent, ledger stops before the entry, no recovery settings left, reachable via the gateway, guardrail active, main untouched and its integrity result unchanged; refusals (existing name, BLOCKED, unknown, non-main source, bad id) leave nothing behind; REST 404 / 400 / 409 and success; MCP listed, refuses BLOCKED, succeeds by timestamp for an entry without capture; the branch deletes like any other	H13, K1, K6

Integration section (<code>scripts/integration_update.sh</code>)	Assertions	Covers
Setup	test releases 0.98.0 (installed) and 0.99.0 built and served by a local fake GitHub with a prerelease v1.0.0 and an incomplete v0.99.5; stack started on 0.98.0 with data on main and a branch	A8, A9
§1–§3 Notice	the notice is start's last line, once, with the banner still printed; <code>FOX_NO_UPDATE_CHECK=1</code> prints nothing; offline prints nothing and doesn't slow start	A8
§4–§6 Before updating	<code>--check</code> reports v0.99.0 and not the prerelease or incomplete release, changes nothing; without a terminal, or with stdin closed, <code>--yes</code> is required; a tampered download is refused ("Nothing was changed"), version and servers unchanged, no partial file left	A9
§7 Update	exits 0 and says done; Blackbox upgraded on the branch; <code>fox</code> is 0.99.0; the three servers restarted and run the new binary; main and branch containers not recreated; main and branch data unchanged; every Blackbox entry from before the update unchanged on main and the branch (hash of the entries), both chains intact, new Blackbox objects present; <code>secrets.json</code> and TLS certificates unchanged; previous engine kept; the gateway serves main with an API key made before the update; control plane answers	A9
§8 Afterwards	<code>--check</code> says up to date; <code>start</code> shows no notice	A8, A9

Not covered by any automated test: most MCP tools, OAuth against a real provider, REST pipeline handlers, SDKs, web UI, SCRAM, storage quotas, non-Mongo imports. `fox vm` is covered on Linux only (macOS and Windows were exercised by hand and compile-checked).

Known issues

Found while building this inventory. Recorded, not fixed — each fix changes existing behaviour and needs explicit approval. Struck-through items have been fixed.

- ~~1. PITR restore broken on fresh installs (hardcoded minioadmin) —~~ **fixed** in Phase 1. (D3, H7)
- ~~2. Destructive DDL override ungated —~~ **fixed** in Phase 1 (superusers and db_admin only). Use of the override is now recorded (H9). (H4, H8)
- ~~3. Agents and MCP run as superuser —~~ **fixed** in Phase 1 (own agent-<id> role; MCP as db_client). (K5, K6, H3)
- ~~4. Integration \$1 and \$5 call http:// endpoints that now serve TLS —~~ **fixed** in 13734a7 .
- ~~5. Integration "bad key" and "append-only" checks gave false failures —~~ **fixed** (test-only; both protections confirmed working in the VM).
- ~~6. fox branch suspend main stops the primary and the gateway won't wake it —~~ **fixed**: suspend refuses main and the HA standby. (C3)
- ~~7. fox ha enable after a failover deletes the live primary (pg-standby); post failover ha disable routes main to a stopped container —~~ **fixed**: after a failover ha enable|disable|failover are refused, fox up brings the promoted standby back instead of the old main, and fox ha fallback returns to main keeping every write. (D4)
- ~~8. backup / restore always target pg-main , even after a failover; PITR always fetches the LATEST base backup —~~ **fixed**: fox backup create backs up whichever container serves main (and says when that is not pg-main), the listing is read from a throwaway container so it works with main stopped, and a restore starts from the newest base backup that precedes the target — a target no backup precedes is refused, naming the oldest. Verified on a real install: a restore to three weeks earlier chose the right backup and came up without the fox schema, which did not exist then. (D2, D3, H13)
- ~~9. A promoted standby archived no WAL, so nothing written while main ran on it could be backed up or restored —~~ **fixed**: the standby runs with the primary's object-storage environment and archive_mode=on , which archives only once it is out of recovery. (D4)
- ~~10. Ledger verify can't detect deletion of the newest rows or a wiped table —~~ **addressed** in Phase 3: checkpoint anchors kept outside the database catch it (H10–H12). Entries written since the last checkpoint are still only chain-checked. (H2)
- ~~11. A release job could replace SHA256SUMS and drop the other job's entries (v0.8.1 lost foxbyte-distro.tar.gz , so the Windows installer skipped verifying it) —~~ **fixed** for future releases by the shared merge script. v0.8.1 itself needs wsl-distro.yml re-run for tag v0.8.1 by a repository admin. (M4, M5)
- ~~12. MinIO images are pulled from Docker Hub, where minio/minio and minio/mc no longer exist — the Windows distro release build and fresh installs failed —~~ **fixed**: pinned quay.io images, with cached old images still used. (D1, A4)
- ~~13. The 2.0 capture kill switch bb.v2 = 'off' could be set by any client for its own session, skipping its capture rows —~~ **fixed**: it is now honoured only database-wide or in a superuser's session. (H9)
- ~~14. The new release notice never appeared against real GitHub: the check's SHA256SUMS request is throttled to 7–75 s, far past its budget —~~ **fixed**: the check is one request (the releases list) and its answer is remembered for 6 hours. (A8)
- ~~15. Policy patterns match the whole query text a client sent (all statements of a multi-statement string, comments and literals included) —~~ **fixed**: rules and risk flags match the statement being run, without comments or literals, falling back to the whole query only when that statement can't be established. (H15, H5)
- ~~16. A blocked command after a Blackbox write in the same transaction hung the client forever (the dblink connection recording the attempt waited on its own session's lock) —~~ **fixed**: the attempt goes unrecorded with a WARNING and the block stands. (H4, H15)
- ~~17. fox ledger revert — to doesn't revert a branch —~~ **fixed**: its help and output now say it is a point-in-time restore of main; H13 branches from just before a change. (H7)
- ~~18. README documents fox branch diff (didn't exist) —~~ **now added** (H18) — and ledger UI filters by table/risk (not in the UI) — **now added**: the Blackbox page filters by table, risk and date range, verifies the chain and expands a row to the statement and session. (L5)
- ~~19. OpenAPI spec disagrees with the code (import fields, status codes, missing params/routes) —~~ **fixed**: the spec was corrected against the handlers, and TestOpenAPIMatchesRoutes fails the build if the two diverge again. (K2)
- ~~20. .sql imports don't stop on error; JSON array imports stop silently at the first bad element —~~ **fixed**: both now fail, listing the failed statements or naming the invalid element. Web Import mentions SQLite (no loader). (I5, I6, I12)
- ~~21. Continuous import cutover is CLI only —~~ **fixed**: GET /api/replication , GET /api/branches/{name}/replication and POST .../replication/cutover , with a **Continuous imports** panel on the Import page. (I8)
- ~~22. The Gateway suspended a branch mid-continuous import, so the import silently stopped about two minutes in —~~ **fixed**: found by watching the new panel — a subscription's apply worker is a background worker, so the branch reported zero client connections and looked idle. The reaper now leaves a replicating branch running. (I8, C3)
- ~~23. Creating a branch from a non-main parent a second time failed: zfs destroy only removed the main@for-<branch> snapshot, leaving the parent's clone snapshot behind —~~ **fixed**: delete removes the clone's own origin snapshot too. (C1, C2)

24. ~~Agent cap and TTL reaper only count running containers; agent DSN only reachable inside the VM~~ — **fixed**: all three count suspended branches, and the DSN is Gateway-routed with a branch-scoped key, so it works from the host without widening what an agent may reach. (K5)
25. ~~Auth defaults point at http://localhost:8080 + http://localhost:5173~~ — **fixed**: the public URL defaults to https://localhost:8080 — what the engine actually serves — and the web origin defaults to the public URL, so a default install's callbacks and post-login redirect land on the running console. Session cookies follow the origin: `Secure` when it is https, and `SameSite=None` only when the console is on a different origin than the API. (G5, G3)
26. ~~fox vm is listed as a command but not implemented~~ — **fixed**: `fox vm status` reports the engine VM's state and size and `fox vm shell` opens a shell in it (Lima on macOS, WSL on Windows); on Linux both say there is no VM. (B1)
27. ~~Windows installer never downloads the prebuilt distro; its preloaded image tag doesn't match the engine's image name; setup summary omits the API key~~ — **fixed**: the installer prefers the verified distro image, the build and the loader use the engine's image name, and the summary prints the connection string with the key. (A2, A4, M5)
28. ~~Web Guide/Docs pages are stale in places~~ — **fixed**: both were read against the code and refreshed (https, `sslmode=require`, the full REST surface, MCP). (L7)
29. ~~The SQL console could never override the guardrail — not even for an admin — so DROP TABLE failed with the same message however it was retried~~ — **fixed**: the console logged every user in as the shared `db_client` role, which is never in `db_admin`, and each run is a new connection so a typed `SET` was lost. It now logs in as the user, and the override travels with the run. (L4, H8)
30. ~~The integration suites run against the install's own main and leave tables there (pit, led_a, repl, rlsdemo, t1, ...), which then show as user tables in the console~~ — **fixed**: worse than tables, they deleted main's Blackbox history, restored main to earlier points and left 8 test accounts with 70 working API keys. They now run only in a throwaway test VM and refuse to run anywhere else. On this install the tables were dropped and the keys revoked; the Blackbox history from before 16 Sep 2026 cannot be recovered. (M1)
31. ~~On a new install the first fox start could leave the control plane and the Agent API down (SQLITE_BUSY creating the auth store)~~ — **fixed**: store creation takes the write lock up front and retries. (G1)
32. ~~An agent's changes through its Gateway DSN lost the session its branch was created with~~ — **fixed**: the Gateway no longer gives an agent key a session of its own. (H16)
33. ~~Tool, kind and session are client-settable, so a client can record itself as the other kind of actor~~ — **fixed** for the kind, which is what attribution rests on: it is derived from the login role (an `agent-<id>` role is an agent, a per-user role is a person), so an agent can no longer be recorded as a human. Tool and session are declared by the client by design and are now labelled that way on the Blackbox page. (H3)
34. ~~The MCP server has no authentication of its own: any client that can start it gets all 16 tools, and its changes are recorded against the shared role~~ — **fixed**: `fox mcp` requires an API key and acts as that account, with no unauthenticated path; a key scoped to one branch is held to that branch. (K6)
35. ~~The console displayed a uuid column as a list of 16 numbers (and an interval as a struct, an inet with quotes, a bytea as mangled text)~~ — **fixed**: values `pgx` decodes into byte arrays or structs are shown in Postgres's own text form. Reported from the UI, where every `id` column looked like `[195,64,152,78,...]`. (L4)
36. ~~setup hung for ever when a container named minio already existed on the machine~~ — **fixed**: the object store is called `objstore`, a name of ours rather than the software's, and the bucket step gives up with a clear message instead of retrying invisibly. Found by a user whose machine had another project's `minio`. (A3, D1)
37. ~~Removing the product was a list of commands to run by hand, and missing one left containers, a pool or a stale binary behind~~ — **fixed**: `fox uninstall` lists what it will remove, asks, then removes it — including installs left by retired product names. (B1)
38. ~~Renaming the product broke behaviour, not just text: the database raised a message carrying the product's name, the SQL console matched that prose, and a test asserted it~~ — **fixed**: messages the database raises carry no product name, and the names written into data carry none either, so a rename cannot reach them. (H4, L4)
39. Storage quotas are ZFS-only and per-branch caps are opt-in. (C6)
40. The Postgres image has no pgvector. (M6)